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# OGC DOCUMENT TITLE

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**COMMUNITY STANDARD**

**APPROVED**

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## ABSTRACT

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<Insert Abstract Text here>



## KEYWORDS

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The following are keywords to be used by search engines and document catalogues.

keyword\_1, keyword\_2, keyword\_3, etc.



# PREFACE

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This document establishes the OGC CRS ontology and its submodules. The definition of elements of coordinate reference systems is an essential part of geospatial data provision. However, until now, coordinate reference systems and their components could not be represented in an OGC-standardized semantic web vocabulary. This document introduces the ontology model, its classes and properties, application examples and can serve as the foundation of a semantic web based coordinate system registry at OGC. Special attention is given to the compatibility of the CRS Ontology vocabulary to other OGC-endorsed Semantic Web standards such as GeoSPARQL and alignments to other data standards are provided as part of this specification.

**NOTE:** Insert Preface Text here. Give OGC specific commentary: describe the technical content, reason for document, history of the document and precursors, and plans for future work.

There are two ways to specify the Preface: “simple clause” or “full clause”

If the Preface does not contain subclauses, it is considered a simple preface clause. This one is entered as text after the `.Preface` label and must be placed between the AsciiDoc document attributes and the first AsciiDoc section title. It should not be give a section title of its own.

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## IV

## SECURITY CONSIDERATIONS

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No security considerations have been made for this Standard.

## V

## SUBMITTERS

---

All questions regarding this submission should be directed to the editor or the submitters:

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| Ghislain Atemezing    | European Union Agency for Railways (ERA) | Yes        |

## VI

## SOURCE OF THE CONTENT FOR THIS OGC DOCUMENT

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## VII

## VALIDITY OF CONTENT

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## VIII

## FUTURE WORK

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**NOTE:** If you need to place any further sections in the preface area use the [.preface] attribute.





## CONTRIBUTORS

---

Additional contributors to this Standard include the following:

Individual name(s), Organization



1

# SCOPE

---



# SCOPE

---

<Insert Scope text here>

**NOTE:** Give the subject of the document and the aspects of that scope covered by the document.



2

# CONFORMANCE

---



## CONFORMANCE

---

<Insert conformance content here>

**NOTE:** Provide a short description of the content approached in subsequent sections and the main subject of the document



3

# NORMATIVE REFERENCES

---

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

*Identification of Common Molecular Subsequences.* Smith, T.F., Waterman, M.S., J. Mol. Biol. 147, 195–197 (1981)

*ZIB Structure Prediction Pipeline: Composing a Complex Biological Workflow through Web Services.* May, P., Ehrlich, H.C., Steinke, T. In: Nagel, W.E., Walter, W.V., Lehner, W. (eds.) Euro-Par 2006. LNCS, vol. 4128, pp. 1148–1158. Springer, Heidelberg (2006)

*The Grid: Blueprint for a New Computing Infrastructure.*, Foster, I., Kesselman, C.. Morgan Kaufmann, San Francisco (1999).

*Grid Information Services for Distributed Resource Sharing.* Czajkowski, K., Fitzgerald, S., Foster, I., Kesselman, C. In: 10th IEEE International Symposium on High Performance Distributed Computing, pp. 181–184. IEEE Press, New York (2001)

The background is a solid dark blue. It features several thin, light yellow lines that intersect to form a network of triangles and polygons. At each of these intersection points, there is a small, solid yellow circle. The overall effect is a modern, geometric pattern.

4

# TERMS AND DEFINITIONS

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This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

## 4.1. example term

---

term used for exemplary purposes

**Note 1 to entry:** An example note.

Example      Here’s an example of an example term.

[SOURCE: ]



5

# CONVENTIONS

---

**NOTE:** This section provides details and examples for any conventions used in the document. Examples of conventions are symbols, abbreviations, use of XML schema, or special notes regarding how to read the document.

## 5.1. Identifiers

---

The normative provisions in this standard are denoted by the URI

<http://www.opengis.net/spec/{standard}/{m.n}>

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

## 5.2. Other conventions

---

<Place any other convention needed with its corresponding title>



6

# CORE

---

This clause establishes the **Core** Requirements class, with IRI /req/core, which has a corresponding Conformance Class, **Core**, with IRI /conf/core.

The Core module establishes a set of classes and properties which define the building blocks of a spatial reference system definition. Some of the definitions are extended in specialized modules related to the Core module.

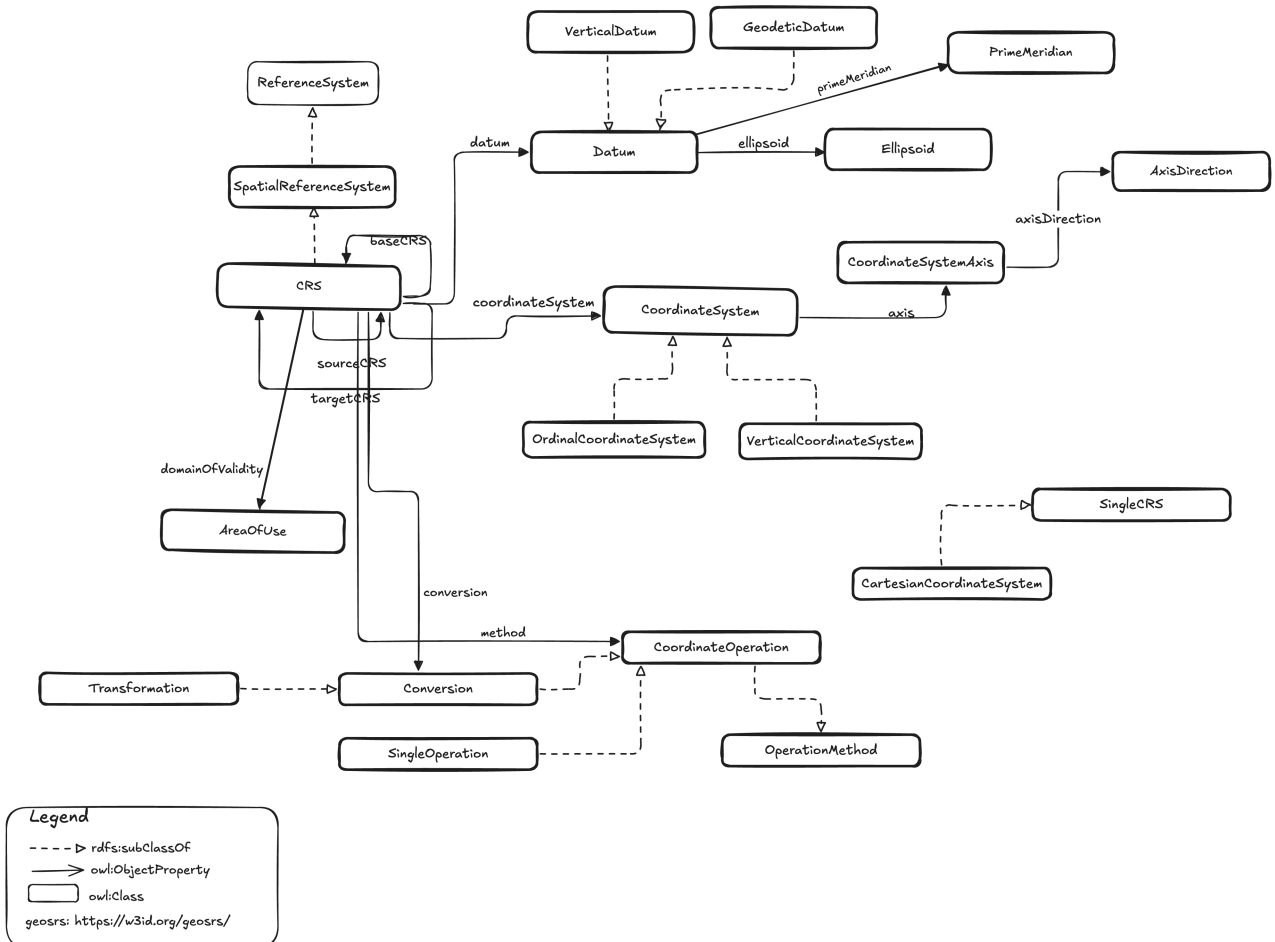


Figure 1

From a base class SpatialReferenceSystem, we define a class for a coordinate system, as the superclass of all spatial reference systems describing locations using coordinates. These SpatialReferenceSystems are described using a Datum and a coordinate system definitions with at least one coordinate axis. Together with several subtypes of coordinate reference system, these definitions complete the Core module.

## REQUIREMENTS CLASS 1: 06-CORE.ADOC EXTENSION

|                   |                                                  |
|-------------------|--------------------------------------------------|
| IDENTIFIER        | /req/core                                        |
| TARGET TYPE       | Implementation Specification                     |
| CONFORMANCE CLASS | Conformance class A.1: /conf/core                |
| REQUIREMENT       | /req/core/Coordinate_Reference_System_Parameters |
|                   | /req/core/Coordinate_Reference_System_Types      |
|                   | /req/core/Coordinate_Reference_System_Properties |

## 6.1. Coordinate Reference System Parameters

### REQUIREMENT 1: COORDINATE REFERENCE SYSTEM PARAMETERS

|            |                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/core/Coordinate_Reference_System_Parameters                                                                                                                                       |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:AreaOfUse, geosrs:Extent, geosrs:GeographicBoundingBox, geosrs:AxesList, geosrs:SingleCRSList to be used in SPARQL graph patterns. |

#### 6.1.1. Class: geosrs:AreaOfUse

Table 1 — geosrs:AreaOfUse

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/AreaOfUse">https://w3id.org/geosrs/srs/AreaOfUse</a> |
| Definition    | Area within which a coordinate operation may be used.                                     |
| Super-classes | geo:Feature[geo:Feature]                                                                  |
| Example       | <a href="#"><u>geosrs:AreaOfUse</u></a>                                                   |

#### 6.1.2. Class: geosrs:Extent

**Table 2 — geosrs:Extent**

|            |                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/srs/Extent">https://w3id.org/geosrs/srs/Extent</a>                                                        |
| Definition | Geographic area or time interval in which the referring object is valid. Cf. ISO 19115-1:2014:2014-04, part 6.6.1 and table B.15 line 335. |

### 6.1.3. Class: geosrs:GeographicBoundingBox

**Table 3 — geosrs:GeographicBoundingBox**

|            |                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/srs/GeographicBoundingBox">https://w3id.org/geosrs/srs/GeographicBoundingBox</a> |
| Definition | Frame delimiting an area of interest. See ISO 19115-1:2014:2014-04, part 6.6.1 and table B.15.1 line 344.         |

### 6.1.4. Class: geosrs:AxesList

**Table 4 — geosrs:AxesList**

|            |                                                                                         |
|------------|-----------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/srs/AxesList">https://w3id.org/geosrs/srs/AxesList</a> |
| Definition | Ordered list of coordinate system axes.                                                 |

### 6.1.5. Class: geosrs:SingleCRSList

**Table 5 — geosrs:SingleCRSList**

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/srs/SingleCRSList">https://w3id.org/geosrs/srs/SingleCRSList</a> |
| Definition | Ordered list of simple reference coordinate systems.                                              |

## 6.2. Coordinate Reference System Properties

---

## REQUIREMENT 2: COORDINATE REFERENCE SYSTEM PROPERTIES

|            |                                                                                                    |
|------------|----------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/core/Coordinate_Reference_System_Properties                                                   |
| STATEMENT  | Implementations shall allow the RDFS properties geosrs:method to be used in SPARQL graph patterns. |

### 6.2.1. Property: geosrs:method

Table 6 – geosrs:method

|        |                                                                             |
|--------|-----------------------------------------------------------------------------|
| URI    | <a href="https://w3id.org/geosrs/method">https://w3id.org/geosrs/method</a> |
| Type   | <a href="#">owl:ObjectProperty</a>                                          |
| Range  | <a href="#">CoordinateOperation</a>                                         |
| Domain | <a href="#">CRS</a>                                                         |

## 6.3. Coordinate Reference System Types

## REQUIREMENT 3: COORDINATE REFERENCE SYSTEM TYPES

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/core/Coordinate_Reference_System_Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:BoundCRS, geosrs:CompoundCRS, geosrs:CRS, geosrs:EngineeringCRS, geosrs:GeocentricCRS, geosrs:GeodeticCRS, geosrs:GeographicCRS, geosrs:ParametricCRS, geosrs:ProjectedCRS, geosrs:SelenographicCRS, geosrs:ReferenceSystem, geosrs:SingleCRS, geosrs:SpatialReferenceSystem, geosrs:SpatioParametricCompoundCRS, geosrs:SpatioParametricTemporalCompoundCRS, geosrs:SpatioTemporalCompoundCRS, geosrs:StaticCRS, geosrs:TemporalCRS, geosrs:VerticalCRS to be used in SPARQL graph patterns. |

Coordinate reference systems are typed according to their area of application, e.g. Geodetic vs. Engineering vs. TemporalCRS and by their ability to contain further

### 6.3.1. Class: geosrs:BoundCRS



**Table 7 — geosrs:BoundCRS**

|               |                                                                                         |
|---------------|-----------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/BoundCRS">https://w3id.org/geosrs/srs/BoundCRS</a> |
| Super-classes | <a href="#">CRS</a>                                                                     |

### 6.3.2. Class: geosrs:CompoundCRS

**Table 8 — geosrs:CompoundCRS**

|               |                                                                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/CompoundCRS">https://w3id.org/geosrs/srs/CompoundCRS</a>                                                                                |
| Definition    | Coordinate reference system using at least two independent single coordinate reference systems. Cf. ISO 19111:2007:2007-07, parts 8.2.3.c, 8.2.4, table 6 and annex B.1.2.4. |
| Super-classes | <a href="#">CRS</a>                                                                                                                                                          |
| Example       | <a href="#">geosrs:CompoundCRS</a>                                                                                                                                           |

### 6.3.3. Class: geosrs:CRS

**Table 9 — geosrs:CRS**

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/CRS">https://w3id.org/geosrs/srs/CRS</a>                                                                                                                                                                                                                                                                                                                                                    |
| Definition    | Depending on the spatial dimension of coordinates (1D, 2D, 3D), this piece of metadata is used for specifying the elements of definition associated to a given set of coordinates: its datum, its ellipsoid, its prime meridian, the type of coordinates (geocentric, geographic, projected,...), the coordinates units of measure, when appropriate the cartographic projection used, the vertical coordinate reference system. |
| Super-classes | <a href="#">SpatialReferenceSystem</a>                                                                                                                                                                                                                                                                                                                                                                                           |

### 6.3.4. Class: geosrs:EngineeringCRS

**Table 10** — geosrs:EngineeringCRS

|               |                                                                                                                                                                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/EngineeringCRS">https://w3id.org/geosrs/srs/EngineeringCRS</a>                                                                                                                                                                              |
| Definition    | A contextually local coordinate reference system which can be divided into two broad categories: — earth-fixed systems applied to engineering activities on or near the surface of the earth; — CRSs on moving platforms such as road vehicles, vessels, aircraft or spacecraft. |
| Super-classes | <a href="#">CRS</a>                                                                                                                                                                                                                                                              |

### 6.3.5. Class: geosrs:GeocentricCRS

**Table 11** — geosrs:GeocentricCRS

|               |                                                                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/GeocentricCRS">https://w3id.org/geosrs/srs/GeocentricCRS</a>                                                                                                                                                                               |
| Definition    | A cartesian coordinate reference system that represents locations in the vicinity of the Earth (including its surface, interior, atmosphere, and surrounding outer space) as X, Y, and Z measurements from its center of mass. Commonly used to track the orbits of satellites. |
| Super-classes | <a href="#">CRS</a>                                                                                                                                                                                                                                                             |
| Example       | <a href="#">geosrs:GeocentricCRS</a>                                                                                                                                                                                                                                            |

### 6.3.6. Class: geosrs:GeodeticCRS

**Table 12** — geosrs:GeodeticCRS

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/GeodeticCRS">https://w3id.org/geosrs/srs/GeodeticCRS</a>                                         |
| Definition    | Coordinate Reference System associated with a geodetic datum. Cf. ISO 19111:2007:2007-07, part 8.2.2.a, table 10 and annex B.1.2.1.a. |
| Super-classes | <a href="#">CRS</a>                                                                                                                   |

### 6.3.7. Class: geosrs:GeographicCRS

**Table 13** — geosrs:GeographicCRS

|               |                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/GeographicCRS">https://w3id.org/geosrs/srs/GeographicCRS</a>    |
| Definition    | Coordinate Reference System that has a geodetic reference frame and an ellipsoidal coordinate system |
| Super-classes | <a href="#">CRS</a>                                                                                  |
| Example       | <a href="#">geosrs:GeographicCRS</a>                                                                 |

### 6.3.8. Class: geosrs:ParametricCRS

**Table 14** — geosrs:ParametricCRS

|               |                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/ParametricCRS">https://w3id.org/geosrs/srs/ParametricCRS</a> |
| Definition    | Coordinate Reference System based on a parametric datum                                           |
| Super-classes | <a href="#">CRS</a>                                                                               |

### 6.3.9. Class: geosrs:ProjectedCRS

**Table 15** — geosrs:ProjectedCRS

|               |                                                                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/ProjectedCRS">https://w3id.org/geosrs/srs/ProjectedCRS</a>                                                                                                     |
| Definition    | Coordinate Reference System derived from a two-dimensional geodetic coordinate reference system by applying a map projection. Cf. ISO 19111:2007:2007-07, part 8.2.3.b, table 11 and annex B.1.2.3. |
| Super-classes | <a href="#">CRS</a>                                                                                                                                                                                 |
| Example       | <a href="#">geosrs:ProjectedCRS</a>                                                                                                                                                                 |

### 6.3.10. Class: geosrs:SelenographicCRS

**Table 16** — geosrs:SelenographicCRS

|               |                                                                                                         |
|---------------|---------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/SelenographicCRS">https://w3id.org/geosrs/srs/SelenographicCRS</a> |
| Definition    | Coordinate Reference System to refer locations on the surface of the Earth's Moon.                      |
| Super-classes | <a href="#">CRS</a>                                                                                     |

### 6.3.11. Class: geosrs:ReferenceSystem

**Table 17** — geosrs:ReferenceSystem

|            |                                                                                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/srs/ReferenceSystem">https://w3id.org/geosrs/srs/ReferenceSystem</a>                                                                                                                                 |
| Definition | An abstract coordinate system, whose origin, orientation and scale are specified in physical space. It is based on a set of reference points, defined as geometric points whose position is identified physically and mathematically. |

### 6.3.12. Class: geosrs:SingleCRS

**Table 18** — geosrs:SingleCRS

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/SingleCRS">https://w3id.org/geosrs/srs/SingleCRS</a>                           |
| Definition    | Coordinate reference system consisting of one coordinate system and one datum. Cf. ISO 19111:2007:2007-07, table 5. |
| Super-classes | <a href="#">CRS</a>                                                                                                 |

### 6.3.13. Class: geosrs:SpatialReferenceSystem

**Table 19** — geosrs:SpatialReferenceSystem

|            |                                                                                                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/srs/SpatialReferenceSystem">https://w3id.org/geosrs/srs/SpatialReferenceSystem</a>                                                                                                           |
| Definition | A spatial reference system (SRS) is a system for establishing spatial position. A spatial reference system can use geographic identifiers (place names, for example), coordinates (in which case it is a coordinate reference |

|               |                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------|
|               | system), or identifiers with structured geometry (in which case it is a discrete global grid system). |
| Super-classes | <a href="#">ReferenceSystem</a>                                                                       |

### 6.3.14. Class: geosrs:SpatioParametricCompoundCRS

**Table 20** — geosrs:SpatioParametricCompoundCRS

|               |                                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/SpatioParametricCompoundCRS">https://w3id.org/geosrs/srs/SpatioParametricCompoundCRS</a>                                                                                       |
| Definition    | A spatio-parametric coordinate reference system is a compound CRS in which one component is a geographic 2D, projected 2D or engineering 2D CRS, supplemented by a parametric CRS to create a three-dimensional CRS |
| Super-classes | <a href="#">CompoundCRS</a>                                                                                                                                                                                         |

### 6.3.15. Class: geosrs:SpatioParametricTemporalCompoundCRS

**Table 21** — geosrs:SpatioParametricTemporalCompoundCRS

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/SpatioParametricTemporalCompoundCRS">https://w3id.org/geosrs/srs/SpatioParametricTemporalCompoundCRS</a> |
| Definition    | Coordinate reference system combining a spatio-parametric reference system with at least one temporal reference system                        |
| Super-classes | <a href="#">SpatioParametricCompoundCRS</a>                                                                                                   |

### 6.3.16. Class: geosrs:SpatioTemporalCompoundCRS

**Table 22** — geosrs:SpatioTemporalCompoundCRS

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/SpatioTemporalCompoundCRS">https://w3id.org/geosrs/srs/SpatioTemporalCompoundCRS</a> |
| Definition    | Coordinate reference system combining a spatial reference system with at least one temporal reference system              |
| Super-classes | <a href="#">CompoundCRS</a>                                                                                               |

### 6.3.17. Class: geosrs:StaticCRS

**Table 23** — geosrs:StaticCRS

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/StaticCRS">https://w3id.org/geosrs/srs/StaticCRS</a> |
| Definition    | Coordinate Reference System that has a static reference frame                             |
| Super-classes | <a href="#">CRS</a>                                                                       |

### 6.3.18. Class: geosrs:TemporalCRS

**Table 24** — geosrs:TemporalCRS

|               |                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/TemporalCRS">https://w3id.org/geosrs/srs/TemporalCRS</a> |
| Definition    | Coordinate Reference System based on a temporal datum                                         |
| Super-classes | <a href="#">CRS</a>                                                                           |

### 6.3.19. Class: geosrs:VerticalCRS

**Table 25** — geosrs:VerticalCRS

|               |                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/srs/VerticalCRS">https://w3id.org/geosrs/srs/VerticalCRS</a>                                                                                                                                                                                                                                                                              |
| Definition    | One-dimensional coordinate reference system associated with a vertical datum and used for recording heights or depths. Ellipsoidal heights are not captured in a vertical coordinate reference system but as part of a 3D coordinates tuple defined in a geodetic 3D coordinate reference system. Cf. ISO 19111:2007:2007-07, parts 8.2.2.b, table 14 and annex B.1.2.1.b. |
| Super-classes | <a href="#">CRS</a>                                                                                                                                                                                                                                                                                                                                                        |
| Example       | <a href="#">geosrs:VerticalCRS</a>                                                                                                                                                                                                                                                                                                                                         |

7

# COORDINATE OPERATION MODULE

---

# COORDINATE OPERATION MODULE

This clause establishes the **Co** Requirements class, with IRI /req/co, which has a corresponding Conformance Class, **Co**, with IRI /conf/co.

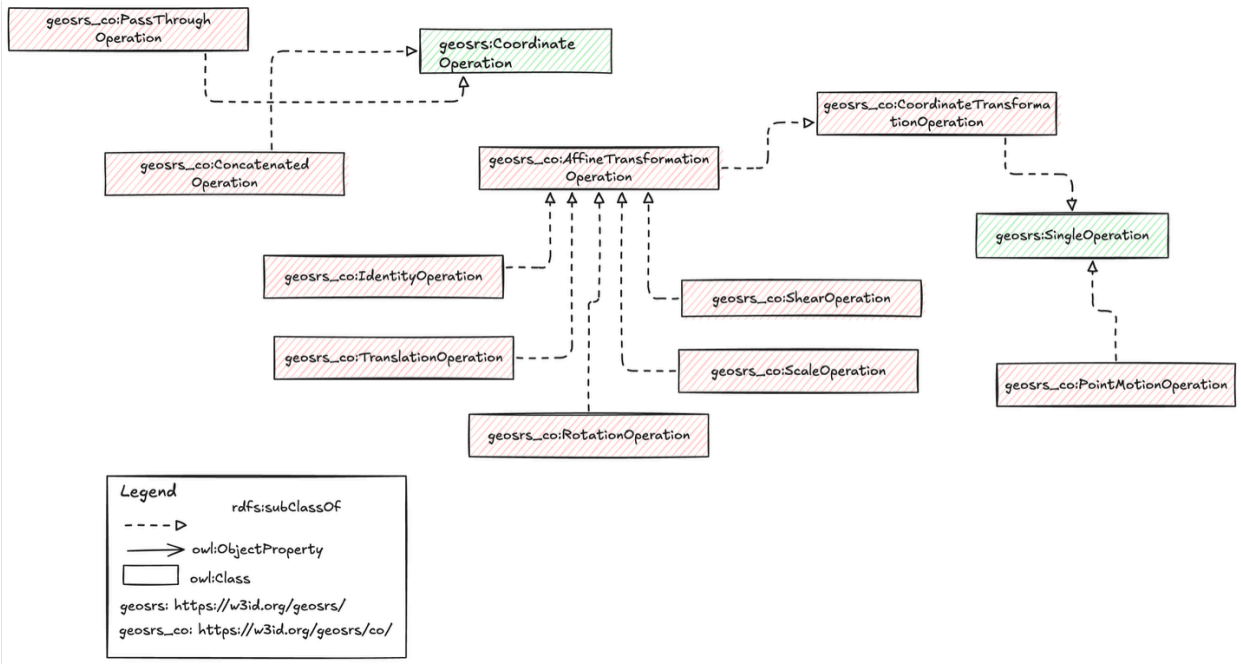


Figure 2

| Requirements Class 2: 07-co_module.adoc Extension |                                         |
|---------------------------------------------------|-----------------------------------------|
| IDENTIFIER                                        | /req/co                                 |
| TARGET TYPE                                       | Implementation Specification            |
| CONFORMANCE CLASS                                 | Conformance class A.2: /conf/co         |
| REQUIREMENT                                       | /req/co/Coordinate_Operation_Methods    |
|                                                   | /req/co/Coordinate_Operation_Parameters |
|                                                   | /req/co/Coordinate_Operation_Categories |
|                                                   | /req/co/Coordinate_Operation_Properties |



# 7.1. Coordinate Operation Categories

| Requirement 4: Coordinate Operation Categories |                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                                     | /req/co/Coordinate_Operation_Categories                                                                                                                                                                                                                                                                    |
| STATEMENT                                      | Implementations shall allow the RDFS classes geosrs:GeographicObject, geosrs:Register Operations, geosrs:ScaleOperation, geosrs:RotationOperation, geosrs:IdentityOperation, geosrs:ShearOperation, geosrs:TranslationOperation, geosrs:AffineTransformationOperation to be used in SPARQL graph patterns. |

## 7.1.1. Class: geosrs:GeographicObject

Table 26 — geosrs:GeographicObject

|               |                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/GeographicObject">https://w3id.org/geosrs/co/GeographicObject</a> |
| Definition    | Identifier of a geographic feature of which the coordinates are used as operation parameters.         |
| Super-classes | <a href="#">Geometry#Geometry</a>                                                                     |

## 7.1.2. Class: geosrs:RegisterOperations

Table 27 — geosrs:RegisterOperations

|            |                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/RegisterOperations">https://w3id.org/geosrs/co/RegisterOperations</a> |
| Definition | Operations supported in the Coordinate Operations package.                                                |

## 7.1.3. Class: geosrs:ScaleOperation

Table 28 — geosrs:ScaleOperation

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/ScaleOperation">https://w3id.org/geosrs/co/ScaleOperation</a> |
| Definition | Scale transformation operation                                                                    |

|               |                                                      |
|---------------|------------------------------------------------------|
| Super-classes | <a href="#"><u>AffineTransformationOperation</u></a> |
|---------------|------------------------------------------------------|

#### 7.1.4. Class: geosrs:RotationOperation

**Table 29** — geosrs:RotationOperation

|               |                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/RotationOperation"><u>https://w3id.org/geosrs/co/RotationOperation</u></a> |
| Definition    | Rotation transformation operation                                                                              |
| Super-classes | <a href="#"><u>AffineTransformationOperation</u></a>                                                           |

#### 7.1.5. Class: geosrs:IdentityOperation

**Table 30** — geosrs:IdentityOperation

|               |                                                                                                                |
|---------------|----------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/IdentityOperation"><u>https://w3id.org/geosrs/co/IdentityOperation</u></a> |
| Definition    | Identity transformation operation                                                                              |
| Super-classes | <a href="#"><u>AffineTransformationOperation</u></a>                                                           |

#### 7.1.6. Class: geosrs:ShearOperation

**Table 31** — geosrs:ShearOperation

|               |                                                                                                          |
|---------------|----------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/ShearOperation"><u>https://w3id.org/geosrs/co/ShearOperation</u></a> |
| Definition    | Shear transformation operation                                                                           |
| Super-classes | <a href="#"><u>AffineTransformationOperation</u></a>                                                     |

#### 7.1.7. Class: geosrs:TranslationOperation

**Table 32** — geosrs:TranslationOperation

|               |                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/TranslationOperation">https://w3id.org/geosrs/co/TranslationOperation</a> |
| Definition    | Translation transformation operation                                                                          |
| Super-classes | <a href="#">AffineTransformationOperation</a>                                                                 |

### 7.1.8. Class: geosrs:AffineTransformationOperation

**Table 33** — geosrs:AffineTransformationOperation

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/AffineTransformationOperation">https://w3id.org/geosrs/co/AffineTransformationOperation</a> |
| Definition    | Affine coordinate transformation operation                                                                                      |
| Super-classes | <a href="#">CoordinateTransformationOperation</a> []                                                                            |

## 7.2. Coordinate Operation Methods

### REQUIREMENT 5: COORDINATE OPERATION METHODS

|            |                                                                                                                                                                                                                                                                                                |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/co/Coordinate_Operation_Methods                                                                                                                                                                                                                                                           |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:CoordinateOperation, geosrs:PassThroughOperation, geosrs:ConcatenatedOperation, geosrs:SingleOperation, geosrs:Transformation, geosrs:Conversion, geosrs:PointMotionOperation, geosrs:OperationMethod to be used in SPARQL graph patterns. |

### 7.2.1. Class: geosrs:PassThroughOperation

**Table 34** — geosrs:PassThroughOperation

|            |                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/PassThroughOperation">https://w3id.org/geosrs/co/PassThroughOperation</a> |
| Definition | Specification of a subset of coordinate tuples that is subject to a coordinate operation                      |

## 7.2.2. Class: geosrs:ConcatenatedOperation

**Table 35** — geosrs:ConcatenatedOperation

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/ConcatenatedOperation">https://w3id.org/geosrs/co/ConcatenatedOperation</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Definition    | <p>Ordered sequence of two or more single coordinate operations. Note: The sequence of coordinate operations is constrained by the requirement that the source coordinate reference system of step (n + 1) shall be the same as the target coordinate reference system of step (n). The source coordinate reference system of the first step and the target coordinate reference system of the last step are the source and target coordinate reference system associated with the concatenated coordinate operation. For a concatenated coordinate operation sequence of n coordinate operations: source CRS (concatenated coordinate operation) .eq. source CRS (coordinate operation step 1) target CRS (coordinate operation step i) .eq. source CRS (coordinate operation step i + 1); i .eq. 1 ... (n - 1) target CRS (concatenated coordinate operation) .eq. target CRS (coordinate operation step n) Instead of a forward coordinate operation, an inverse coordinate operation may be used for one or more of the coordinate operation steps mentioned above, if the inverse coordinate operation is uniquely defined by the forward coordinate operation method.</p> |
| Super-classes | <a href="#">CoordinateOperation</a>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |

## 7.2.3. Class: geosrs:PointMotionOperation

**Table 36** — geosrs:PointMotionOperation

|            |                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/PointMotionOperation">https://w3id.org/geosrs/co/PointMotionOperation</a>                                                                                                      |
| Definition | <p>Mathematical operation that describes the change of coordinate values within one coordinate reference system due to the motion of the point between one coordinate epoch and another coordinate epoch Note:</p> |

|               |                                                                               |
|---------------|-------------------------------------------------------------------------------|
|               | In this document the motion is due to tectonic plate movement or deformation. |
| Super-classes | <a href="#">SingleOperation</a>                                               |

### 7.3. Coordinate Operation Parameters

| REQUIREMENT 6: COORDINATE OPERATION PARAMETERS |                                                                                                                                                                                                                                                                                                                                                            |
|------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                                     | /req/co/Coordinate_Operation_Parameters                                                                                                                                                                                                                                                                                                                    |
| STATEMENT                                      | Implementations shall allow the RDFS classes <code>geosrs:GeneralOperationParameter</code> , <code>geosrs:OperationParameterGroup</code> , <code>geosrs:OperationParameter</code> , <code>geosrs:GeneralParameterValue</code> , <code>geosrs:ParameterValueGroup</code> , <code>geosrs:OperationParameterValue</code> to be used in SPARQL graph patterns. |

#### 7.3.1. Class: `geosrs:OperationParameterGroup`

**Table 37** — `geosrs:OperationParameterGroup`

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/co/OperationParameterGroup">https://w3id.org/geosrs/co/OperationParameterGroup</a> |
| Definition    | Definition of a group of related parameters used by a coordinate operation method.                                  |
| Super-classes | <a href="#">GeneralOperationParameter</a>                                                                           |

#### 7.3.2. Class: `geosrs:ParameterValueGroup`

**Table 38** — `geosrs:ParameterValueGroup`

|            |                                                                                                                                                                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/ParameterValueGroup">https://w3id.org/geosrs/co/ParameterValueGroup</a>                                                                                                                                                                                                |
| Definition | Group of related parameter values. Note: The same group can be repeated more than once in a coordinate operation or higher level <code>ParameterValueGroup</code> , if those instances contain different values of one or more <code>ParameterValues</code> which suitably distinguish among those groups. |

## 7.4. Coordinate Operation Properties

### REQUIREMENT 7: COORDINATE OPERATION PROPERTIES

|            |                                                                                                                                                                                                                             |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | <code>/req/co/Coordinate_Operation_Properties</code>                                                                                                                                                                        |
| STATEMENT  | Implementations shall allow the RDFS properties <code>geosrs:derivingConversion</code> , <code>geosrs:parameter</code> , <code>geosrs:sourceCRS</code> , <code>geosrs:targetCRS</code> to be used in SPARQL graph patterns. |

#### 7.4.1. Property: `geosrs:derivingConversion`

**Table 39** — `geosrs:derivingConversion`

|            |                                                                                                           |
|------------|-----------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/derivingConversion">https://w3id.org/geosrs/co/derivingConversion</a> |
| Type       | <a href="#">owl:ObjectProperty</a>                                                                        |
| Definition | Relates a derived CRS to a conversion                                                                     |
| Range      | <a href="#">Conversion</a>                                                                                |
| Domain     | <a href="#">DerivedCRS</a>                                                                                |

#### 7.4.2. Property: `geosrs:parameter`

**Table 40** — `geosrs:parameter`

|            |                                                                                         |
|------------|-----------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/parameter">https://w3id.org/geosrs/co/parameter</a> |
| Type       | <a href="#">owl:ObjectProperty</a>                                                      |
| Definition | Value of the datum-defining parameter                                                   |
| Range      | <a href="#">OperationParameter</a>                                                      |

|        |                                   |
|--------|-----------------------------------|
| Domain | <a href="#"><u>Conversion</u></a> |
|--------|-----------------------------------|

### 7.4.3. Property: geosrs:sourceCRS

**Table 41** — geosrs:sourceCRS

|            |                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/sourceCRS"><u>https://w3id.org/geosrs/co/sourceCRS</u></a>                                                             |
| Type       | <a href="#"><u>owl:ObjectProperty</u></a>                                                                                                                  |
| Definition | The coordinate reference system associated to the data used as input of a given operation. Cf. ISO 19111:2007:2007-07, table 42, named association Source. |
| Range      | <a href="#"><u>CRS</u></a>                                                                                                                                 |
| Domain     | <a href="#"><u>CoordinateOperation</u></a>                                                                                                                 |
| Example    | <a href="#"><u>geosrs:sourceCRS</u></a>                                                                                                                    |

### 7.4.4. Property: geosrs:targetCRS

**Table 42** — geosrs:targetCRS

|            |                                                                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/co/targetCRS"><u>https://w3id.org/geosrs/co/targetCRS</u></a>                                                                  |
| Type       | <a href="#"><u>owl:ObjectProperty</u></a>                                                                                                                       |
| Definition | The coordinate reference system associated to the data obtained as output of a given operation. Cf. ISO 19111:2007:2007-07, table 42, named association Target. |
| Range      | <a href="#"><u>CRS</u></a>                                                                                                                                      |
| Domain     | <a href="#"><u>CoordinateOperation</u></a>                                                                                                                      |



8

# COORDINATE SYSTEM MODULE

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This clause establishes the **CS** Requirements class, with IRI /req/cs, which has a corresponding Conformance Class, **CS**, with IRI /conf/cs.

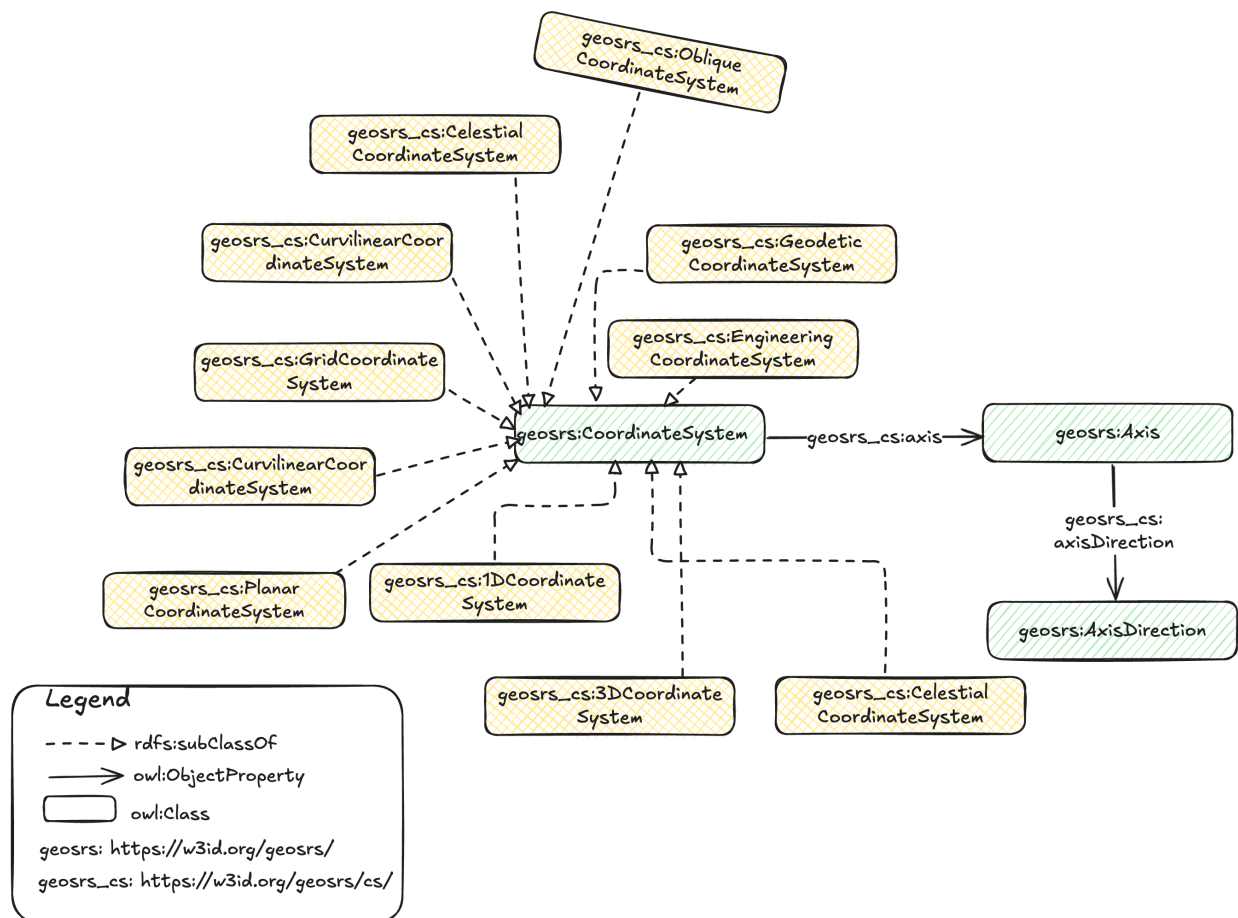


Figure 3

The coordinate system module introduces different types of coordinate systems which are distinguished in geospatial science and applications. Coordinate systems are distinguished by their area of use, i.e planetary or interstellar and by their multidimensionality.

### REQUIREMENTS CLASS 3: 08-CS\_MODULE.ADOC EXTENSION

|                   |                                 |
|-------------------|---------------------------------|
| IDENTIFIER        | /req/cs                         |
| TARGET TYPE       | Implementation Specification    |
| CONFORMANCE CLASS | Conformance class A.3: /conf/cs |

## REQUIREMENTS CLASS 3: 08-CS\_MODULE.ADOC EXTENSION

|             |                                      |
|-------------|--------------------------------------|
| REQUIREMENT | /req/cs/Temporal_Coordinate_Systems  |
|             | /req/cs/3D_Coordinate_Systems        |
|             | /req/cs/Coordinate_System_Types      |
|             | /req/cs/Celestial_Coordinate_Systems |
|             | /req/cs/Coordinate_System_Components |
|             | /req/cs/Coordinate_System_Properties |

## 8.1. 3D Coordinate Systems

### REQUIREMENT 8: 3D COORDINATE SYSTEMS

|            |                                                                                                                                                                                                                                                       |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/cs/3D_Coordinate_Systems                                                                                                                                                                                                                         |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:3DCoordinateSystem, geosrs:ConicalCoordinateSystem, geosrs:CylindricalCoordinateSystem, geosrs:EllipsoidalCoordinateSystem, geosrs:SphericalCoordinateSystem to be used in SPARQL graph patterns. |

#### 8.1.1. Class: geosrs:3DCoordinateSystem

The class geosrs:3DCoordinateSystem describes a coordinate system in three dimensions. These coordinate systems are common for 3D representations or 2D representations with a time aspect.

**Table 43** — geosrs:3DCoordinateSystem

|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/3DCoordinateSystem">https://w3id.org/geosrs/cs/3DCoordinateSystem</a> |
| Definition    | Non-repeating sequence of coordinate system axes that spans a given coordinate space in three dimensions  |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                          |
| Example       | <a href="#">geosrs:3DCoordinateSystem</a>                                                                 |

### 8.1.2. Class: geosrs:ConicalCoordinateSystem

Table 44 — geosrs:ConicalCoordinateSystem

|               |                                                                                                                                                                                                                                                                  |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/ConicalCoordinateSystem">https://w3id.org/geosrs/cs/ConicalCoordinateSystem</a>                                                                                                                                              |
| Definition    | A conical coordinate system is a three-dimensional orthogonal coordinate system consisting of concentric spheres (described by their radius <i>r</i> ) and by two families of perpendicular cones, aligned along the <i>z</i> - and <i>x</i> -axes, respectively |
| Super-classes | <a href="#">OrthogonalCoordinateSystem</a>                                                                                                                                                                                                                       |

### 8.1.3. Class: geosrs:CylindricalCoordinateSystem

Table 45 — geosrs:CylindricalCoordinateSystem

|               |                                                                                                                                            |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/CylindricalCoordinateSystem">https://w3id.org/geosrs/cs/CylindricalCoordinateSystem</a>                |
| Definition    | Three-dimensional coordinate system in Euclidean space in which position is specified by two linear coordinates and one angular coordinate |
| Super-classes | <a href="#">3DCoordinateSystem</a>                                                                                                         |

## 8.2. Celestial Coordinate Systems

| REQUIREMENT 9: CELESTIAL COORDINATE SYSTEMS |                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                                  | /req/cs/Celestial_Coordinate_Systems                                                                                                                                                                                                                                                                                 |
| STATEMENT                                   | Implementations shall allow the RDFS classes geosrs:CelestialCoordinateSystem, geosrs:EclipticCoordinateSystem, geosrs:EquatorialCoordinateSystem, geosrs:GalacticCoordinateSystem, geosrs:HorizontalCoordinateSystem, geosrs:PerifocalCoordinateSystem, geosrs:SuperGalacticCS to be used in SPARQL graph patterns. |

### 8.2.1. Class: geosrs:CelestialCoordinateSystem

**Table 46** — geosrs:CelestialCoordinateSystem

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/CelestialCoordinateSystem">https://w3id.org/geosrs/cs/CelestialCoordinateSystem</a> |
| Definition    | A coordinate system for specifying positions of celestial objects relative to physical reference points                 |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                        |

### 8.2.2. Class: geosrs:EclipticCoordinateSystem

**Table 47** — geosrs:EclipticCoordinateSystem

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/EclipticCoordinateSystem">https://w3id.org/geosrs/cs/EclipticCoordinateSystem</a> |
| Definition    | An ecliptic coordinate system is used for representing the apparent positions and orbits of solar system objects.     |
| Super-classes | <a href="#">CelestialCoordinateSystem</a>                                                                             |

### 8.2.3. Class: geosrs:EquatorialCoordinateSystem

**Table 48** — geosrs:EquatorialCoordinateSystem

|               |                                                                                                                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/EquatorialCoordinateSystem">https://w3id.org/geosrs/cs/EquatorialCoordinateSystem</a>                                                                                                                            |
| Definition    | A celestial coordinate system in which an object's position on the celestial sphere is described in terms of its north-south declination and east-west right ascension, measured relative to the celestial equator and vernal equinox, respectively. |
| Super-classes | <a href="#">CelestialCoordinateSystem</a>                                                                                                                                                                                                            |

### 8.2.4. Class: geosrs:GalacticCoordinateSystem

**Table 49** — geosrs:GalacticCoordinateSystem

|               |                                                                                                                                                                                                                                          |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/GalacticCoordinateSystem">https://w3id.org/geosrs/cs/GalacticCoordinateSystem</a>                                                                                                                    |
| Definition    | A coordinate system with the Sun as its center, the primary direction aligned with the approximate center of the Milky Way Galaxy, and the fundamental plane parallel to an approximation of the galactic plane but offset to its north. |
| Super-classes | <a href="#">CelestialCoordinateSystem</a> <a href="#">3DCoordinateSystem</a>                                                                                                                                                             |

## 8.2.5. Class: geosrs:HorizontalCoordinateSystem

**Table 50** — geosrs:HorizontalCoordinateSystem

|               |                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/HorizontalCoordinateSystem">https://w3id.org/geosrs/cs/HorizontalCoordinateSystem</a>        |
| Definition    | A horizontal coordinate system is a celestial coordinate system that uses the observer's local horizon as the fundamental plane. |
| Super-classes | <a href="#">CelestialCoordinateSystem</a>                                                                                        |

## 8.2.6. Class: geosrs:PerifocalCoordinateSystem

**Table 51** — geosrs:PerifocalCoordinateSystem

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/PerifocalCoordinateSystem">https://w3id.org/geosrs/cs/PerifocalCoordinateSystem</a> |
| Definition    | A frame of reference centered at the focus of the orbit, i. e. the celestial body about which the orbit is centered.    |
| Super-classes | <a href="#">CelestialCoordinateSystem</a>                                                                               |

## 8.2.7. Class: geosrs:SuperGalacticCS

**Table 52** — geosrs:SuperGalacticCS

|            |                                                                                                              |
|------------|--------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/cs/SuperGalacticCS">https://w3id.org/geosrs/cs/SuperGalacticCS</a>          |
| Definition | A reference frame for the supercluster of galaxies that contains the Milky Way galaxy, referenced to a local |

|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
|               | relatively flat collection of galaxy clusters used to define the supergalactic plane. |
| Super-classes | <a href="#">CelestialCoordinateSystem</a> <a href="#">3DCoordinateSystem</a>          |

### 8.3. Coordinate System Components

| Requirement 10: Coordinate System Components |                                                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                                   | /req/cs/Coordinate_System_Components                                                                                       |
| STATEMENT                                    | Implementations shall allow the RDFS classes <code>geosrs:CoordinateSystemAxis</code> to be used in SPARQL graph patterns. |

### 8.4. Coordinate System Properties

| Requirement 11: Coordinate System Properties |                                                                                                                                                   |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                                   | /req/cs/Coordinate_System_Properties                                                                                                              |
| STATEMENT                                    | Implementations shall allow the RDFS properties <code>geosrs:axis</code> , <code>geosrs:axisDirection</code> to be used in SPARQL graph patterns. |

#### 8.4.1. Property: `geosrs:axis`

Table 53 — `geosrs:axis`

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/cs/axis">https://w3id.org/geosrs/cs/axis</a> |
| Type       | <a href="#">owl:ObjectProperty</a>                                            |
| Definition | The property relates a coordinate system to one of its axis                   |
| Range      | <a href="#">Axis</a>                                                          |

|        |                                  |
|--------|----------------------------------|
| Domain | <a href="#">CoordinateSystem</a> |
|--------|----------------------------------|

### 8.4.2. Property: geosrs:axisDirection

Table 54 — geosrs:axisDirection

|            |                                                                                                             |
|------------|-------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/cs/axisDirection">https://w3id.org/geosrs/cs/axisDirection</a>             |
| Type       | <a href="#">owl:ObjectProperty</a>                                                                          |
| Definition | The direction of an axis. Cf. ISO 19111:2007:2007-07, table 27, attribute coordinate system axis direction. |
| Range      | <a href="#">AxisDirection</a>                                                                               |
| Domain     | <a href="#">Axis</a>                                                                                        |
| Example    | <a href="#">geosrs:axisDirection</a>                                                                        |

## 8.5. Coordinate System Types

| Requirement 12: Coordinate System Types |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|-----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                              | /req/cs/Coordinate_System_Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| STATEMENT                               | Implementations shall allow the RDFS classes geosrs:CoordinateSystem, geosrs:AffineCoordinateSystem, geosrs:BarycentricCoordinateSystem, geosrs:CartesianCoordinateSystem, geosrs:CurvilinearCoordinateSystem, geosrs:EngineeringCoordinateSystem, geosrs:GeodeticCoordinateSystem, geosrs:GeographicalCoordinateSystem, geosrs:GridCoordinateSystem, geosrs:HexagonalCoordinateSystem, geosrs:LinearCoordinateSystem, geosrs:LocalCoordinateSystem, geosrs:ObliqueCoordinateSystem, geosrs:OrdinalCoordinateSystem, geosrs:OrthogonalCoordinateSystem, geosrs:ParametricCoordinateSystem, geosrs:PlanarCoordinateSystem, geosrs:PolarCoordinateSystem, geosrs:VerticalCoordinateSystem to be used in SPARQL graph patterns. |

### 8.5.1. Class: geosrs:AffineCoordinateSystem

**Table 55** — geosrs:AffineCoordinateSystem

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/AffineCoordinateSystem">https://w3id.org/geosrs/cs/AffineCoordinateSystem</a> |
| Definition    | Coordinate system in Euclidean space with straight axes that are not necessarily mutually perpendicular           |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                  |

### 8.5.2. Class: geosrs:BarycentricCoordinateSystem

**Table 56** — geosrs:BarycentricCoordinateSystem

|               |                                                                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/BarycentricCoordinateSystem">https://w3id.org/geosrs/cs/BarycentricCoordinateSystem</a>                                                               |
| Definition    | A coordinate system in which the location of a point is specified by reference to a simplex (a triangle for points in a plane, a tetrahedron for points in three-dimensional space, etc.) |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                                                                                          |

### 8.5.3. Class: geosrs:CurvilinearCoordinateSystem

**Table 57** — geosrs:CurvilinearCoordinateSystem

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/CurvilinearCoordinateSystem">https://w3id.org/geosrs/cs/CurvilinearCoordinateSystem</a> |
| Definition    | A coordinate system for the Euclidean space in which the coordinate lines may be curved                                     |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                            |

### 8.5.4. Class: geosrs:EngineeringCoordinateSystem

**Table 58** — geosrs:EngineeringCoordinateSystem

|            |                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/cs/EngineeringCoordinateSystem">https://w3id.org/geosrs/cs/EngineeringCoordinateSystem</a>                                       |
| Definition | Coordinate system used by an engineering coordinate reference system, one of an affine coordinate system, a Cartesian coordinate system, a cylindrical coordinate |



|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
|               | system, a linear coordinate sytem, an ordinal coordinate system, a polar coordinate system or a spherical coordinate system |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                            |

### 8.5.5. Class: geosrs:GeodeticCoordinateSystem

**Table 59** — geosrs:GeodeticCoordinateSystem

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/GeodeticCoordinateSystem">https://w3id.org/geosrs/cs/GeodeticCoordinateSystem</a> |
| Definition    | Coordinate system used by a Geodetic CRS, one of a Cartesian coordinate system or a spherical coordinate system.      |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                      |

### 8.5.6. Class: geosrs:GeographicalCoordinateSystem

**Table 60** — geosrs:GeographicalCoordinateSystem

|               |                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/GeographicalCoordinateSystem">https://w3id.org/geosrs/cs/GeographicalCoordinateSystem</a>  |
| Definition    | Spherical or geodetic coordinate system for measuring and communicating positions directly on Earth as latitude and longitude. |
| Super-classes | <a href="#">SphericalCoordinateSystem</a> <a href="#">GeodeticCoordinateSystem</a>                                             |

### 8.5.7. Class: geosrs:GridCoordinateSystem

**Table 61** — geosrs:GridCoordinateSystem

|               |                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/GridCoordinateSystem">https://w3id.org/geosrs/cs/GridCoordinateSystem</a> |
| Definition    | A grid coordinate system identifies areas within a grid.                                                      |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                              |

## 8.5.8. Class: geosrs:HexagonalCoordinateSystem

**Table 62** — geosrs:HexagonalCoordinateSystem

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/HexagonalCoordinateSystem">https://w3id.org/geosrs/cs/HexagonalCoordinateSystem</a> |
| Definition    | A hexagonal coordinate system identifies areas within a hexagonal lattice.                                              |
| Super-classes | <a href="#">GridCoordinateSystem</a>                                                                                    |

## 8.5.9. Class: geosrs:LocalCoordinateSystem

**Table 63** — geosrs:LocalCoordinateSystem

|               |                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/LocalCoordinateSystem">https://w3id.org/geosrs/cs/LocalCoordinateSystem</a> |
| Definition    | Coordinate system with a point of local reference.                                                              |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                |

## 8.5.10. Class: geosrs:ObliqueCoordinateSystem

**Table 64** — geosrs:ObliqueCoordinateSystem

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/ObliqueCoordinateSystem">https://w3id.org/geosrs/cs/ObliqueCoordinateSystem</a> |
| Definition    | A plane coordinate system whose axes are not perpendicular.                                                         |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                    |

## 8.5.11. Class: geosrs:OrthogonalCoordinateSystem

**Table 65** — geosrs:OrthogonalCoordinateSystem

|     |                                                                                                                           |
|-----|---------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/cs/OrthogonalCoordinateSystem">https://w3id.org/geosrs/cs/OrthogonalCoordinateSystem</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Definition    | A orthogonal coordinate system is a system of curvilinear coordinates in which each family of surfaces intersects the others at right angles. |
| Super-classes | <a href="#">CurvilinearCoordinateSystem</a>                                                                                                   |

## 8.5.12. Class: geosrs:PlanarCoordinateSystem

**Table 66** — geosrs:PlanarCoordinateSystem

|               |                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/PlanarCoordinateSystem">https://w3id.org/geosrs/cs/PlanarCoordinateSystem</a>                                |
| Definition    | A two-dimensional measurement system that locates features on a plane based on their distance from an origin (0,0) along two perpendicular axes. |
| Super-classes | <a href="#">CoordinateSystem</a>                                                                                                                 |
| Example       | <a href="#">geosrs:PlanarCoordinateSystem</a>                                                                                                    |

## 8.6. Temporal Coordinate Systems

### REQUIREMENT 13: TEMPORAL COORDINATE SYSTEMS

|            |                                                                                                                                                                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/cs/Temporal_Coordinate_Systems                                                                                                                                                                                                                                                                                                                     |
| STATEMENT  | Implementations shall allow the RDFS classes <a href="#">geosrs:1DCoordinateSystem</a> , <a href="#">geosrs:DateTimeTemporalCoordinateSystem</a> , <a href="#">geosrs:TemporalCountCoordinateSystem</a> , <a href="#">geosrs:TemporalCoordinateSystem</a> , <a href="#">geosrs:TemporalMeasureCoordinateSystem</a> to be used in SPARQL graph patterns. |

### 8.6.1. Class: geosrs:1DCoordinateSystem

The class [geosrs:1DCoordinateSystem](#) describes a coordinate system with only one dimension. Often, these definitions include temporal coordinate systems which only represent time using one coordinate system axis.

**Table 67** — geosrs:1DCoordinateSystem

|     |                                                                                                           |
|-----|-----------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/cs/1DCoordinateSystem">https://w3id.org/geosrs/cs/1DCoordinateSystem</a> |
|-----|-----------------------------------------------------------------------------------------------------------|

|               |                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------|
| Definition    | Non-repeating sequence of coordinate system axes that spans a given coordinate space in one dimension |
| Super-classes | <a href="#"><u>CoordinateSystem</u></a>                                                               |

## 8.6.2. Class: geosrs:DateTimeTemporalCoordinateSystem

**Table 68** — geosrs:DateTimeTemporalCoordinateSystem

|               |                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/DateTimeTemporalCoordinateSystem"><u>https://w3id.org/geosrs/cs/DateTimeTemporalCoordinateSystem</u></a> |
| Definition    | One-dimensional coordinate system used to record time in dateTime representation as defined in ISO 8601.                                     |
| Super-classes | <a href="#"><u>TemporalCoordinateSystem</u></a>                                                                                              |

## 8.6.3. Class: geosrs:TemporalCountCoordinateSystem

**Table 69** — geosrs:TemporalCountCoordinateSystem

|               |                                                                                                                                        |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/TemporalCountCoordinateSystem"><u>https://w3id.org/geosrs/cs/TemporalCountCoordinateSystem</u></a> |
| Definition    | One-dimensional coordinate system used to record time as an integer count.                                                             |
| Super-classes | <a href="#"><u>TemporalCoordinateSystem</u></a>                                                                                        |

## 8.6.4. Class: geosrs:TemporalCoordinateSystem

**Table 70** — geosrs:TemporalCoordinateSystem

|               |                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/TemporalCoordinateSystem"><u>https://w3id.org/geosrs/cs/TemporalCoordinateSystem</u></a> |
| Definition    | One-dimensional coordinate system where the axis is time.                                                                    |
| Super-classes | <a href="#"><u>1DCoordinateSystem</u></a>                                                                                    |

## 8.6.5. Class: geosrs:TemporalMeasureCoordinateSystem

**Table 71** — geosrs:TemporalMeasureCoordinateSystem

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/cs/TemporalMeasureCoordinateSystem">https://w3id.org/geosrs/cs/TemporalMeasureCoordinateSystem</a> |
| Definition    | One-dimensional coordinate system used to record a time as a real number.                                                           |
| Super-classes | <a href="#">TemporalCoordinateSystem</a>                                                                                            |

9

# DATUM MODULE

---

This clause establishes the **Datum** Requirements class, with IRI /req/datum, which has a corresponding Conformance Class, **Datum**, with IRI /conf/datum.

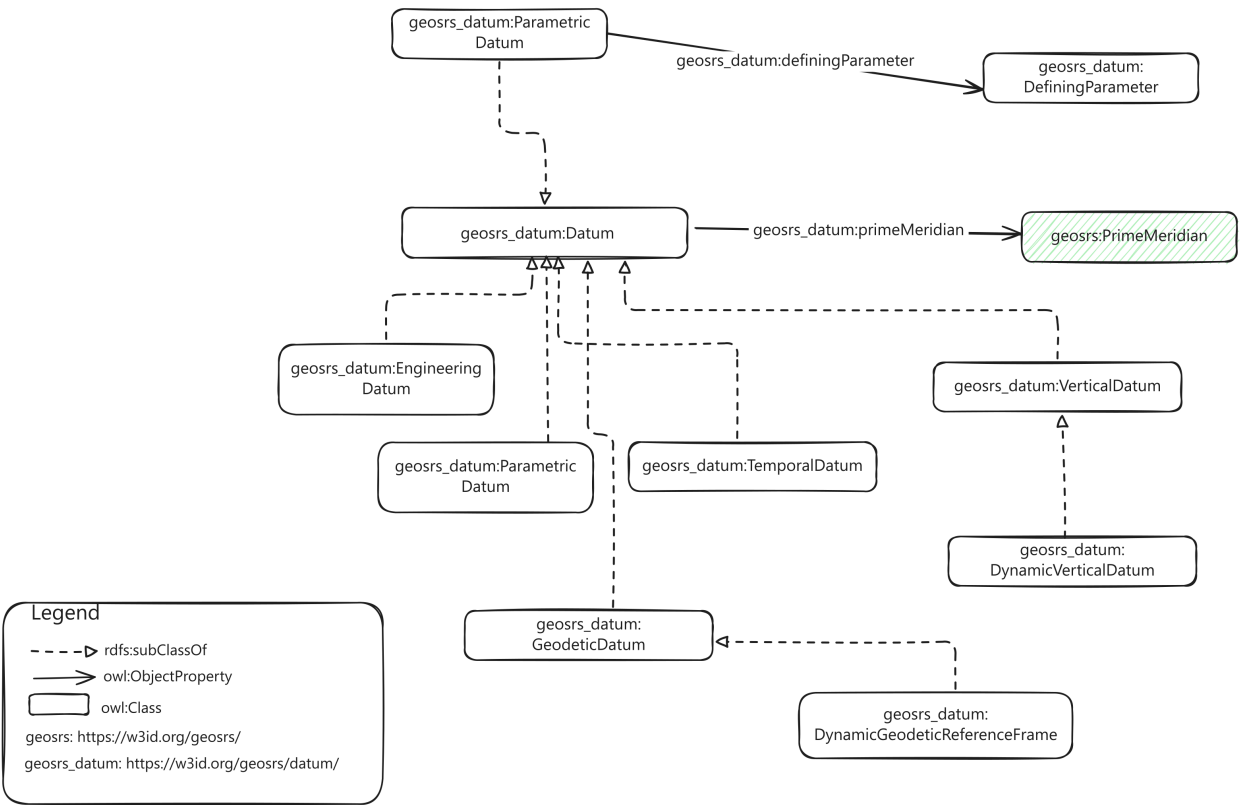


Figure 4

| Requirements Class 4: 09-DATUM_MODULE.ADOC Extension |                                    |
|------------------------------------------------------|------------------------------------|
| IDENTIFIER                                           | /req/datum                         |
| TARGET TYPE                                          | Implementation Specification       |
| CONFORMANCE CLASS                                    | Conformance class A.4: /conf/datum |
| REQUIREMENT                                          | /req/datum/Datum_Types             |
|                                                      | /req/datum/Datum_Parameters        |
|                                                      | /req/datum/Spheroid_Types          |
|                                                      | /req/datum/Spheroid_Properties     |

## 9.1. Datum Parameters

### REQUIREMENT 14: DATUM PARAMETERS

|            |                                                                                                                                  |
|------------|----------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/datum/Datum_Parameters                                                                                                      |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:PrimeMeridian, geosrs:DefiningParameter to be used in SPARQL graph patterns. |

#### 9.1.1. Class: geosrs:DefiningParameter

Table 72 — geosrs:DefiningParameter

|            |                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/DefiningParameter">https://w3id.org/geosrs/datum/DefiningParameter</a>                                                                                       |
| Definition | Parameter value, an ordered sequence of values, or a reference to a file of parameter values that define a paramtric datum. Cf. ISO 19111:2019 Geographic information — Referencing by coordinates. |

## 9.2. Datum Properties

### REQUIREMENT 15: DATUM PROPERTIES

|            |                                                                                                                                                            |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/datum/Datum_Properties                                                                                                                                |
| STATEMENT  | Implementations shall allow the RDFS properties geosrs:datumDefiningParameter, geosrs:ellipsoid, geosrs:primeMeridian to be used in SPARQL graph patterns. |



### 9.2.1. Property: geosrs:datumDefiningParameter

**Table 73** — geosrs:datumDefiningParameter

|            |                                                                                                                         |
|------------|-------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/datumDefiningParameter">https://w3id.org/geosrs/datum/datumDefiningParameter</a> |
| Type       | <a href="#">owl:ObjectProperty</a>                                                                                      |
| Definition | Parameter used to define the parametric datum                                                                           |
| Range      | <a href="#">DefiningParameter</a>                                                                                       |
| Domain     | <a href="#">ParametricDatum</a>                                                                                         |

### 9.2.2. Property: geosrs:ellipsoid

**Table 74** — geosrs:ellipsoid

|            |                                                                                               |
|------------|-----------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/ellipsoid">https://w3id.org/geosrs/datum/ellipsoid</a> |
| Type       | <a href="#">owl:ObjectProperty</a>                                                            |
| Definition | The properties relates a datum to its ellipsoid definition                                    |
| Range      | <a href="#">Ellipsoid</a>                                                                     |
| Domain     | <a href="#">Datum</a>                                                                         |
| Example    | <a href="#">geosrs:ellipsoid</a>                                                              |

### 9.2.3. Property: geosrs:primeMeridian

**Table 75** — geosrs:primeMeridian

|      |                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------|
| URI  | <a href="https://w3id.org/geosrs/datum/primeMeridian">https://w3id.org/geosrs/datum/primeMeridian</a> |
| Type | <a href="#">owl:ObjectProperty</a>                                                                    |

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| Definition | The prime meridian used by a geodetic datum. Cf. ISO 19111:2007:2007-07, table 34, association role prime Meridian. |
| Range      | <a href="#">PrimeMeridian</a>                                                                                       |
| Domain     | <a href="#">Datum</a>                                                                                               |
| Example    | <a href="#">geosrs:primeMeridian</a>                                                                                |

## 9.3. Datum Types

### REQUIREMENT 16: DATUM TYPES

**IDENTIFIER**    `/req/datum/Datum_Types`

**STATEMENT**

Implementations shall allow the RDFS classes `geosrs:Datum`, `geosrs:GeodeticDatum`, `geosrs:DynamicGeodeticReferenceFrame`, `geosrs:VerticalDatum`, `geosrs:DynamicVerticalDatum`, `geosrs:ParametricDatum`, `geosrs:EngineeringDatum`, `geosrs:TemporalDatum`, `geosrs:DatumEnsemble` to be used in SPARQL graph patterns.

### 9.3.1. Class: `geosrs:DynamicGeodeticReferenceFrame`

**Table 76** — `geosrs:DynamicGeodeticReferenceFrame`

|               |                                                                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/datum/DynamicGeodeticReferenceFrame">https://w3id.org/geosrs/datum/DynamicGeodeticReferenceFrame</a>                                                                             |
| Definition    | Geodetic reference frame in which some of the parameters describe time evolution of defining station coordinates<br>Example: defining station coordinates having linear velocities to account for crustal motion. |
| Super-classes | <a href="#">GeodeticDatum</a>                                                                                                                                                                                     |

### 9.3.2. Class: `geosrs:DynamicVerticalDatum`

**Table 77** — geosrs:DynamicVerticalDatum

|               |                                                                                                                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/datum/DynamicVerticalDatum">https://w3id.org/geosrs/datum/DynamicVerticalDatum</a>                                                                                                                                                 |
| Definition    | Vertical reference frame in which some of the defining parameters have time dependencyExample: Defining station heights have velocity to account for post-glacial isostatic rebound motion. Cf. ISO 19111:2019 Geographic information — Referencing by coordinates. |
| Super-classes | <a href="#">VerticalDatum</a>                                                                                                                                                                                                                                       |
| Example       | <a href="#">geosrs:DynamicVerticalDatum</a>                                                                                                                                                                                                                         |

### 9.3.3. Class: geosrs:ParametricDatum

**Table 78** — geosrs:ParametricDatum

|               |                                                                                                                                                                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/datum/ParametricDatum">https://w3id.org/geosrs/datum/ParametricDatum</a>                                                                                                                                                                 |
| Definition    | Textual description and/or a set of parameters identifying a particular reference surface used as the origin of a parametric coordinate system, including its position with respect to the Earth. Cf. ISO 19111:2019 Geographic information — Referencing by coordinates. |
| Super-classes | <a href="#">Datum</a>                                                                                                                                                                                                                                                     |

### 9.3.4. Class: geosrs:EngineeringDatum

**Table 79** — geosrs:EngineeringDatum

|               |                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/datum/EngineeringDatum">https://w3id.org/geosrs/datum/EngineeringDatum</a>                                                                                                                                                                                                                                                                         |
| Definition    | Definition of the origin and orientation of an engineering coordinate reference systemNote: The origin can be fixed with respect to the Earth (such as a defined point at a construction site), or be a defined point on a moving vehicle (such as on a ship or satellite), or a defined point of an image. Cf. ISO 19111:2019 Geographic information — Referencing by coordinates. |
| Super-classes | <a href="#">Datum</a>                                                                                                                                                                                                                                                                                                                                                               |

### 9.3.5. Class: geosrs:TemporalDatum

Table 80 — geosrs:TemporalDatum

|               |                                                                                                                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/datum/TemporalDatum">https://w3id.org/geosrs/datum/TemporalDatum</a>                                                                                              |
| Definition    | Definition of the relationship of a temporal coordinate system to an objectNote: The object is normally time on the Earth. Cf. ISO 19111:2019 Geographic information — Referencing by coordinates. |
| Super-classes | <a href="#">Datum</a>                                                                                                                                                                              |

### 9.3.6. Class: geosrs:DatumEnsemble

Table 81 — geosrs:DatumEnsemble

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/DatumEnsemble">https://w3id.org/geosrs/datum/DatumEnsemble</a>                                                                                                                                                                                                                                                                                                                                                                                             |
| Definition | A collection of two or more datums (or if geodetic or vertical, a collection of two or more reference frames) that are realizations of one Conventional Reference System and which for all but the highest accuracy requirements may be considered to be insignificantly different from each other. Note: Within the datum ensemble every frame or datum is constrained to be a realization of the same reference system. Cf. ISO 19111:2019 Geographic information — Referencing by coordinates. |

## 9.4. Spheroid Properties

| REQUIREMENT 17: SPHEROID PROPERTIES |                                                                                                                                                                                                 |
|-------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                          | /req/datum/Spheroid_Properties                                                                                                                                                                  |
| STATEMENT                           | Implementations shall allow the RDFS properties geosrs:eccentricity, geosrs:inverseFlattening, geosrs:isSphere, geosrs:semiMajorAxis, geosrs:semiMinorAxis to be used in SPARQL graph patterns. |

### 9.4.1. Property: geosrs:eccentricity

**Table 82** — geosrs:eccentricity

|            |                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/eccentricity">https://w3id.org/geosrs/datum/eccentricity</a> |
| Type       | <a href="#">owl:DatatypeProperty</a>                                                                |
| Definition | A measure of how much an ellipse deviates from a perfect circle.                                    |
| Range      | <a href="#">xsd:double</a>                                                                          |
| Domain     | <a href="#">Ellipsoid</a>                                                                           |
| Example    | <a href="#">geosrs:eccentricity</a>                                                                 |

### 9.4.2. Property: geosrs:inverseFlattening

**Table 83** — geosrs:inverseFlattening

|            |                                                                                                                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/inverseFlattening">https://w3id.org/geosrs/datum/inverseFlattening</a>                                                                                           |
| Type       | <a href="#">owl:DatatypeProperty</a>                                                                                                                                                                    |
| Definition | Indicates the inverse flattening value of an ellipsoid, expressed as a number or a ratio (percentage rate, parts per million, etc.). Cf. ISO 19111:2007:2007-07, table 37, attribute inverse flattening |
| Range      | <a href="#">xsd:double</a>                                                                                                                                                                              |
| Domain     | <a href="#">Ellipsoid</a>                                                                                                                                                                               |
| Example    | <a href="#">geosrs:inverseFlattening</a>                                                                                                                                                                |

### 9.4.3. Property: geosrs:isSphere

**Table 84** — geosrs:isSphere

|            |                                                                                                                          |
|------------|--------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/isSphere">https://w3id.org/geosrs/datum/isSphere</a>                              |
| Type       | <a href="#">owl:DatatypeProperty</a>                                                                                     |
| Definition | Indicates whether the ellipsoid is a sphere. Cf. ISO 19111:2007:2007-07, table 37, attribute ellipsoid=sphere indicator. |
| Range      | <a href="#">xsd:boolean</a>                                                                                              |
| Domain     | <a href="#">Ellipsoid</a>                                                                                                |
| Example    | <a href="#">geosrs:isSphere</a>                                                                                          |

#### 9.4.4. Property: geosrs:semiMajorAxis

**Table 85** — geosrs:semiMajorAxis

|            |                                                                                                                                         |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/semiMajorAxis">https://w3id.org/geosrs/datum/semiMajorAxis</a>                                   |
| Type       | <a href="#">owl:DatatypeProperty</a>                                                                                                    |
| Definition | Indicates the length of the semi major axis of an ellipsoid. Cf. ISO 19111:2007:2007-07, table 36, attribute length of semi-major axis. |
| Range      | <a href="#">xsd:double</a>                                                                                                              |
| Domain     | <a href="#">Ellipsoid</a>                                                                                                               |
| Example    | <a href="#">geosrs:semiMajorAxis</a>                                                                                                    |

#### 9.4.5. Property: geosrs:semiMinorAxis

**Table 86** — geosrs:semiMinorAxis

|      |                                                                                                       |
|------|-------------------------------------------------------------------------------------------------------|
| URI  | <a href="https://w3id.org/geosrs/datum/semiMinorAxis">https://w3id.org/geosrs/datum/semiMinorAxis</a> |
| Type | <a href="#">owl:DatatypeProperty</a>                                                                  |

|            |                                                                                                                                            |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Definition | Indicates the length of the semi minor axis of an ellipsoid.<br>Cf. ISO 19111:2007:2007-07, table 37, attribute length of semi-minor axis. |
| Range      | <a href="#">xsd:double</a>                                                                                                                 |
| Domain     | <a href="#">Ellipsoid</a>                                                                                                                  |
| Example    | <a href="#">geosrs:semiMinorAxis</a>                                                                                                       |

## 9.5. Spheroid Types

### REQUIREMENT 18: SPHEROID TYPES

|            |                                                                                                                                                               |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | <a href="#">/req/datum/Spheroid_Types</a>                                                                                                                     |
| STATEMENT  | Implementations shall allow the RDFS classes <a href="#">geosrs:Ellipsoid</a> , <a href="#">geosrs:TriaxialEllipsoid</a> to be used in SPARQL graph patterns. |

### 9.5.1. Class: [geosrs:TriaxialEllipsoid](#)

**Table 87** — [geosrs:TriaxialEllipsoid](#)

|            |                                                                                                                 |
|------------|-----------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/datum/TriaxialEllipsoid">https://w3id.org/geosrs/datum/TriaxialEllipsoid</a>   |
| Definition | Surface of an analytic ellipsoid defined by three axes of different length. Also referred as scalene ellipsoid. |

10

# SRS APPLICATION MODULE

---



This clause establishes the **SRSAPP** Requirements class, with IRI /req/srsapp, which has a corresponding Conformance Class, **SRSAPP**, with IRI /conf/srsapp.

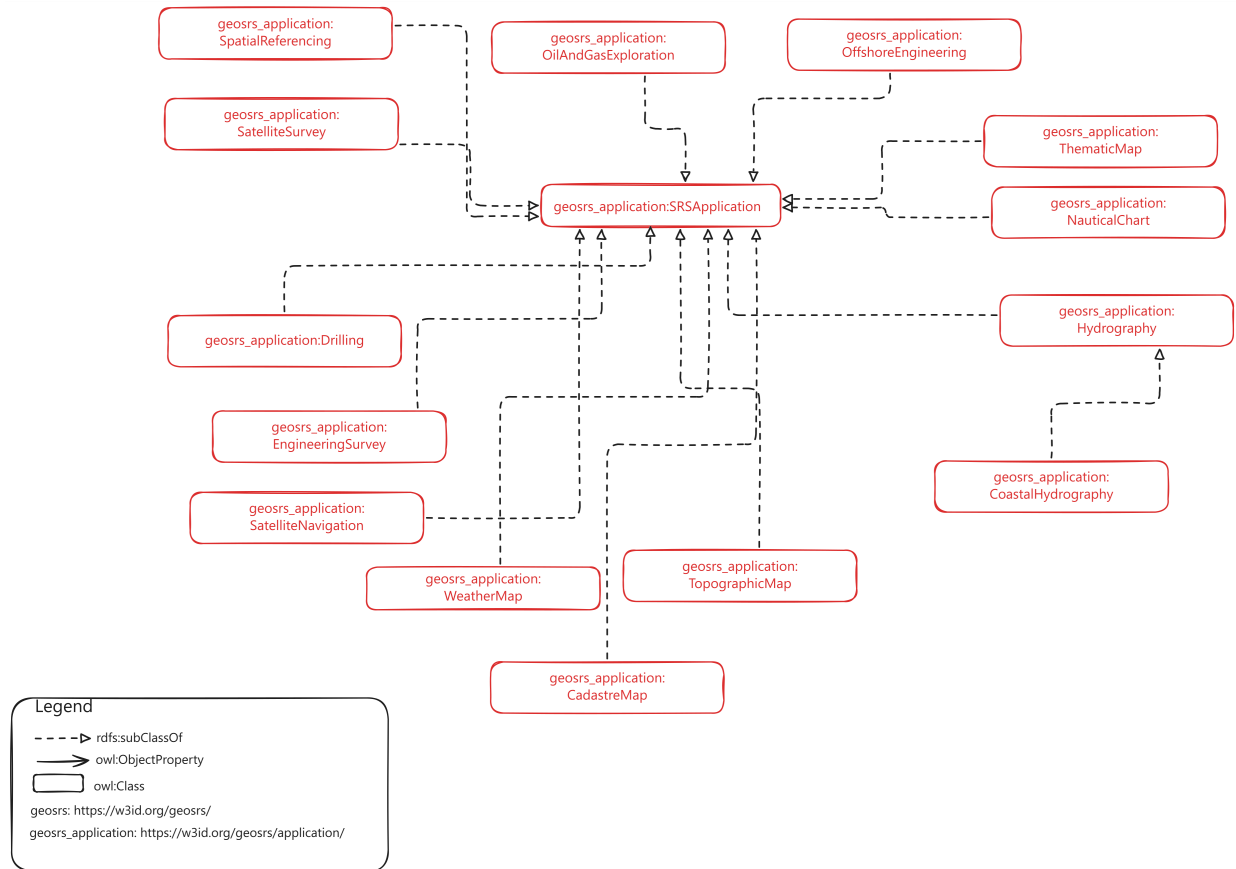


Figure 5

## REQUIREMENTS CLASS 5: 10-SRSAPPLICATION\_MODULE.ADOC EXTENSION

|                   |                                             |
|-------------------|---------------------------------------------|
| IDENTIFIER        | /req/srsapplication                         |
| TARGET TYPE       | Implementation Specification                |
| CONFORMANCE CLASS | Conformance class A.5: /conf/srsapplication |
| REQUIREMENT       | /req/srsapplication/SRS_Application_Types   |
|                   | /req/srsapplication/Map_Types               |

# 10.1. Map Types

## REQUIREMENT 19: MAP TYPES

**IDENTIFIER** /req/srsapplication/Map\_Types

**STATEMENT** Implementations shall allow the RDFS classes geosrs:CadastreMap, geosrs:NauticalChart, geosrs:ThematicMap, geosrs:TopographicMap, geosrs:WeatherMap to be used in SPARQL graph patterns.

### 10.1.1. Class: geosrs:CadastreMap

Table 88 — geosrs:CadastreMap

|               |                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/CadastreMap">https://w3id.org/geosrs/application/CadastreMap</a> |
| Definition    | A map displaying a cadastre.                                                                                  |
| Super-classes | <a href="#">SRSApplication</a>                                                                                |
| Example       | <a href="#">geosrs:CadastreMap</a>                                                                            |

### 10.1.2. Class: geosrs:NauticalChart

Table 89 — geosrs:NauticalChart

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/NauticalChart">https://w3id.org/geosrs/application/NauticalChart</a> |
| Definition    | A graphic representation of a sea area and adjacent coastal regions.                                              |
| Super-classes | <a href="#">SRSApplication</a>                                                                                    |

### 10.1.3. Class: geosrs:ThematicMap

Table 90 — geosrs:ThematicMap

|     |                                                                                                               |
|-----|---------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/application/ThematicMap">https://w3id.org/geosrs/application/ThematicMap</a> |
|-----|---------------------------------------------------------------------------------------------------------------|

|               |                                                |
|---------------|------------------------------------------------|
| Definition    | A map used to highlight a specific phenomenon. |
| Super-classes | <a href="#">SRSApplication</a>                 |

### 10.1.4. Class: geosrs:TopographicMap

**Table 91** — geosrs:TopographicMap

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/TopographicMap">https://w3id.org/geosrs/application/TopographicMap</a> |
| Definition    | A type of map characterized by large-scale detail and quantitative representation of relief.                        |
| Super-classes | <a href="#">SRSApplication</a>                                                                                      |
| Example       | <code>geosrs:TopographicMap</code>                                                                                  |

### 10.1.5. Class: geosrs:WeatherMap

**Table 92** — geosrs:WeatherMap

|               |                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/WeatherMap">https://w3id.org/geosrs/application/WeatherMap</a> |
| Definition    | A map for showing the local direction in which weather systems are moving.                                  |
| Super-classes | <a href="#">SRSApplication</a>                                                                              |

## 10.2. SRS Application Types

### REQUIREMENT 20: SRS APPLICATION TYPES

**IDENTIFIER**    /req/srsapplication/SRS\_Application\_Types

**STATEMENT**

Implementations shall allow the RDFS classes `geosrs:SRSApplication`, `geosrs:SpatialReferencing`, `geosrs:EngineeringSurvey`, `geosrs:SatelliteSurvey`, `geosrs:SatelliteNavigation`, `geosrs:CoastalHydrography`, `geosrs:OffshoreEngineering`, `geosrs:Hydrography`, `geosrs:Drilling`, `geosrs:OilAndGasExploration` to be used in SPARQL graph patterns.

### 10.2.1. Class: geosrs:SRSApplication

**Table 93** — geosrs:SRSApplication

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/application/SRSApplication">https://w3id.org/geosrs/application/SRSApplication</a> |
| Definition | An application for which a spatial reference system is used.                                                        |

### 10.2.2. Class: geosrs:SpatialReferencing

**Table 94** — geosrs:SpatialReferencing

|               |                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/SpatialReferencing">https://w3id.org/geosrs/application/SpatialReferencing</a>                                                        |
| Definition    | Spatial referencing is the process of assigning real-world coordinates to data so that it can be located on the Earth's surface and used in a geographic information system (GIS). |
| Super-classes | <a href="#">SRSApplication</a>                                                                                                                                                     |

### 10.2.3. Class: geosrs:EngineeringSurvey

**Table 95** — geosrs:EngineeringSurvey

|               |                                                                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/EngineeringSurvey">https://w3id.org/geosrs/application/EngineeringSurvey</a>                                                     |
| Definition    | An engineering survey is the branch of surveying that provides the precise measurements and data needed to plan, build, and maintain engineering and infrastructure projects. |
| Super-classes | <a href="#">SRSApplication</a>                                                                                                                                                |
| Example       | <a href="#">geosrs:EngineeringSurvey</a>                                                                                                                                      |

### 10.2.4. Class: geosrs:SatelliteSurvey

**Table 96** — geosrs:SatelliteSurvey

|               |                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/SatelliteSurvey">https://w3id.org/geosrs/application/SatelliteSurvey</a>              |
| Definition    | A remote sensing survey conducted from Earth-orbiting satellites, collecting imagery and other data without direct ground contact. |
| Super-classes | <a href="#">SRSAApplication</a>                                                                                                    |

### 10.2.5. Class: geosrs:SatelliteNavigation

**Table 97** — geosrs:SatelliteNavigation

|               |                                                                                                                                                                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/SatelliteNavigation">https://w3id.org/geosrs/application/SatelliteNavigation</a>                                                                                                                                                                              |
| Definition    | Satellite navigation is a system that uses satellites to provide autonomous geo-spatial positioning. It allows small electronic receivers to determine their location (longitude, latitude, and altitude) to high precision using time signals transmitted along a line of sight by radio from satellites. |
| Super-classes | <a href="#">SRSAApplication</a>                                                                                                                                                                                                                                                                            |

### 10.2.6. Class: geosrs:CoastalHydrography

**Table 98** — geosrs:CoastalHydrography

|               |                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/CoastalHydrography">https://w3id.org/geosrs/application/CoastalHydrography</a>    |
| Definition    | Hydrographic surveying & monitoring focused on nearshore waters, where navigation safety and coastal change are most critical. |
| Super-classes | <a href="#">Hydrography</a>                                                                                                    |
| Example       | <a href="#">geosrs:CoastalHydrography</a>                                                                                      |

### 10.2.7. Class: geosrs:OffshoreEngineering

**Table 99** — geosrs:OffshoreEngineering

|               |                                                                                                                                                                                                                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/OffshoreEngineering">https://w3id.org/geosrs/application/OffshoreEngineering</a>                                                                                                                                                                                                         |
| Definition    | Offshore engineering (also called ocean engineering or marine engineering in some contexts) is the branch of engineering concerned with the design, construction, installation, and maintenance of structures and systems in the ocean environment, such as oil and gas platforms, subsea pipelines, and renewable energy facilities. |
| Super-classes | <a href="#">SRSAApplication</a>                                                                                                                                                                                                                                                                                                       |
| Example       | <a href="#">geosrs:OffshoreEngineering</a>                                                                                                                                                                                                                                                                                            |

## 10.2.8. Class: geosrs:Hydrography

**Table 100** — geosrs:Hydrography

|               |                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/Hydrography">https://w3id.org/geosrs/application/Hydrography</a>                                                                                                                                                                                                                                                                                                                           |
| Definition    | Hydrography is the branch of applied sciences which deals with the measurement and description of the physical features of oceans, seas, coastal areas, lakes, and rivers, as well as the prediction of their change over time, for the primary purpose of safety of navigation and in support of all other marine activities, including economic development, security and defense, scientific research, and environmental protection. |
| Super-classes | <a href="#">SRSAApplication</a>                                                                                                                                                                                                                                                                                                                                                                                                         |
| Example       | <a href="#">geosrs:Hydrography</a>                                                                                                                                                                                                                                                                                                                                                                                                      |

## 10.2.9. Class: geosrs:Drilling

**Table 101** — geosrs:Drilling

|            |                                                                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/application/Drilling">https://w3id.org/geosrs/application/Drilling</a>                                                                        |
| Definition | Drilling is the process of creating holes in the ground (or other solid materials) using specialized tools, widely applied in energy, construction, mining, and manufacturing. |

|               |                                 |
|---------------|---------------------------------|
| Super-classes | <a href="#">SRSAApplication</a> |
| Example       | <a href="#">geosrs:Drilling</a> |

## 10.2.10. Class: [geosrs:OilAndGasExploration](#)

**Table 102** — [geosrs:OilAndGasExploration](#)

|               |                                                                                                                                                                                                             |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/application/OilAndGasExploration">https://w3id.org/geosrs/application/OilAndGasExploration</a>                                                                             |
| Definition    | Oil and natural gas exploration is the search for underground or underwater reservoirs containing hydrocarbons, using geological and geophysical methods, followed by drilling to confirm and produce them. |
| Super-classes | <a href="#">SRSAApplication</a>                                                                                                                                                                             |



11

# PROJECTIONS MODULE

---



This clause establishes the **PROJ** Requirements class, with IRI /req/proj, which has a corresponding Conformance Class, **PROJ**, with IRI /conf/proj.

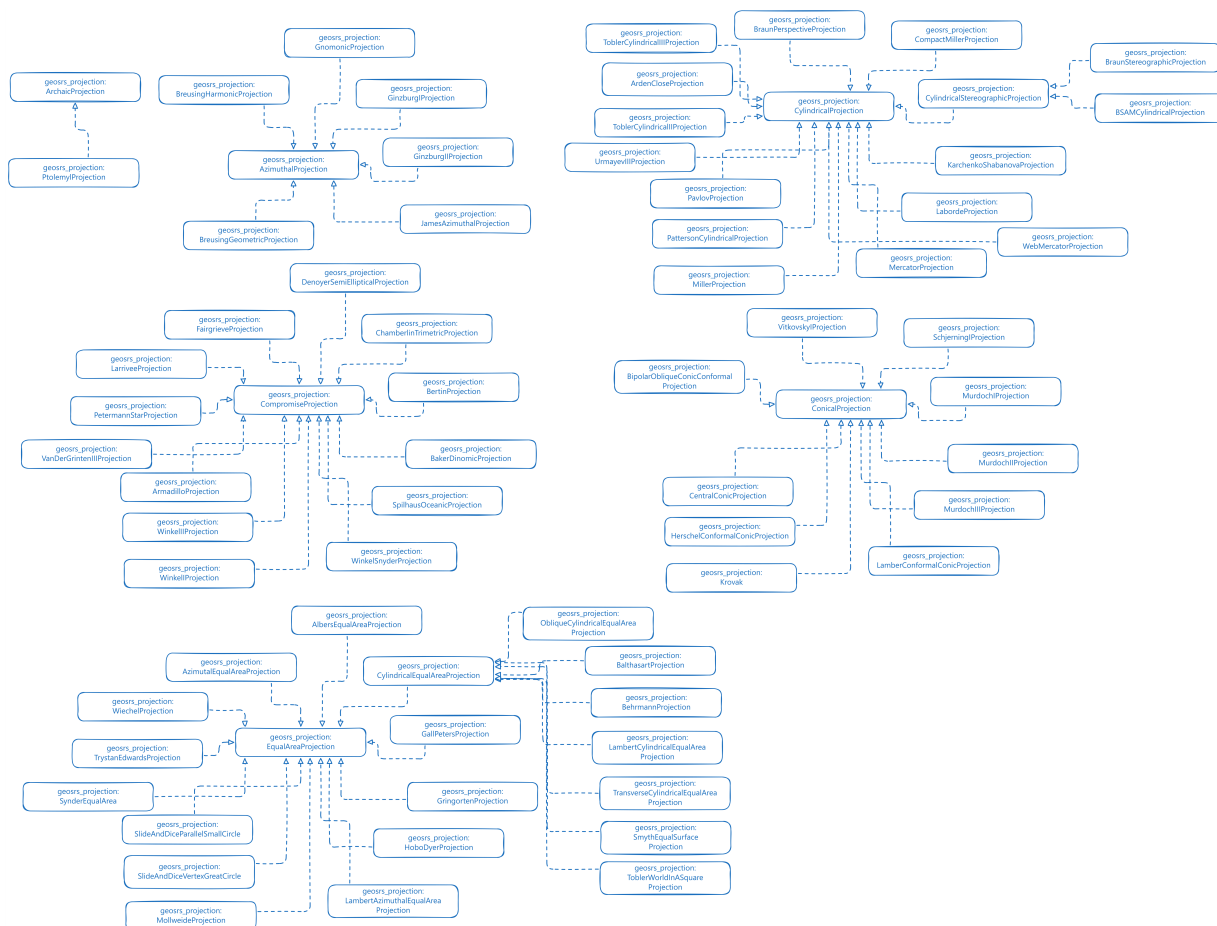


Figure 6

## REQUIREMENTS CLASS 6: 11-PROJECTIONS\_MODULE.ADOC EXTENSION

|                   |                                          |
|-------------------|------------------------------------------|
| IDENTIFIER        | /req/projections                         |
| TARGET TYPE       | Implementation Specification             |
| CONFORMANCE CLASS | Conformance class A.6: /conf/projections |
| REQUIREMENT       | /req/projections/Lenticular_Projections  |
|                   | /req/projections/Conformal_Projections   |

REQUIREMENTS CLASS 6: 11-PROJECTIONS\_MODULE.ADOC EXTENSION

|  |                                                 |
|--|-------------------------------------------------|
|  | /req/projections/Minimum_Error_Projections      |
|  | /req/projections/Pseudo_Azimuthal_Projections   |
|  | /req/projections/Equal_Area_Projections         |
|  | /req/projections/Pseudo_Conical_Projections     |
|  | /req/projections/Globular_Projections           |
|  | /req/projections/Pseudo_Cylindrical_Projections |
|  | /req/projections/Archaic_Projections            |
|  | /req/projections/Cylindrical_Projections        |
|  | /req/projections/Compromise_Projections         |
|  | /req/projections/Polyhedral_Projections         |
|  | /req/projections/Equidistant_Projections        |
|  | /req/projections/Azimuthal_Projections          |
|  | /req/projections/Conical_Projections            |
|  | /req/projections/Perspective_Projections        |
|  | /req/projections/Stereographic_Projections      |
|  | /req/projections/Polyconic_Projections          |
|  | /req/projections/Projection                     |

11.1. Archaic Projections

REQUIREMENT 21: ARCHAIC PROJECTIONS

|            |                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/projections/Archaic_Projections                                                                                                   |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:ArchaicProjection, geosrs:Ptolemy IProjection to be used in SPARQL graph patterns. |

### 11.1.1. Class: geosrs:ArchaicProjection

Table 103 — geosrs:ArchaicProjection

|     |                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/ArchaicProjection">https://w3id.org/geosrs/projection/ArchaicProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------|

### 11.1.2. Class: geosrs:PtolemyIProjection

Table 104 — geosrs:PtolemyIProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PtolemyIProjection">https://w3id.org/geosrs/projection/PtolemyIProjection</a> |
| Super-classes | <a href="#">ArchaicProjection</a>                                                                                         |

## 11.2. Azimuthal Projections

### REQUIREMENT 22: AZIMUTHAL PROJECTIONS

**IDENTIFIER** /req/projections/Azimuthal\_Projections

**STATEMENT** Implementations shall allow the RDFS classes geosrs:AzimuthalProjection, geosrs:BreusingGeometricProjection, geosrs:BreusingHarmonicProjection, geosrs:GinzburgIProjection, geosrs:GinzburgIIProjection, geosrs:GnomonicProjection, geosrs:JamesAzimuthalProjection to be used in SPARQL graph patterns.

### 11.2.1. Class: geosrs:AzimuthalProjection

Table 105 — geosrs:AzimuthalProjection

|     |                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/AzimuthalProjection">https://w3id.org/geosrs/projection/AzimuthalProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------|

### 11.2.2. Class: geosrs:BreusingGeometricProjection

**Table 106** — geosrs:BreusingGeometricProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BreusingGeometricProjection">https://w3id.org/geosrs/projection/BreusingGeometricProjection</a> |
| Super-classes | <a href="#">AzimuthalProjection</a>                                                                                                         |

### 11.2.3. Class: geosrs:BreusingHarmonicProjection

**Table 107** — geosrs:BreusingHarmonicProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BreusingHarmonicProjection">https://w3id.org/geosrs/projection/BreusingHarmonicProjection</a> |
| Super-classes | <a href="#">AzimuthalProjection</a>                                                                                                       |

### 11.2.4. Class: geosrs:GinzburgIIProjection

**Table 108** — geosrs:GinzburgIIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GinzburgIIProjection">https://w3id.org/geosrs/projection/GinzburgIIProjection</a> |
| Super-classes | <a href="#">AzimuthalProjection</a>                                                                                           |

### 11.2.5. Class: geosrs:GinzburgIProjection

**Table 109** — geosrs:GinzburgIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GinzburgIProjection">https://w3id.org/geosrs/projection/GinzburgIProjection</a> |
| Super-classes | <a href="#">AzimuthalProjection</a>                                                                                         |

### 11.2.6. Class: geosrs:GnomonicProjection

**Table 110** — geosrs:GnomonicProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GnomonicProjection">https://w3id.org/geosrs/projection/GnomonicProjection</a> |
| Super-classes | <a href="#">AzimuthalProjection</a>                                                                                       |

### 11.2.7. Class: geosrs:JamesAzimuthalProjection

Table 111 — geosrs:JamesAzimuthalProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/JamesAzimuthalProjection">https://w3id.org/geosrs/projection/JamesAzimuthalProjection</a> |
| Super-classes | <a href="#">AzimuthalProjection</a>                                                                                                   |

## 11.3. Compromise Projections

### REQUIREMENT 23: COMPROMISE PROJECTIONS

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/projections/Compromise_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:ArmadilloProjection, geosrs:BakerDinomicProjection, geosrs:BertinProjection, geosrs:ChamberlinTrimetricProjection, geosrs:DenoyerSemiEllipticalProjection, geosrs:FairgrieveProjection, geosrs:LarriveeProjection, geosrs:PetermannStarProjection, geosrs:SpilhausOceanicProjection, geosrs:VanDerGrintenIIIProjection, geosrs:WinkelIIProjection, geosrs:WinkelIProjection, geosrs:WinkelSnyderProjection to be used in SPARQL graph patterns. |

### 11.3.1. Class: geosrs:ArmadilloProjection

Table 112 — geosrs:ArmadilloProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ArmadilloProjection">https://w3id.org/geosrs/projection/ArmadilloProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                        |

### 11.3.2. Class: geosrs:BakerDinomicProjection

Table 113 — geosrs:BakerDinomicProjection

|     |                                                                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/BakerDinomicProjection">https://w3id.org/geosrs/projection/BakerDinomicProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|

|               |                                             |
|---------------|---------------------------------------------|
| Super-classes | <a href="#"><u>CompromiseProjection</u></a> |
|---------------|---------------------------------------------|

### 11.3.3. Class: geosrs:BertinProjection

**Table 114** — geosrs:BertinProjection

|               |                                                                                                                              |
|---------------|------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BertinProjection"><u>https://w3id.org/geosrs/projection/BertinProjection</u></a> |
| Super-classes | <a href="#"><u>CompromiseProjection</u></a>                                                                                  |

### 11.3.4. Class: geosrs:ChamberlinTrimetricProjection

**Table 115** — geosrs:ChamberlinTrimetricProjection

|               |                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ChamberlinTrimetricProjection"><u>https://w3id.org/geosrs/projection/ChamberlinTrimetricProjection</u></a> |
| Super-classes | <a href="#"><u>CompromiseProjection</u></a>                                                                                                            |

### 11.3.5. Class: geosrs:DenoyerSemiEllipticalProjection

**Table 116** — geosrs:DenoyerSemiEllipticalProjection

|               |                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/DenoyerSemiEllipticalProjection"><u>https://w3id.org/geosrs/projection/DenoyerSemiEllipticalProjection</u></a> |
| Super-classes | <a href="#"><u>CompromiseProjection</u></a>                                                                                                                |

### 11.3.6. Class: geosrs:FairgrieveProjection

**Table 117** — geosrs:FairgrieveProjection

|               |                                                                                                                                      |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FairgrieveProjection"><u>https://w3id.org/geosrs/projection/FairgrieveProjection</u></a> |
| Super-classes | <a href="#"><u>CompromiseProjection</u></a>                                                                                          |

### 11.3.7. Class: geosrs:LarriveeProjection

Table 118 — geosrs:LarriveeProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LarriveeProjection">https://w3id.org/geosrs/projection/LarriveeProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                      |

### 11.3.8. Class: geosrs:PetermannStarProjection

Table 119 — geosrs:PetermannStarProjection

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PetermannStarProjection">https://w3id.org/geosrs/projection/PetermannStarProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                                |

### 11.3.9. Class: geosrs:SpilhausOceanicProjection

Table 120 — geosrs:SpilhausOceanicProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/SpilhausOceanicProjection">https://w3id.org/geosrs/projection/SpilhausOceanicProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                                    |

### 11.3.10. Class: geosrs:VanDerGrintenIIIProjection

Table 121 — geosrs:VanDerGrintenIIIProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/VanDerGrintenIIIProjection">https://w3id.org/geosrs/projection/VanDerGrintenIIIProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                                      |

### 11.3.11. Class: geosrs:WinkelIIIProjection

**Table 122** — geosrs:WinkelIIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WinkelIIIProjection">https://w3id.org/geosrs/projection/WinkelIIIProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                        |

**11.3.12. Class: geosrs:WinkelIIIProjection**

**Table 123** — geosrs:WinkelIIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WinkelIIIProjection">https://w3id.org/geosrs/projection/WinkelIIIProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                        |

**11.3.13. Class: geosrs:WinkelSnyderProjection**

**Table 124** — geosrs:WinkelSnyderProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WinkelSnyderProjection">https://w3id.org/geosrs/projection/WinkelSnyderProjection</a> |
| Super-classes | <a href="#">CompromiseProjection</a>                                                                                              |

**11.4. Conformal Projections**

| Requirement 24: Conformal Projections |                                                                                                                                                                                                                                                                                                                                                                                   |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                            | /req/projections/Conformal_Projections                                                                                                                                                                                                                                                                                                                                            |
| STATEMENT                             | Implementations shall allow the RDFS classes geosrs:AdamsProjection, geosrs:AdamsWorldInASquareIProjection, geosrs:AdamsWorldInASquareIProjection, geosrs:AugustEpicycloidalProjection, geosrs:CoxConformalProjection, geosrs:EisenlohrProjection, geosrs:GS50Projection, geosrs:PeirceQuincuncialProjection, geosrs:StereographicProjection to be used in SPARQL graph patterns. |



### 11.4.1. Class: geosrs:AdamsProjection

Table 125 — geosrs:AdamsProjection

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AdamsProjection">https://w3id.org/geosrs/projection/AdamsProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                 |

### 11.4.2. Class: geosrs:AdamsWorldInASquareIIProjection

Table 126 — geosrs:AdamsWorldInASquareIIProjection

|               |                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AdamsWorldInASquareIIProjection">https://w3id.org/geosrs/projection/AdamsWorldInASquareIIProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                                                 |

### 11.4.3. Class: geosrs:AdamsWorldInASquareIProjection

Table 127 — geosrs:AdamsWorldInASquareIProjection

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AdamsWorldInASquareIProjection">https://w3id.org/geosrs/projection/AdamsWorldInASquareIProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                                               |

### 11.4.4. Class: geosrs:AugustEpicycloidalProjection

Table 128 — geosrs:AugustEpicycloidalProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AugustEpicycloidalProjection">https://w3id.org/geosrs/projection/AugustEpicycloidalProjection</a> |
| Definition    | A projection in which every angle between two curves that cross each other on a celestial body is preserved in the image of the projection    |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                                           |

### 11.4.5. Class: geosrs:CoxConformalProjection

Table 129 — geosrs:CoxConformalProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CoxConformalProjection">https://w3id.org/geosrs/projection/CoxConformalProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                               |

### 11.4.6. Class: geosrs:EisenlohrProjection

Table 130 — geosrs:EisenlohrProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/EisenlohrProjection">https://w3id.org/geosrs/projection/EisenlohrProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                         |

### 11.4.7. Class: geosrs:GS50Projection

Table 131 — geosrs:GS50Projection

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GS50Projection">https://w3id.org/geosrs/projection/GS50Projection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                               |

### 11.4.8. Class: geosrs:PeirceQuincuncialProjection

Table 132 — geosrs:PeirceQuincuncialProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PeirceQuincuncialProjection">https://w3id.org/geosrs/projection/PeirceQuincuncialProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                                         |

### 11.4.9. Class: geosrs:StereographicProjection

**Table 133** — geosrs:StereographicProjection

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/StereographicProjection">https://w3id.org/geosrs/projection/StereographicProjection</a> |
| Super-classes | <a href="#">ConformalProjection</a>                                                                                                 |
| Example       | <a href="#">geosrs:StereographicProjection</a>                                                                                      |

## 11.5. Conical Projections

### REQUIREMENT 25: CONICAL PROJECTIONS

**IDENTIFIER**    /req/projections/Conical\_Projections

**STATEMENT**    Implementations shall allow the RDFS classes geosrs:BipolarObliqueConicConformalProjection, geosrs:CentralConicProjection, geosrs:HerschelConformalConicProjection, geosrs:Krovak, geosrs:LambertConformalConicProjection, geosrs:MurdochIIIProjection, geosrs:MurdochIIProjection, geosrs:MurdochIProjection, geosrs:SchjerningIProjection, geosrs:VitkovskylProjection to be used in SPARQL graph patterns.

### 11.5.1. Class: geosrs:BipolarObliqueConicConformalProjection

**Table 134** — geosrs:BipolarObliqueConicConformalProjection

|               |                                                                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BipolarObliqueConicConformalProjection">https://w3id.org/geosrs/projection/BipolarObliqueConicConformalProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                                                                 |

### 11.5.2. Class: geosrs:CentralConicProjection

**Table 135** — geosrs:CentralConicProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CentralConicProjection">https://w3id.org/geosrs/projection/CentralConicProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                                 |

### 11.5.3. Class: geosrs:HerschelConformalConicProjection

Table 136 — geosrs:HerschelConformalConicProjection

|               |                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HerschelConformalConicProjection">https://w3id.org/geosrs/projection/HerschelConformalConicProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                                                     |

### 11.5.4. Class: geosrs:Krovak

Table 137 — geosrs:Krovak

|               |                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Krovak">https://w3id.org/geosrs/projection/Krovak</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                 |
| Example       | <a href="#">geosrs:Krovak</a>                                                                     |

### 11.5.5. Class: geosrs:LambertConformalConicProjection

Table 138 — geosrs:LambertConformalConicProjection

|               |                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LambertConformalConicProjection">https://w3id.org/geosrs/projection/LambertConformalConicProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                                                   |

### 11.5.6. Class: geosrs:MurdochIIIProjection

Table 139 — geosrs:MurdochIIIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MurdochIIIProjection">https://w3id.org/geosrs/projection/MurdochIIIProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                             |

### 11.5.7. Class: geosrs:MurdochIIProjection

**Table 140** — geosrs:MurdochIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MurdochIIProjection">https://w3id.org/geosrs/projection/MurdochIIProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                           |

### 11.5.8. Class: geosrs:MurdochIProjection

**Table 141** — geosrs:MurdochIProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MurdochIProjection">https://w3id.org/geosrs/projection/MurdochIProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                         |

### 11.5.9. Class: geosrs:SchjernerIProjection

**Table 142** — geosrs:SchjernerIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/SchjernerIProjection">https://w3id.org/geosrs/projection/SchjernerIProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                             |

### 11.5.10. Class: geosrs:VitkovskyIProjection

**Table 143** — geosrs:VitkovskyIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/VitkovskyIProjection">https://w3id.org/geosrs/projection/VitkovskyIProjection</a> |
| Super-classes | <a href="#">ConicalProjection</a>                                                                                             |

# 11.6. Cylindrical Projections

## REQUIREMENT 26: CYLINDRICAL PROJECTIONS

IDENTIFIER /req/projections/Cylindrical\_Projections

STATEMENT Implementations shall allow the RDFS classes geosrs:ArdenCloseProjection, geosrs:BSAMCylindricalProjection, geosrs:BalthasartProjection, geosrs:BehrmannProjection, geosrs:BraunPerspectiveProjection, geosrs:BraunStereographicProjection, geosrs:CompactMillerProjection, geosrs:CylindricalProjection, geosrs:CylindricalStereographicProjection, geosrs:KarchenkoShabanovaProjection, geosrs:LabordeProjection, geosrs:MercatorProjection, geosrs:MillerProjection, geosrs:PattersonCylindricalProjection, geosrs:PavlovProjection, geosrs:ToblerCylindricalIIIProjection, geosrs:ToblerCylindricalIIProjection, geosrs:TransverseMercatorProjection, geosrs:UrmayevIIIIProjection, geosrs:WebMercatorProjection to be used in SPARQL graph patterns.

### 11.6.1. Class: geosrs:ArdenCloseProjection

Table 144 – geosrs:ArdenCloseProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ArdenCloseProjection">https://w3id.org/geosrs/projection/ArdenCloseProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                         |

### 11.6.2. Class: geosrs:BSAMCylindricalProjection

Table 145 – geosrs:BSAMCylindricalProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BSAMCylindricalProjection">https://w3id.org/geosrs/projection/BSAMCylindricalProjection</a> |
| Super-classes | <a href="#">CylindricalStereographicProjection</a>                                                                                      |

### 11.6.3. Class: geosrs:BalthasartProjection

Table 146 – geosrs:BalthasartProjection

|     |                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/BalthasartProjection">https://w3id.org/geosrs/projection/BalthasartProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------------|

|               |                                                                                          |
|---------------|------------------------------------------------------------------------------------------|
| Definition    | A cylindrical equal-area projection that uses a standard parallel of $\phi_s=50$ degrees |
| Super-classes | <a href="#"><u>CylindricalEqualArea</u></a>                                              |

#### 11.6.4. Class: geosrs:BehrmannProjection

**Table 147** — geosrs:BehrmannProjection

|               |                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BehrmannProjection"><u>https://w3id.org/geosrs/projection/BehrmannProjection</u></a> |
| Definition    | A cylindrical equal-area map projection with standard parallels set at 30° north and south                                       |
| Super-classes | <a href="#"><u>CylindricalEqualArea</u></a>                                                                                      |

#### 11.6.5. Class: geosrs:BraunPerspectiveProjection

**Table 148** — geosrs:BraunPerspectiveProjection

|               |                                                                                                                                                  |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BraunPerspectiveProjection"><u>https://w3id.org/geosrs/projection/BraunPerspectiveProjection</u></a> |
| Super-classes | <a href="#"><u>CylindricalProjection</u></a>                                                                                                     |

#### 11.6.6. Class: geosrs:BraunStereographicProjection

**Table 149** — geosrs:BraunStereographicProjection

|               |                                                                                                                                                      |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BraunStereographicProjection"><u>https://w3id.org/geosrs/projection/BraunStereographicProjection</u></a> |
| Super-classes | <a href="#"><u>CylindricalStereographicProjection</u></a>                                                                                            |

#### 11.6.7. Class: geosrs:CompactMillerProjection

**Table 150** — geosrs:CompactMillerProjection

|     |                                                                                                                                            |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/CompactMillerProjection"><u>https://w3id.org/geosrs/projection/CompactMillerProjection</u></a> |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                       |
|---------------|---------------------------------------|
| Super-classes | <a href="#">CylindricalProjection</a> |
|---------------|---------------------------------------|

### 11.6.8. Class: geosrs:CylindricalProjection

**Table 151** — geosrs:CylindricalProjection

|     |                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/CylindricalProjection">https://w3id.org/geosrs/projection/CylindricalProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------------|

### 11.6.9. Class: geosrs:CylindricalStereographicProjection

**Table 152** — geosrs:CylindricalStereographicProjection

|               |                                                                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CylindricalStereographicProjection">https://w3id.org/geosrs/projection/CylindricalStereographicProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                                                     |

### 11.6.10. Class: geosrs:KarchenkoShabanovaProjection

**Table 153** — geosrs:KarchenkoShabanovaProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/KarchenkoShabanovaProjection">https://w3id.org/geosrs/projection/KarchenkoShabanovaProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                                         |

### 11.6.11. Class: geosrs:LabordeProjection

**Table 154** — geosrs:LabordeProjection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LabordeProjection">https://w3id.org/geosrs/projection/LabordeProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                   |
| Example       | <a href="#">geosrs:LabordeProjection</a>                                                                                |



### 11.6.12. Class: geosrs:MercatorProjection

Table 155 — geosrs:MercatorProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MercatorProjection">https://w3id.org/geosrs/projection/MercatorProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                     |
| Example       | <a href="#">geosrs:MercatorProjection</a>                                                                                 |

### 11.6.13. Class: geosrs:MillerProjection

Table 156 — geosrs:MillerProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MillerProjection">https://w3id.org/geosrs/projection/MillerProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                 |

### 11.6.14. Class: geosrs:PattersonCylindricalProjection

Table 157 — geosrs:PattersonCylindricalProjection

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PattersonCylindricalProjection">https://w3id.org/geosrs/projection/PattersonCylindricalProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                                             |

### 11.6.15. Class: geosrs:PavlovProjection

Table 158 — geosrs:PavlovProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PavlovProjection">https://w3id.org/geosrs/projection/PavlovProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                 |

### 11.6.16. Class: geosrs:ToblerCylindricalIIIProjection

Table 159 — geosrs:ToblerCylindricalIIIProjection

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ToblerCylindricalIIIProjection">https://w3id.org/geosrs/projection/ToblerCylindricalIIIProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                                             |

### 11.6.17. Class: geosrs:ToblerCylindricalProjection

Table 160 — geosrs:ToblerCylindricalProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ToblerCylindricalProjection">https://w3id.org/geosrs/projection/ToblerCylindricalProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                                       |

### 11.6.18. Class: geosrs:TransverseMercatorProjection

Table 161 — geosrs:TransverseMercatorProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/TransverseMercatorProjection">https://w3id.org/geosrs/projection/TransverseMercatorProjection</a> |
| Super-classes | <a href="#">MercatorProjection</a>                                                                                                            |

### 11.6.19. Class: geosrs:UrmayevIIIProjection

Table 162 — geosrs:UrmayevIIIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/UrmayevIIIProjection">https://w3id.org/geosrs/projection/UrmayevIIIProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                         |

### 11.6.20. Class: geosrs:WebMercatorProjection

**Table 163** — geosrs:WebMercatorProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WebMercatorProjection">https://w3id.org/geosrs/projection/WebMercatorProjection</a> |
| Super-classes | <a href="#">CylindricalProjection</a>                                                                                           |

## 11.7. Equal Area Projections

### REQUIREMENT 27: EQUAL AREA PROJECTIONS

**IDENTIFIER** /req/projections/Equal\_Area\_Projections

**STATEMENT**

Implementations shall allow the RDFS classes geosrs:AlbersEqualAreaProjection, geosrs:AzimuthalEqualAreaProjection, geosrs:CylindricalEqualArea, geosrs:EqualAreaProjection, geosrs:GallPetersProjection, geosrs:HoboDyerProjection, geosrs:LambertAzimuthalEqualArea, geosrs:LambertCylindricalEqualAreaProjection, geosrs:ObliqueCylindricalEqualAreaProjection, geosrs:SlideAndDiceParallelSmallCircle, geosrs:SliceAndDiceVertexGreatCircle, geosrs:SmythEqualSurfaceProjection, geosrs:SnyderEqualArea, geosrs:ToblerWorldInASquareProjection, geosrs:TransverseCylindricalEqualAreaProjection, geosrs:TrystanEdwardsProjection, geosrs:WiechelProjection to be used in SPARQL graph patterns.

### 11.7.1. Class: geosrs:AlbersEqualAreaProjection

**Table 164** — geosrs:AlbersEqualAreaProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AlbersEqualAreaProjection">https://w3id.org/geosrs/projection/AlbersEqualAreaProjection</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                                     |
| Example       | <code>geosrs:AlbersEqualAreaProjection</code>                                                                                           |

### 11.7.2. Class: geosrs:AzimuthalEqualAreaProjection

**Table 165** — geosrs:AzimuthalEqualAreaProjection

|     |                                                                                                                                               |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/AzimuthalEqualAreaProjection">https://w3id.org/geosrs/projection/AzimuthalEqualAreaProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                     |
|---------------|-------------------------------------|
| Super-classes | <a href="#">EqualAreaProjection</a> |
|---------------|-------------------------------------|

### 11.7.3. Class: geosrs:CylindricalEqualArea

**Table 166** — geosrs:CylindricalEqualArea

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CylindricalEqualArea">https://w3id.org/geosrs/projection/CylindricalEqualArea</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                           |
| Example       | <a href="#">geosrs:CylindricalEqualArea</a>                                                                                   |

### 11.7.4. Class: geosrs:EqualAreaProjection

**Table 167** — geosrs:EqualAreaProjection

|     |                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/EqualAreaProjection">https://w3id.org/geosrs/projection/EqualAreaProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------|

### 11.7.5. Class: geosrs:GallPetersProjection

**Table 168** — geosrs:GallPetersProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GallPetersProjection">https://w3id.org/geosrs/projection/GallPetersProjection</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                           |

### 11.7.6. Class: geosrs:HoboDyerProjection

**Table 169** — geosrs:HoboDyerProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HoboDyerProjection">https://w3id.org/geosrs/projection/HoboDyerProjection</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                       |

### 11.7.7. Class: geosrs:LambertAzimuthalEqualArea

Table 170 — geosrs:LambertAzimuthalEqualArea

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LambertAzimuthalEqualArea">https://w3id.org/geosrs/projection/LambertAzimuthalEqualArea</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                                     |

### 11.7.8. Class: geosrs:LambertCylindricalEqualAreaProjection

Table 171 — geosrs:LambertCylindricalEqualAreaProjection

|               |                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LambertCylindricalEqualAreaProjection">https://w3id.org/geosrs/projection/LambertCylindricalEqualAreaProjection</a> |
| Super-classes | <a href="#">CylindricalEqualArea</a>                                                                                                                            |

### 11.7.9. Class: geosrs:ObliqueCylindricalEqualAreaProjection

Table 172 — geosrs:ObliqueCylindricalEqualAreaProjection

|               |                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ObliqueCylindricalEqualAreaProjection">https://w3id.org/geosrs/projection/ObliqueCylindricalEqualAreaProjection</a> |
| Super-classes | <a href="#">CylindricalEqualArea</a>                                                                                                                            |

### 11.7.10. Class: geosrs:SlideAndDiceParallelSmallCircle

Table 173 — geosrs:SlideAndDiceParallelSmallCircle

|            |                                                                                                                                                                                                                                                                                                                    |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/projection/SlideAndDiceParallelSmallCircle">https://w3id.org/geosrs/projection/SlideAndDiceParallelSmallCircle</a>                                                                                                                                                                |
| Definition | The Parallel Small Circle version of the equa-area projection method defined for polyhedral globes by van Leeuwen and Strebe. van Leeuwen, D., & Strebe, D. (2006). A “Slice-and-Dice” Approach to Area Equivalence in Polyhedral Map Projections. Cartography and Geographic Information Science, 33(4), 269–286. |

|               |                                     |
|---------------|-------------------------------------|
| Super-classes | <a href="#">EqualAreaProjection</a> |
|---------------|-------------------------------------|

### 11.7.11. Class: geosrs:SliceAndDiceVertexGreatCircle

**Table 174** — geosrs:SliceAndDiceVertexGreatCircle

|               |                                                                                                                                                                                                                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/SliceAndDiceVertexGreatCircle">https://w3id.org/geosrs/projection/SliceAndDiceVertexGreatCircle</a>                                                                                                                                                                                    |
| Definition    | The Vertex-oriented Great Circle version of the equal-area projection method defined for polyhedral globes by van Leeuwen and Strebe. van Leeuwen, D., & Strebe, D. (2006). A "Slice-and-Dice" Approach to Area Equivalence in Polyhedral Map Projections. <i>Cartography and Geographic Information Science</i> , 33(4), 269–286. |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                                                                                                                                                                                                                                |

### 11.7.12. Class: geosrs:SmythEqualSurfaceProjection

**Table 175** — geosrs:SmythEqualSurfaceProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/SmythEqualSurfaceProjection">https://w3id.org/geosrs/projection/SmythEqualSurfaceProjection</a> |
| Super-classes | <a href="#">CylindricalEqualArea</a>                                                                                                        |

### 11.7.13. Class: geosrs:SnyderEqualArea

**Table 176** — geosrs:SnyderEqualArea

|               |                                                                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/SnyderEqualArea">https://w3id.org/geosrs/projection/SnyderEqualArea</a>                                                                                           |
| Definition    | Equal area projection for polyhedral globes, used frequently in Discrete Global Grid Systems. Snyder, J.P. (1992). "An Equal-Area Map Projection for Polyhedral Globes". <i>Cartographica</i> . 29 (1): 10–21 |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                                                                                                           |

### 11.7.14. Class: geosrs:ToblerWorldInASquareProjection

Table 177 — geosrs:ToblerWorldInASquareProjection

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ToblerWorldInASquareProjection">https://w3id.org/geosrs/projection/ToblerWorldInASquareProjection</a> |
| Super-classes | <a href="#">CylindricalEqualArea</a>                                                                                                              |

### 11.7.15. Class: geosrs:TransverseCylindricalEqualAreaProjection

Table 178 — geosrs:TransverseCylindricalEqualAreaProjection

|               |                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/TransverseCylindricalEqualAreaProjection">https://w3id.org/geosrs/projection/TransverseCylindricalEqualAreaProjection</a> |
| Super-classes | <a href="#">CylindricalEqualArea</a>                                                                                                                                  |

### 11.7.16. Class: geosrs:TrystanEdwardsProjection

Table 179 — geosrs:TrystanEdwardsProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/TrystanEdwardsProjection">https://w3id.org/geosrs/projection/TrystanEdwardsProjection</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                                   |

### 11.7.17. Class: geosrs:WiechelProjection

Table 180 — geosrs:WiechelProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WichelProjection">https://w3id.org/geosrs/projection/WichelProjection</a> |
| Super-classes | <a href="#">EqualAreaProjection</a>                                                                                   |

# 11.8. Equidistant Projections

## REQUIREMENT 28: EQUIDISTANT PROJECTIONS

**IDENTIFIER**     /req/projections/Equidistant\_Projections

**STATEMENT**     Implementations shall allow the RDFS classes geosrs:AzimuthalEquidistantProjection, geosrs:BerghausStarProjection, geosrs:CassiniProjection, geosrs:EquidistantConicProjection, geosrs:EquidistantCylindricalProjection, geosrs:EquidistantProjection, geosrs:EquirectangularProjection, geosrs:ObliquePlateCarreeProjection, geosrs:PlateCarreeProjection, geosrs:TwoPointEquidistantProjection to be used in SPARQL graph patterns.

### 11.8.1. Class: geosrs:AzimuthalEquidistantProjection

Table 181 — geosrs:AzimuthalEquidistantProjection

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AzimuthalEquidistantProjection">https://w3id.org/geosrs/projection/AzimuthalEquidistantProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                                             |
| Example       | <a href="#">geosrs:AzimuthalEquidistantProjection</a>                                                                                             |

### 11.8.2. Class: geosrs:BerghausStarProjection

Table 182 — geosrs:BerghausStarProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BerghausStarProjection">https://w3id.org/geosrs/projection/BerghausStarProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                             |

### 11.8.3. Class: geosrs:CassiniProjection

Table 183 — geosrs:CassiniProjection

|     |                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/CassiniProjection">https://w3id.org/geosrs/projection/CassiniProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------|



|               |                                                                                                    |
|---------------|----------------------------------------------------------------------------------------------------|
| Definition    | A map projection first described in an approximate form by César-François Cassini de Thury in 1745 |
| Super-classes | <a href="#">EquidistantProjection</a>                                                              |
| Example       | <a href="#">geosrs:CassiniProjection</a>                                                           |

#### 11.8.4. Class: geosrs:EquidistantConicProjection

**Table 184** — geosrs:EquidistantConicProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/EquidistantConicProjection">https://w3id.org/geosrs/projection/EquidistantConicProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                                     |

#### 11.8.5. Class: geosrs:EquidistantCylindricalProjection

**Table 185** — geosrs:EquidistantCylindricalProjection

|               |                                                                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/EquidistantCylindricalProjection">https://w3id.org/geosrs/projection/EquidistantCylindricalProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                                                 |
| Example       | <a href="#">geosrs:EquidistantCylindricalProjection</a>                                                                                               |

#### 11.8.6. Class: geosrs:EquidistantProjection

**Table 186** — geosrs:EquidistantProjection

|     |                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/EquidistantProjection">https://w3id.org/geosrs/projection/EquidistantProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------------|

#### 11.8.7. Class: geosrs:EquirectangularProjection

**Table 187** — geosrs:EquirectangularProjection

|     |                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/EquirectangularProjection">https://w3id.org/geosrs/projection/EquirectangularProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|

Super-classes

[EquidistantProjection](#)

### 11.8.8. Class: geosrs:ObliquePlateCarreeProjection

Table 188 — geosrs:ObliquePlateCarreeProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ObliquePlateCarreeProjection">https://w3id.org/geosrs/projection/ObliquePlateCarreeProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                                         |

### 11.8.9. Class: geosrs:PlateCarreeProjection

Table 189 — geosrs:PlateCarreeProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PlateCarreeProjection">https://w3id.org/geosrs/projection/PlateCarreeProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                           |

### 11.8.10. Class: geosrs:TwoPointEquidistantProjection

Table 190 — geosrs:TwoPointEquidistantProjection

|               |                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/TwoPointEquidistantProjection">https://w3id.org/geosrs/projection/TwoPointEquidistantProjection</a> |
| Super-classes | <a href="#">EquidistantProjection</a>                                                                                                           |

## 11.9. Globular Projections

### REQUIREMENT 29: GLOBULAR PROJECTIONS

|            |                                                                                                                                                                                       |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/projections/Globular_Projections                                                                                                                                                 |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:ApianGlobularIProjection, geosrs:BaconGlobularProjection, geosrs:FournierGlobularIProjection to be used in SPARQL graph patterns. |

### 11.9.1. Class: geosrs:ApianGlobularIProjection

Table 191 — geosrs:ApianGlobularIProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ApianGlobularIProjection">https://w3id.org/geosrs/projection/ApianGlobularIProjection</a> |
| Super-classes | <a href="#">GlobularProjection</a>                                                                                                    |

### 11.9.2. Class: geosrs:BaconGlobularProjection

Table 192 — geosrs:BaconGlobularProjection

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BaconGlobularProjection">https://w3id.org/geosrs/projection/BaconGlobularProjection</a> |
| Super-classes | <a href="#">GlobularProjection</a>                                                                                                  |

### 11.9.3. Class: geosrs:FournierGlobularIProjection

Table 193 — geosrs:FournierGlobularIProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FournierGlobularIProjection">https://w3id.org/geosrs/projection/FournierGlobularIProjection</a> |
| Super-classes | <a href="#">GlobularProjection</a>                                                                                                          |

## 11.10. Lenticular Projections

#### REQUIREMENT 30: LENTICULAR PROJECTIONS

**IDENTIFIER** /req/projections/Lenticular\_Projections

**STATEMENT** Implementations shall allow the RDFS classes geosrs:A4Projection, geosrs:BriesemeisterProjection, geosrs:CiricIProjection, geosrs:CupolaProjection, geosrs:DedistortProjection, geosrs:DietrichKitadaProjection, geosrs:FranculaIIIProjection, geosrs:FranculaIVProjection, geosrs:FranculaIXProjection, geosrs:FranculaVIIIProjection, geosrs:FranculaVProjection, geosrs:FranculaXIIIProjection, geosrs:

## REQUIREMENT 30: LENTICULAR PROJECTIONS

FranculaXIIProjection, geosrs:FranculaXIVProjection, geosrs:HamusoidalProjection, geosrs:KissProjection to be used in SPARQL graph patterns.

### 11.10.1. Class: geosrs:A4Projection

Table 194 — geosrs:A4Projection

|               |                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/A4Projection">https://w3id.org/geosrs/projection/A4Projection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                          |

### 11.10.2. Class: geosrs:BriesemeisterProjection

Table 195 — geosrs:BriesemeisterProjection

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BriesemeisterProjection">https://w3id.org/geosrs/projection/BriesemeisterProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                                |

### 11.10.3. Class: geosrs:CiricIProjection

Table 196 — geosrs:CiricIProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CiricIProjection">https://w3id.org/geosrs/projection/CiricIProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                  |

### 11.10.4. Class: geosrs:CupolaProjection

Table 197 — geosrs:CupolaProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CupolaProjection">https://w3id.org/geosrs/projection/CupolaProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                  |

### 11.10.5. Class: geosrs:DedistortProjection

Table 198 — geosrs:DedistortProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/DedistortProjection">https://w3id.org/geosrs/projection/DedistortProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                        |

### 11.10.6. Class: geosrs:DietrichKitadaProjection

Table 199 — geosrs:DietrichKitadaProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/DietrichKitadaProjection">https://w3id.org/geosrs/projection/DietrichKitadaProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                                  |

### 11.10.7. Class: geosrs:FranculaIIIProjection

Table 200 — geosrs:FranculaIIIProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaIIIProjection">https://w3id.org/geosrs/projection/FranculaIIIProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                            |

### 11.10.8. Class: geosrs:FranculaIVProjection

Table 201 — geosrs:FranculaIVProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaIVProjection">https://w3id.org/geosrs/projection/FranculaIVProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                          |

### 11.10.9. Class: geosrs:FranculaIXProjection

**Table 202** — geosrs:FranculaIXProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaIXProjection">https://w3id.org/geosrs/projection/FranculaIXProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                          |

### 11.10.10. Class: geosrs:FranculaVIIIProjection

**Table 203** — geosrs:FranculaVIIIProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaVIIIProjection">https://w3id.org/geosrs/projection/FranculaVIIIProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                              |

### 11.10.11. Class: geosrs:FranculaVProjection

**Table 204** — geosrs:FranculaVProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaVProjection">https://w3id.org/geosrs/projection/FranculaVProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                        |

### 11.10.12. Class: geosrs:FranculaXIIIProjection

**Table 205** — geosrs:FranculaXIIIProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaXIIIProjection">https://w3id.org/geosrs/projection/FranculaXIIIProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                              |

### 11.10.13. Class: geosrs:FranculaXIIProjection

**Table 206** — geosrs:FranculaXIIProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaXIIProjection">https://w3id.org/geosrs/projection/FranculaXIIProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                            |

### 11.10.14. Class: geosrs:FranculaXIVProjection

Table 207 — geosrs:FranculaXIVProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FranculaXIVProjection">https://w3id.org/geosrs/projection/FranculaXIVProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                            |

### 11.10.15. Class: geosrs:HamusoidalProjection

Table 208 — geosrs:HamusoidalProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HamusoidalProjection">https://w3id.org/geosrs/projection/HamusoidalProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                                          |

### 11.10.16. Class: geosrs:KissProjection

Table 209 — geosrs:KissProjection

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/KissProjection">https://w3id.org/geosrs/projection/KissProjection</a> |
| Super-classes | <a href="#">LenticularProjection</a>                                                                              |

## 11.11. Minimum Error Projections

#### REQUIREMENT 31: MINIMUM ERROR PROJECTIONS

|            |                                                                                                         |
|------------|---------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/projections/Minimum_Error_Projections                                                              |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:AiryProjection to be used in SPARQL graph patterns. |

### 11.11.1. Class: geosrs:AiryProjection

Table 210 — geosrs:AiryProjection

|               |                                                                                                                                                            |
|---------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AiryProjection">https://w3id.org/geosrs/projection/AiryProjection</a>                                          |
| Definition    | An azimuthal minimum error projection for the region within the small or great circle defined by an angular distance, from the tangency point of the plane |
| Super-classes | <a href="#">MinimumErrorProjection</a>                                                                                                                     |
| Example       | <a href="#">geosrs:AiryProjection</a>                                                                                                                      |

## 11.12. Perspective Projections

### REQUIREMENT 32: PERSPECTIVE PROJECTIONS

|            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/projections/Perspective_Projections                                                                                                                                                                                                                                                                                                                                                                                                                            |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:CentralCylindricalProjection, geosrs:GeneralVerticalPerspectiveProjection, geosrs:GilbertTwoWorldPerspectiveProjection, geosrs:LaHireProjection, geosrs:LorgnaProjection, geosrs:LowryProjection, geosrs:OrthographicProjection, geosrs:PerspectiveConicProjection, geosrs:PerspectiveProjection, geosrs:TiltedPerspectiveProjection, geosrs:VerticalPerspectiveProjection to be used in SPARQL graph patterns. |

### 11.12.1. Class: geosrs:CentralCylindricalProjection

Table 211 — geosrs:CentralCylindricalProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CentralCylindricalProjection">https://w3id.org/geosrs/projection/CentralCylindricalProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                                         |
| Example       | <a href="#">geosrs:CentralCylindricalProjection</a>                                                                                           |



### 11.12.2. Class: geosrs:GeneralVerticalPerspectiveProjection

Table 212 — geosrs:GeneralVerticalPerspectiveProjection

|               |                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GeneralVerticalPerspectiveProjection">https://w3id.org/geosrs/projection/GeneralVerticalPerspectiveProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                                                         |

### 11.12.3. Class: geosrs:GilbertTwoWorldPerspectiveProjection

Table 213 — geosrs:GilbertTwoWorldPerspectiveProjection

|               |                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GilbertTwoWorldPerspectiveProjection">https://w3id.org/geosrs/projection/GilbertTwoWorldPerspectiveProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                                                         |

### 11.12.4. Class: geosrs:LaHireProjection

Table 214 — geosrs:LaHireProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LaHireProjection">https://w3id.org/geosrs/projection/LaHireProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                 |

### 11.12.5. Class: geosrs:LorgnaProjection

Table 215 — geosrs:LorgnaProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LorgnaProjection">https://w3id.org/geosrs/projection/LorgnaProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                 |

### 11.12.6. Class: geosrs:LowryProjection

**Table 216** — geosrs:LowryProjection

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LowryProjection">https://w3id.org/geosrs/projection/LowryProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                               |

### 11.12.7. Class: geosrs:OrthographicProjection

**Table 217** — geosrs:OrthographicProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/OrthographicProjection">https://w3id.org/geosrs/projection/OrthographicProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                             |

### 11.12.8. Class: geosrs:PerspectiveConicProjection

**Table 218** — geosrs:PerspectiveConicProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PerspectiveConicProjection">https://w3id.org/geosrs/projection/PerspectiveConicProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                                     |

### 11.12.9. Class: geosrs:PerspectiveProjection

**Table 219** — geosrs:PerspectiveProjection

|     |                                                                                                                                 |
|-----|---------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/PerspectiveProjection">https://w3id.org/geosrs/projection/PerspectiveProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------------|

### 11.12.10. Class: geosrs:TiltedPerspectiveProjection

**Table 220** — geosrs:TiltedPerspectiveProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/TiltedPerspectiveProjection">https://w3id.org/geosrs/projection/TiltedPerspectiveProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                                       |

### 11.12.11. Class: geosrs:VerticalPerspectiveProjection

Table 221 — geosrs:VerticalPerspectiveProjection

|               |                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/VerticalPerspectiveProjection">https://w3id.org/geosrs/projection/VerticalPerspectiveProjection</a> |
| Super-classes | <a href="#">PerspectiveProjection</a>                                                                                                           |

## 11.13. Polyconic Projections

### REQUIREMENT 33: POLYCONIC PROJECTIONS

**IDENTIFIER** /req/projections/Polyconic\_Projections

**STATEMENT** Implementations shall allow the RDFS classes geosrs:GinzburgIVProjection, geosrs:GinzburgIXProjection, geosrs:GinzburgVIPProjection, geosrs:GinzburgVProjection, geosrs:GottWagnerProjection, geosrs:HillEucyclicProjection, geosrs:LagrangeProjection, geosrs:LaskowskiProjection, geosrs:PolyconicProjection, geosrs:RectangularPolyconicProjection, geosrs:StabiusWernerIIIProjection, geosrs:StabiusWernerIIProjection, geosrs:VanDerGrintenIIProjection, geosrs:VanDerGrintenIProjection, geosrs:VanDerGrintenIVProjection, geosrs:WagnerIXProjection, geosrs:WagnerVIIIProjection, geosrs:WagnerVIIProjection to be used in SPARQL graph patterns.

### 11.13.1. Class: geosrs:GinzburgIVProjection

Table 222 — geosrs:GinzburgIVProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GinzburgIVProjection">https://w3id.org/geosrs/projection/GinzburgIVProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                           |

### 11.13.2. Class: geosrs:GinzburgIXProjection

Table 223 — geosrs:GinzburgIXProjection

|     |                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/GinzburgIXProjection">https://w3id.org/geosrs/projection/GinzburgIXProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------------|

|               |                                     |
|---------------|-------------------------------------|
| Super-classes | <a href="#">PolyconicProjection</a> |
|---------------|-------------------------------------|

### 11.13.3. Class: geosrs:GinzburgVIProjection

**Table 224** — geosrs:GinzburgVIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GinzburgVIProjection">https://w3id.org/geosrs/projection/GinzburgVIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                           |

### 11.13.4. Class: geosrs:GinzburgVProjection

**Table 225** — geosrs:GinzburgVProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GinzburgVProjection">https://w3id.org/geosrs/projection/GinzburgVProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                         |

### 11.13.5. Class: geosrs:GottWagnerProjection

**Table 226** — geosrs:GottWagnerProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GottWagnerProjection">https://w3id.org/geosrs/projection/GottWagnerProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                           |

### 11.13.6. Class: geosrs:HillEucyclicProjection

**Table 227** — geosrs:HillEucyclicProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HillEucyclicProjection">https://w3id.org/geosrs/projection/HillEucyclicProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                               |

### 11.13.7. Class: geosrs:LagrangeProjection

Table 228 — geosrs:LagrangeProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LagrangeProjection">https://w3id.org/geosrs/projection/LagrangeProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                       |

### 11.13.8. Class: geosrs:LaskowskiProjection

Table 229 — geosrs:LaskowskiProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LaskowskiProjection">https://w3id.org/geosrs/projection/LaskowskiProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                         |

### 11.13.9. Class: geosrs:PolyconicProjection

Table 230 — geosrs:PolyconicProjection

|     |                                                                                                                             |
|-----|-----------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/PolyconicProjection">https://w3id.org/geosrs/projection/PolyconicProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------|

### 11.13.10. Class: geosrs:RectangularPolyconicProjection

Table 231 — geosrs:RectangularPolyconicProjection

|               |                                                                                                                                                   |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/RectangularPolyconicProjection">https://w3id.org/geosrs/projection/RectangularPolyconicProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                                               |

### 11.13.11. Class: geosrs:StabiusWernerIIIProjection

**Table 232** — geosrs:StabiusWernerIIIProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/StabiusWernerIIIProjection">https://w3id.org/geosrs/projection/StabiusWernerIIIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                                       |

### 11.13.12. Class: geosrs:StabiusWernerIProjection

**Table 233** — geosrs:StabiusWernerIProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/StabiusWernerIProjection">https://w3id.org/geosrs/projection/StabiusWernerIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                                   |

### 11.13.13. Class: geosrs:VanDerGrintenIIProjection

**Table 234** — geosrs:VanDerGrintenIIProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/VanDerGrintenIIProjection">https://w3id.org/geosrs/projection/VanDerGrintenIIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                                     |

### 11.13.14. Class: geosrs:VanDerGrintenIProjection

**Table 235** — geosrs:VanDerGrintenIProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/VanDerGrintenIProjection">https://w3id.org/geosrs/projection/VanDerGrintenIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                                   |

### 11.13.15. Class: geosrs:VanDerGrintenIVProjection

**Table 236** — geosrs:VanDerGrintenIVProjection

|     |                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/VanDerGrintenIVProjection">https://w3id.org/geosrs/projection/VanDerGrintenIVProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|

|               |                                     |
|---------------|-------------------------------------|
| Super-classes | <a href="#">PolyconicProjection</a> |
|---------------|-------------------------------------|

### 11.13.16. Class: geosrs:WagnerIXProjection

**Table 237** — geosrs:WagnerIXProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerIXProjection">https://w3id.org/geosrs/projection/WagnerIXProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                       |

### 11.13.17. Class: geosrs:WagnerVIIIProjection

**Table 238** — geosrs:WagnerVIIIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerVIIIProjection">https://w3id.org/geosrs/projection/WagnerVIIIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                           |

### 11.13.18. Class: geosrs:WagnerVIIProjection

**Table 239** — geosrs:WagnerVIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerVIIProjection">https://w3id.org/geosrs/projection/WagnerVIIProjection</a> |
| Super-classes | <a href="#">PolyconicProjection</a>                                                                                         |

## 11.14. Polyhedral Projections

| Requirement 34: Polyhedral Projections |                                                                                                                                                                                                                                                                                            |
|----------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                             | /req/projections/Polyhedral_Projections                                                                                                                                                                                                                                                    |
| STATEMENT                              | Implementations shall allow the RDFS classes geosrs:AuthaGraphProjection, geosrs:CahillKeyes Projection, geosrs:CollignonButterflyProjection, geosrs:DodecahedralProjection, geosrs:Dymaxion Projection, geosrs:GnomonicButterflyProjection, geosrs:GnomonicCubedSphereProjection, geosrs: |

## REQUIREMENT 34: POLYHEDRAL PROJECTIONS

GnomonicIcosahedronProjection, geosrs:GuyouProjection, geosrs:IcosahedronProjection, geosrs:LeeProjection, geosrs:MyrahedalProjection, geosrs:OctantProjection, geosrs:PolyhedralProjection, geosrs:QuadrilateralizedSphericalCubeProjection, geosrs:WatermanButterflyProjection to be used in SPARQL graph patterns.

### 11.14.1. Class: geosrs:AuthaGraphProjection

Table 240 — geosrs:AuthaGraphProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AuthaGraphProjection">https://w3id.org/geosrs/projection/AuthaGraphProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                          |

### 11.14.2. Class: geosrs:CahillKeyesProjection

Table 241 — geosrs:CahillKeyesProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CahillKeyesProjection">https://w3id.org/geosrs/projection/CahillKeyesProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                            |

### 11.14.3. Class: geosrs:CollignonButterflyProjection

Table 242 — geosrs:CollignonButterflyProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CollignonButterflyProjection">https://w3id.org/geosrs/projection/CollignonButterflyProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                                          |

### 11.14.4. Class: geosrs:DodecahedralProjection

Table 243 — geosrs:DodecahedralProjection

|     |                                                                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/DodecahedralProjection">https://w3id.org/geosrs/projection/DodecahedralProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|



|               |                                      |
|---------------|--------------------------------------|
| Super-classes | <a href="#">PolyhedralProjection</a> |
|---------------|--------------------------------------|

### 11.14.5. Class: geosrs:DymaxionProjection

**Table 244** — geosrs:DymaxionProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/DymaxionProjection">https://w3id.org/geosrs/projection/DymaxionProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                      |

### 11.14.6. Class: geosrs:GnomonicButterflyProjection

**Table 245** — geosrs:GnomonicButterflyProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GnomonicButterflyProjection">https://w3id.org/geosrs/projection/GnomonicButterflyProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                                        |

### 11.14.7. Class: geosrs:GnomonicCubedSphereProjection

**Table 246** — geosrs:GnomonicCubedSphereProjection

|               |                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GnomonicCubedSphereProjection">https://w3id.org/geosrs/projection/GnomonicCubedSphereProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                                            |

### 11.14.8. Class: geosrs:GnomonicIcosahedronProjection

**Table 247** — geosrs:GnomonicIcosahedronProjection

|               |                                                                                                                                                 |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GnomonicIcosahedronProjection">https://w3id.org/geosrs/projection/GnomonicIcosahedronProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                                            |

### 11.14.9. Class: geosrs:GuyouProjection

Table 248 — geosrs:GuyouProjection

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GuyouProjection">https://w3id.org/geosrs/projection/GuyouProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                |

### 11.14.10. Class: geosrs:IcosahedralProjection

Table 249 — geosrs:IcosahedralProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/IcosahedralProjection">https://w3id.org/geosrs/projection/IcosahedralProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                            |

### 11.14.11. Class: geosrs:LeeProjection

Table 250 — geosrs:LeeProjection

|               |                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LeeProjection">https://w3id.org/geosrs/projection/LeeProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                            |

### 11.14.12. Class: geosrs:MyrahedralProjection

Table 251 — geosrs:MyrahedralProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MyrahedralProjection">https://w3id.org/geosrs/projection/MyrahedralProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                          |

### 11.14.13. Class: geosrs:OctantProjection

**Table 252** — geosrs:OctantProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/OctantProjection">https://w3id.org/geosrs/projection/OctantProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                  |

#### 11.14.14. Class: geosrs:PolyhedralProjection

**Table 253** — geosrs:PolyhedralProjection

|     |                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/PolyhedralProjection">https://w3id.org/geosrs/projection/PolyhedralProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------------|

#### 11.14.15. Class: geosrs:QuadrilateralizedSphericalCubeProjection

**Table 254** — geosrs:QuadrilateralizedSphericalCubeProjection

|               |                                                                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/QuadrilateralizedSphericalCubeProjection">https://w3id.org/geosrs/projection/QuadrilateralizedSphericalCubeProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                                                                  |

#### 11.14.16. Class: geosrs:WatermanButterflyProjection

**Table 255** — geosrs:WatermanButterflyProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WatermanButterflyProjection">https://w3id.org/geosrs/projection/WatermanButterflyProjection</a> |
| Super-classes | <a href="#">PolyhedralProjection</a>                                                                                                        |

### 11.15. Projection

---

#### REQUIREMENT 35: PROJECTION

|            |                             |
|------------|-----------------------------|
| IDENTIFIER | /req/projections/Projection |
|------------|-----------------------------|

## REQUIREMENT 35: PROJECTION

|           |                                                                                                                  |
|-----------|------------------------------------------------------------------------------------------------------------------|
| STATEMENT | Implementations shall allow the RDFS classes <code>geosrs:Projection</code> to be used in SPARQL graph patterns. |
|-----------|------------------------------------------------------------------------------------------------------------------|

### 11.15.1. Class: `geosrs:Projection`

Table 256 — `geosrs:Projection`

|               |                                                                                                           |
|---------------|-----------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Projection">https://w3id.org/geosrs/projection/Projection</a> |
| Super-classes | <a href="#">Conversion</a>                                                                                |

## 11.16. Pseudo Azimuthal Projections

## REQUIREMENT 36: PSEUDO AZIMUTHAL PROJECTIONS

|            |                                                                                                                                                                                                                                                                                                                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | <code>/req/projections/Pseudo_Azimuthal_Projections</code>                                                                                                                                                                                                                                                                                                                                      |
| STATEMENT  | Implementations shall allow the RDFS classes <code>geosrs:AitoffObliqueProjection</code> , <code>geosrs:AitoffProjection</code> , <code>geosrs:BartholomewProjection</code> , <code>geosrs:HammerProjection</code> , <code>geosrs:PseudoAzimuthalProjection</code> , <code>geosrs:Strebe1995Projection</code> , <code>geosrs:WinkelTripelProjection</code> to be used in SPARQL graph patterns. |

### 11.16.1. Class: `geosrs:AitoffObliqueProjection`

Table 257 — `geosrs:AitoffObliqueProjection`

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AitoffObliqueProjection">https://w3id.org/geosrs/projection/AitoffObliqueProjection</a> |
| Super-classes | <a href="#">PseudoAzimuthalProjection</a>                                                                                           |

### 11.16.2. Class: `geosrs:AitoffProjection`

**Table 258** — geosrs:AitoffProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AitoffProjection">https://w3id.org/geosrs/projection/AitoffProjection</a> |
| Definition    | A modified azimuthal projection whose graticule takes the form of an ellipse                                          |
| Super-classes | <a href="#">PseudoAzimuthalProjection</a>                                                                             |

### 11.16.3. Class: geosrs:BartholomewProjection

**Table 259** — geosrs:BartholomewProjection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BartholomewProjection">https://w3id.org/geosrs/projection/BartholomewProjection</a> |
| Super-classes | <a href="#">WinkelTripelProjection</a>                                                                                          |

### 11.16.4. Class: geosrs:HammerProjection

**Table 260** — geosrs:HammerProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HammerProjection">https://w3id.org/geosrs/projection/HammerProjection</a> |
| Super-classes | <a href="#">PseudoAzimuthalProjection</a>                                                                             |

### 11.16.5. Class: geosrs:PseudoAzimuthalProjection

**Table 261** — geosrs:PseudoAzimuthalProjection

|     |                                                                                                                                         |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/PseudoAzimuthalProjection">https://w3id.org/geosrs/projection/PseudoAzimuthalProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------|

### 11.16.6. Class: geosrs:Strebe1995Projection

**Table 262** — geosrs:Strebe1995Projection

|     |                                                                                                                               |
|-----|-------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/Strebe1995Projection">https://w3id.org/geosrs/projection/Strebe1995Projection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------------|

Super-classes

[PseudoAzimuthalProjection](#)

### 11.16.7. Class: geosrs:WinkelTripelProjection

Table 263 — geosrs:WinkelTripelProjection

|     |                                                                                                                                   |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/WinkelTripelProjection">https://w3id.org/geosrs/projection/WinkelTripelProjection</a> |
|-----|-----------------------------------------------------------------------------------------------------------------------------------|

|               |                                           |
|---------------|-------------------------------------------|
| Super-classes | <a href="#">PseudoAzimuthalProjection</a> |
|---------------|-------------------------------------------|

## 11.17. Pseudo Conical Projections

### REQUIREMENT 37: PSEUDO CONICAL PROJECTIONS

|            |                                             |
|------------|---------------------------------------------|
| IDENTIFIER | /req/projections/Pseudo_Conical_Projections |
|------------|---------------------------------------------|

|           |                                                                                                                                                                                                                                                                                                                                    |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| STATEMENT | Implementations shall allow the RDFS classes geosrs:AmericanPolyconicProjection, geosrs:BonneProjection, geosrs:BottomleyProjection, geosrs:NicolosiGlobularProjection, geosrs:PseudoConicalProjection, geosrs:PtolemyIIProjection, geosrs:StabiusWernerIIProjection, geosrs:WernerProjection to be used in SPARQL graph patterns. |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

### 11.17.1. Class: geosrs:AmericanPolyconicProjection

Table 264 — geosrs:AmericanPolyconicProjection

|     |                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/AmericanPolyconicProjection">https://w3id.org/geosrs/projection/AmericanPolyconicProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|

|               |                                         |
|---------------|-----------------------------------------|
| Super-classes | <a href="#">PseudoConicalProjection</a> |
|---------------|-----------------------------------------|

|         |                                                    |
|---------|----------------------------------------------------|
| Example | <a href="#">geosrs:AmericanPolyconicProjection</a> |
|---------|----------------------------------------------------|

### 11.17.2. Class: geosrs:BonneProjection

**Table 265** — geosrs:BonneProjection

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BonneProjection">https://w3id.org/geosrs/projection/BonneProjection</a> |
| Super-classes | <a href="#">PseudoConicalProjection</a>                                                                             |

### 11.17.3. Class: geosrs:BottomleyProjection

**Table 266** — geosrs:BottomleyProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BottomleyProjection">https://w3id.org/geosrs/projection/BottomleyProjection</a> |
| Super-classes | <a href="#">PseudoConicalProjection</a>                                                                                     |

### 11.17.4. Class: geosrs:NicolosiGlobularProjection

**Table 267** — geosrs:NicolosiGlobularProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/NicolosiGlobularProjection">https://w3id.org/geosrs/projection/NicolosiGlobularProjection</a> |
| Super-classes | <a href="#">PseudoConicalProjection</a>                                                                                                   |

### 11.17.5. Class: geosrs:PseudoConicalProjection

**Table 268** — geosrs:PseudoConicalProjection

|     |                                                                                                                                     |
|-----|-------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/PseudoConicalProjection">https://w3id.org/geosrs/projection/PseudoConicalProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------------------|

### 11.17.6. Class: geosrs:PtolemyIIProjection

**Table 269** — geosrs:PtolemyIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PtolemyIIProjection">https://w3id.org/geosrs/projection/PtolemyIIProjection</a> |
| Super-classes | <a href="#">PseudoConicalProjection</a>                                                                                     |

### 11.17.7. Class: geosrs:StabiusWernerIIProjection

Table 270 — geosrs:StabiusWernerIIProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/StabiusWernerIIProjection">https://w3id.org/geosrs/projection/StabiusWernerIIProjection</a> |
| Super-classes | <a href="#">BonneProjection</a>                                                                                                         |

### 11.17.8. Class: geosrs:WernerProjection

Table 271 — geosrs:WernerProjection

|               |                                                                                                                       |
|---------------|-----------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WernerProjection">https://w3id.org/geosrs/projection/WernerProjection</a> |
| Super-classes | <a href="#">PseudoConicalProjection</a>                                                                               |

## 11.18. Pseudo Cylindrical Projections

### REQUIREMENT 38: PSEUDO CYLINDRICAL PROJECTIONS

**IDENTIFIER** /req/projections/Pseudo\_Cylindrical\_Projections

**STATEMENT**

Implementations shall allow the RDFS classes geosrs:ApianIIProjection, geosrs:AtlantisProjection, geosrs:BaranyIIIProjection, geosrs:BaranyIIProjection, geosrs:BaranyIProjection, geosrs:BaranyiIVProjection, geosrs:BoggsEumorphicProjection, geosrs:BromleyProjection, geosrs:CabotProjection, geosrs:CollignonProjection, geosrs:CrasterParabolicProjection, geosrs:DeakinMinimumErrorProjection, geosrs:Eckert1Projection, geosrs:Eckert2Projection, geosrs:Eckert3Projection, geosrs:Eckert4Projection, geosrs:Eckert5Projection, geosrs:Eckert6Projection, geosrs:EqualEarthProjection, geosrs:FaheyProjection, geosrs:FoucautProjection, geosrs:FoucautSinusoidalProjection, geosrs:FournierIIProjection, geosrs:GinzburgVIIIProjection, geosrs:GoodeHomolosineProjection, geosrs:HEALPixProjection, geosrs:HatanoAsymmetricalEqualAreaProjection, geosrs:HufnagelProjection, geosrs:Kavrayskiy7Projection, geosrs:LoximuthalProjection, geosrs:MayrProjection, geosrs:McBrydeThomasFlatPolarParabolicProjection, geosrs:McBrydeThomasFlatPolarQuarticProjection, geosrs:McBrydeThomasFlatPolarSinusoidalProjection, geosrs:McBrydeThomasIIProjection, geosrs:McBrydeThomasIProjection, geosrs:NaturalEarth2Projection, geosrs:NaturalEarthProjection, geosrs:NellHammerProjection, geosrs:NellProjection, geosrs:OrteliusOvalProjection, geosrs:PseudoCylindricalProjection, geosrs:PutninsP1Projection, geosrs:PutninsP2Projection, geosrs:PutninsP3Projection, geosrs:PutninsP5Projection, geosrs:PutninsP6Projection, geosrs:QuarticAuthalicProjection,



## REQUIREMENT 38: PSEUDO CYLINDRICAL PROJECTIONS

geosrs:RobinsonProjection, geosrs:SinusoidalProjection, geosrs:TheTimesProjection, geosrs:ToblerG1Projection, geosrs:ToblerHyperellipticalProjection, geosrs:WagnerIIIProjection, geosrs:WagnerIIProjection, geosrs:WagnerIProjection, geosrs:WagnerIVProjection, geosrs:WagnerVProjection, geosrs:WagnerVProjection, geosrs:WerenskioldIProjection, geosrs:PutninsP3'Projection, geosrs:PutninsP4'Projection, geosrs:PutninsP5'Projection, geosrs:PutninsP6'Projection to be used in SPARQL graph patterns.

### 11.18.1. Class: geosrs:ApianIIProjection

Table 272 — geosrs:ApianIIProjection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ApianIIProjection">https://w3id.org/geosrs/projection/ApianIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.2. Class: geosrs:AtlantisProjection

Table 273 — geosrs:AtlantisProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/AtlantisProjection">https://w3id.org/geosrs/projection/AtlantisProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

### 11.18.3. Class: geosrs:BaranyIIIProjection

Table 274 — geosrs:BaranyIIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BaranyIIIProjection">https://w3id.org/geosrs/projection/BaranyIIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

### 11.18.4. Class: geosrs:BaranyIIProjection

Table 275 — geosrs:BaranyIIProjection

|     |                                                                                                                           |
|-----|---------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/BaranyIIProjection">https://w3id.org/geosrs/projection/BaranyIIProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------|

|               |                                                    |
|---------------|----------------------------------------------------|
| Super-classes | <a href="#"><u>PseudoCylindricalProjection</u></a> |
|---------------|----------------------------------------------------|

### 11.18.5. Class: geosrs:BaranyilProjection

**Table 276** — geosrs:BaranyilProjection

|               |                                                                                                                                  |
|---------------|----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BaranyilProjection"><u>https://w3id.org/geosrs/projection/BaranyilProjection</u></a> |
| Super-classes | <a href="#"><u>PseudoCylindricalProjection</u></a>                                                                               |

### 11.18.6. Class: geosrs:BaranyilVProjection

**Table 277** — geosrs:BaranyilVProjection

|               |                                                                                                                                    |
|---------------|------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BaranyilVProjection"><u>https://w3id.org/geosrs/projection/BaranyilVProjection</u></a> |
| Super-classes | <a href="#"><u>PseudoCylindricalProjection</u></a>                                                                                 |

### 11.18.7. Class: geosrs:BoggsEumorphicProjection

**Table 278** — geosrs:BoggsEumorphicProjection

|               |                                                                                                                                              |
|---------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BoggsEumorphicProjection"><u>https://w3id.org/geosrs/projection/BoggsEumorphicProjection</u></a> |
| Super-classes | <a href="#"><u>PseudoCylindricalProjection</u></a>                                                                                           |

### 11.18.8. Class: geosrs:BromleyProjection

**Table 279** — geosrs:BromleyProjection

|               |                                                                                                                                |
|---------------|--------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/BromleyProjection"><u>https://w3id.org/geosrs/projection/BromleyProjection</u></a> |
| Super-classes | <a href="#"><u>PseudoCylindricalProjection</u></a>                                                                             |

### 11.18.9. Class: geosrs: CabotProjection

Table 280 — geosrs: CabotProjection

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CabotProjection">https://w3id.org/geosrs/projection/CabotProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                         |

### 11.18.10. Class: geosrs: CollignonProjection

Table 281 — geosrs: CollignonProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CollignonProjection">https://w3id.org/geosrs/projection/CollignonProjection</a> |
| Definition    | An equal-area pseudocylindrical projection that maps the sphere onto a triangle or diamond                                  |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

### 11.18.11. Class: geosrs: CrasterParabolicProjection

Table 282 — geosrs: CrasterParabolicProjection

|               |                                                                                                                                           |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/CrasterParabolicProjection">https://w3id.org/geosrs/projection/CrasterParabolicProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                               |

### 11.18.12. Class: geosrs: DeakinMinimumErrorProjection

Table 283 — geosrs: DeakinMinimumErrorProjection

|               |                                                                                                                                               |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/DeakinMinimumErrorProjection">https://w3id.org/geosrs/projection/DeakinMinimumErrorProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                                   |

### 11.18.13. Class: geosrs:Eckert1Projection

Table 284 — geosrs:Eckert1Projection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Eckert1Projection">https://w3id.org/geosrs/projection/Eckert1Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.14. Class: geosrs:Eckert2Projection

Table 285 — geosrs:Eckert2Projection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Eckert2Projection">https://w3id.org/geosrs/projection/Eckert2Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.15. Class: geosrs:Eckert3Projection

Table 286 — geosrs:Eckert3Projection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Eckert3Projection">https://w3id.org/geosrs/projection/Eckert3Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.16. Class: geosrs:Eckert4Projection

Table 287 — geosrs:Eckert4Projection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Eckert4Projection">https://w3id.org/geosrs/projection/Eckert4Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.17. Class: geosrs:Eckert5Projection

**Table 288** — geosrs:Eckert5Projection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Eckert5Projection">https://w3id.org/geosrs/projection/Eckert5Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.18. Class: geosrs:Eckert6Projection

**Table 289** — geosrs:Eckert6Projection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Eckert6Projection">https://w3id.org/geosrs/projection/Eckert6Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.19. Class: geosrs:EqualEarthProjection

**Table 290** — geosrs:EqualEarthProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/EqualEarthProjection">https://w3id.org/geosrs/projection/EqualEarthProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |
| Example       | <a href="#">geosrs:EqualEarthProjection</a>                                                                                   |

### 11.18.20. Class: geosrs:FaheyProjection

**Table 291** — geosrs:FaheyProjection

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FaheyProjection">https://w3id.org/geosrs/projection/FaheyProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                         |

### 11.18.21. Class: geosrs:FoucautProjection

**Table 292** — geosrs:FoucautProjection

|     |                                                                                                                         |
|-----|-------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/FoucautProjection">https://w3id.org/geosrs/projection/FoucautProjection</a> |
|-----|-------------------------------------------------------------------------------------------------------------------------|

|               |                                             |
|---------------|---------------------------------------------|
| Super-classes | <a href="#">PseudoCylindricalProjection</a> |
|---------------|---------------------------------------------|

### 11.18.22. Class: geosrs:FoucautSinusoidalProjection

**Table 293** — geosrs:FoucautSinusoidalProjection

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FoucautSinusoidalProjection">https://w3id.org/geosrs/projection/FoucautSinusoidalProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                                 |

### 11.18.23. Class: geosrs:FournierIIProjection

**Table 294** — geosrs:FournierIIProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/FournierIIProjection">https://w3id.org/geosrs/projection/FournierIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

### 11.18.24. Class: geosrs:GinzburgVIIIProjection

**Table 295** — geosrs:GinzburgVIIIProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GinzburgVIIIProjection">https://w3id.org/geosrs/projection/GinzburgVIIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                       |

### 11.18.25. Class: geosrs:GoodeHomolosineProjection

**Table 296** — geosrs:GoodeHomolosineProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GoodeHomolosineProjection">https://w3id.org/geosrs/projection/GoodeHomolosineProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                             |

## 11.18.26. Class: geosrs:HEALPixProjection

Table 297 — geosrs:HEALPixProjection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HEALPixProjection">https://w3id.org/geosrs/projection/HEALPixProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

## 11.18.27. Class: geosrs:HatanoAsymmetricalEqualAreaProjection

Table 298 — geosrs:HatanoAsymmetricalEqualAreaProjection

|               |                                                                                                                                                                 |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HatanoAsymmetricalEqualAreaProjection">https://w3id.org/geosrs/projection/HatanoAsymmetricalEqualAreaProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                                                     |

## 11.18.28. Class: geosrs:HufnagelProjection

Table 299 — geosrs:HufnagelProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/HufnagelProjection">https://w3id.org/geosrs/projection/HufnagelProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

## 11.18.29. Class: geosrs:Kavrayskiy7Projection

Table 300 — geosrs:Kavrayskiy7Projection

|               |                                                                                                                                 |
|---------------|---------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/Kavrayskiy7Projection">https://w3id.org/geosrs/projection/Kavrayskiy7Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                     |

## 11.18.30. Class: geosrs:LoximuthalProjection

**Table 301** — geosrs:LoximuthalProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/LoximuthalProjection">https://w3id.org/geosrs/projection/LoximuthalProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

### 11.18.31. Class: geosrs:MayrProjection

**Table 302** — geosrs:MayrProjection

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MayrProjection">https://w3id.org/geosrs/projection/MayrProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                       |

### 11.18.32. Class: geosrs:McBrydeThomasFlatPolarParabolicProjection

**Table 303** — geosrs:McBrydeThomasFlatPolarParabolicProjection

|               |                                                                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/McBrydeThomasFlatPolarParabolicProjection">https://w3id.org/geosrs/projection/McBrydeThomasFlatPolarParabolicProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                                                             |

### 11.18.33. Class: geosrs:McBrydeThomasFlatPolarQuarticProjection

**Table 304** — geosrs:McBrydeThomasFlatPolarQuarticProjection

|               |                                                                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/McBrydeThomasFlatPolarQuarticProjection">https://w3id.org/geosrs/projection/McBrydeThomasFlatPolarQuarticProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                                                         |

### 11.18.34. Class: geosrs:McBrydeThomasFlatPolarSinusoidalProjection

**Table 305** — geosrs:McBrydeThomasFlatPolarSinusoidalProjection

|     |                                                                                                                                                                           |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/McBrydeThomasFlatPolarSinusoidalProjection">https://w3id.org/geosrs/projection/McBrydeThomasFlatPolarSinusoidalProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|



|               |                                             |
|---------------|---------------------------------------------|
| Super-classes | <a href="#">PseudoCylindricalProjection</a> |
|---------------|---------------------------------------------|

### 11.18.35. Class: geosrs:McBrydeThomasIIProjection

**Table 306** — geosrs:McBrydeThomasIIProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/McBrydeThomasIIProjection">https://w3id.org/geosrs/projection/McBrydeThomasIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                             |

### 11.18.36. Class: geosrs:McBrydeThomasIProjection

**Table 307** — geosrs:McBrydeThomasIProjection

|               |                                                                                                                                       |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/McBrydeThomasIProjection">https://w3id.org/geosrs/projection/McBrydeThomasIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                           |

### 11.18.37. Class: geosrs:NaturalEarth2Projection

**Table 308** — geosrs:NaturalEarth2Projection

|               |                                                                                                                                     |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/NaturalEarth2Projection">https://w3id.org/geosrs/projection/NaturalEarth2Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                         |

### 11.18.38. Class: geosrs:NaturalEarthProjection

**Table 309** — geosrs:NaturalEarthProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/NaturalEarthProjection">https://w3id.org/geosrs/projection/NaturalEarthProjection</a> |
| Definition    | A pseudocylindrical map projection designed by Tom Patterson and introduced in 2008                                               |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                       |

### 11.18.39. Class: geosrs:NellHammerProjection

Table 310 — geosrs:NellHammerProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/NellHammerProjection">https://w3id.org/geosrs/projection/NellHammerProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

### 11.18.40. Class: geosrs:NellProjection

Table 311 — geosrs:NellProjection

|               |                                                                                                                   |
|---------------|-------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/NellProjection">https://w3id.org/geosrs/projection/NellProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                       |

### 11.18.41. Class: geosrs:OrteliusOvalProjection

Table 312 — geosrs:OrteliusOvalProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/OrteliusOvalProjection">https://w3id.org/geosrs/projection/OrteliusOvalProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                       |

### 11.18.42. Class: geosrs:PseudoCylindricalProjection

Table 313 — geosrs:PseudoCylindricalProjection

|     |                                                                                                                                             |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/projection/PseudoCylindricalProjection">https://w3id.org/geosrs/projection/PseudoCylindricalProjection</a> |
|-----|---------------------------------------------------------------------------------------------------------------------------------------------|

### 11.18.43. Class: geosrs:PutninsP1Projection

**Table 314** — geosrs:PutninsP1Projection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP1Projection">https://w3id.org/geosrs/projection/PutninsP1Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

#### 11.18.44. Class: geosrs:PutninsP2Projection

**Table 315** — geosrs:PutninsP2Projection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP2Projection">https://w3id.org/geosrs/projection/PutninsP2Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

#### 11.18.45. Class: geosrs:PutninsP3Projection

**Table 316** — geosrs:PutninsP3Projection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP3Projection">https://w3id.org/geosrs/projection/PutninsP3Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

#### 11.18.46. Class: geosrs:PutninsP5Projection

**Table 317** — geosrs:PutninsP5Projection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP5Projection">https://w3id.org/geosrs/projection/PutninsP5Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

#### 11.18.47. Class: geosrs:PutninsP6Projection

**Table 318** — geosrs:PutninsP6Projection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP6Projection">https://w3id.org/geosrs/projection/PutninsP6Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

## 11.18.48. Class: geosrs:QuarticAuthalicProjection

Table 319 — geosrs:QuarticAuthalicProjection

|               |                                                                                                                                         |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/QuarticAuthalicProjection">https://w3id.org/geosrs/projection/QuarticAuthalicProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                             |

## 11.18.49. Class: geosrs:RobinsonProjection

Table 320 — geosrs:RobinsonProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/RobinsonProjection">https://w3id.org/geosrs/projection/RobinsonProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

## 11.18.50. Class: geosrs:SinusoidalProjection

Table 321 — geosrs:SinusoidalProjection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/SinusoidalProjection">https://w3id.org/geosrs/projection/SinusoidalProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

## 11.18.51. Class: geosrs:TheTimesProjection

Table 322 — geosrs:TheTimesProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/TheTimesProjection">https://w3id.org/geosrs/projection/TheTimesProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

## 11.18.52. Class: geosrs:ToblerG1Projection

**Table 323** — geosrs:ToblerG1Projection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ToblerG1Projection">https://w3id.org/geosrs/projection/ToblerG1Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

### 11.18.53. Class: geosrs:ToblerHyperellipticalProjection

**Table 324** — geosrs:ToblerHyperellipticalProjection

|               |                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/ToblerHyperellipticalProjection">https://w3id.org/geosrs/projection/ToblerHyperellipticalProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                                         |

### 11.18.54. Class: geosrs:WagnerIIIProjection

**Table 325** — geosrs:WagnerIIIProjection

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerIIIProjection">https://w3id.org/geosrs/projection/WagnerIIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                 |

### 11.18.55. Class: geosrs:WagnerIIProjection

**Table 326** — geosrs:WagnerIIProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerIIProjection">https://w3id.org/geosrs/projection/WagnerIIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

### 11.18.56. Class: geosrs:WagnerIProjection

**Table 327** — geosrs:WagnerIProjection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerIProjection">https://w3id.org/geosrs/projection/WagnerIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.57. Class: geosrs:WagnerIVProjection

Table 328 — geosrs:WagnerIVProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerIVProjection">https://w3id.org/geosrs/projection/WagnerIVProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

### 11.18.58. Class: geosrs:WagnerVIProjection

Table 329 — geosrs:WagnerVIProjection

|               |                                                                                                                           |
|---------------|---------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerVIProjection">https://w3id.org/geosrs/projection/WagnerVIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                               |

### 11.18.59. Class: geosrs:WagnerVProjection

Table 330 — geosrs:WagnerVProjection

|               |                                                                                                                         |
|---------------|-------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WagnerVProjection">https://w3id.org/geosrs/projection/WagnerVProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                             |

### 11.18.60. Class: geosrs:WerenskioldIProjection

Table 331 — geosrs:WerenskioldIProjection

|               |                                                                                                                                   |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/WerenskioldIProjection">https://w3id.org/geosrs/projection/WerenskioldIProjection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                       |

### 11.18.61. Class: geosrs:PutninsP3'Projection

**Table 332** — geosrs:PutninsP3'Projection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP3'Projection">https://w3id.org/geosrs/projection/PutninsP3'Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

### 11.18.62. Class: geosrs:PutninsP4'Projection

**Table 333** — geosrs:PutninsP4'Projection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP4'Projection">https://w3id.org/geosrs/projection/PutninsP4'Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

### 11.18.63. Class: geosrs:PutninsP5'Projection

**Table 334** — geosrs:PutninsP5'Projection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP5'Projection">https://w3id.org/geosrs/projection/PutninsP5'Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

### 11.18.64. Class: geosrs:PutninsP6'Projection

**Table 335** — geosrs:PutninsP6'Projection

|               |                                                                                                                               |
|---------------|-------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/PutninsP6'Projection">https://w3id.org/geosrs/projection/PutninsP6'Projection</a> |
| Super-classes | <a href="#">PseudoCylindricalProjection</a>                                                                                   |

## 11.19. Stereographic Projections

---

## REQUIREMENT 39: STEREOGRAPHIC PROJECTIONS

|            |                                                                                                                                                                                                                                        |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/projections/Stereographic_Projections                                                                                                                                                                                             |
| STATEMENT  | Implementations shall allow the RDFS classes <code>geosrs:GallStereographicProjection</code> , <code>geosrs:MillerOblatedStereographicProjection</code> , <code>geosrs:RoussilheProjection</code> to be used in SPARQL graph patterns. |

### 11.19.1. Class: `geosrs:GallStereographicProjection`

Table 336 — `geosrs:GallStereographicProjection`

|               |                                                                                                                                             |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/GallStereographicProjection">https://w3id.org/geosrs/projection/GallStereographicProjection</a> |
| Super-classes | <a href="#">StereographicProjection</a>                                                                                                     |

### 11.19.2. Class: `geosrs:MillerOblatedStereographicProjection`

Table 337 — `geosrs:MillerOblatedStereographicProjection`

|               |                                                                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/MillerOblatedStereographicProjection">https://w3id.org/geosrs/projection/MillerOblatedStereographicProjection</a> |
| Super-classes | <a href="#">StereographicProjection</a>                                                                                                                       |

### 11.19.3. Class: `geosrs:RoussilheProjection`

Table 338 — `geosrs:RoussilheProjection`

|               |                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/projection/RoussilheProjection">https://w3id.org/geosrs/projection/RoussilheProjection</a> |
| Super-classes | <a href="#">StereographicProjection</a>                                                                                     |





12

# PLANET MODULE

---

This clause establishes the **PLANET** Requirements class, with IRI /req/planet, which has a corresponding Conformance Class, **PLANET**, with IRI /conf/planet.

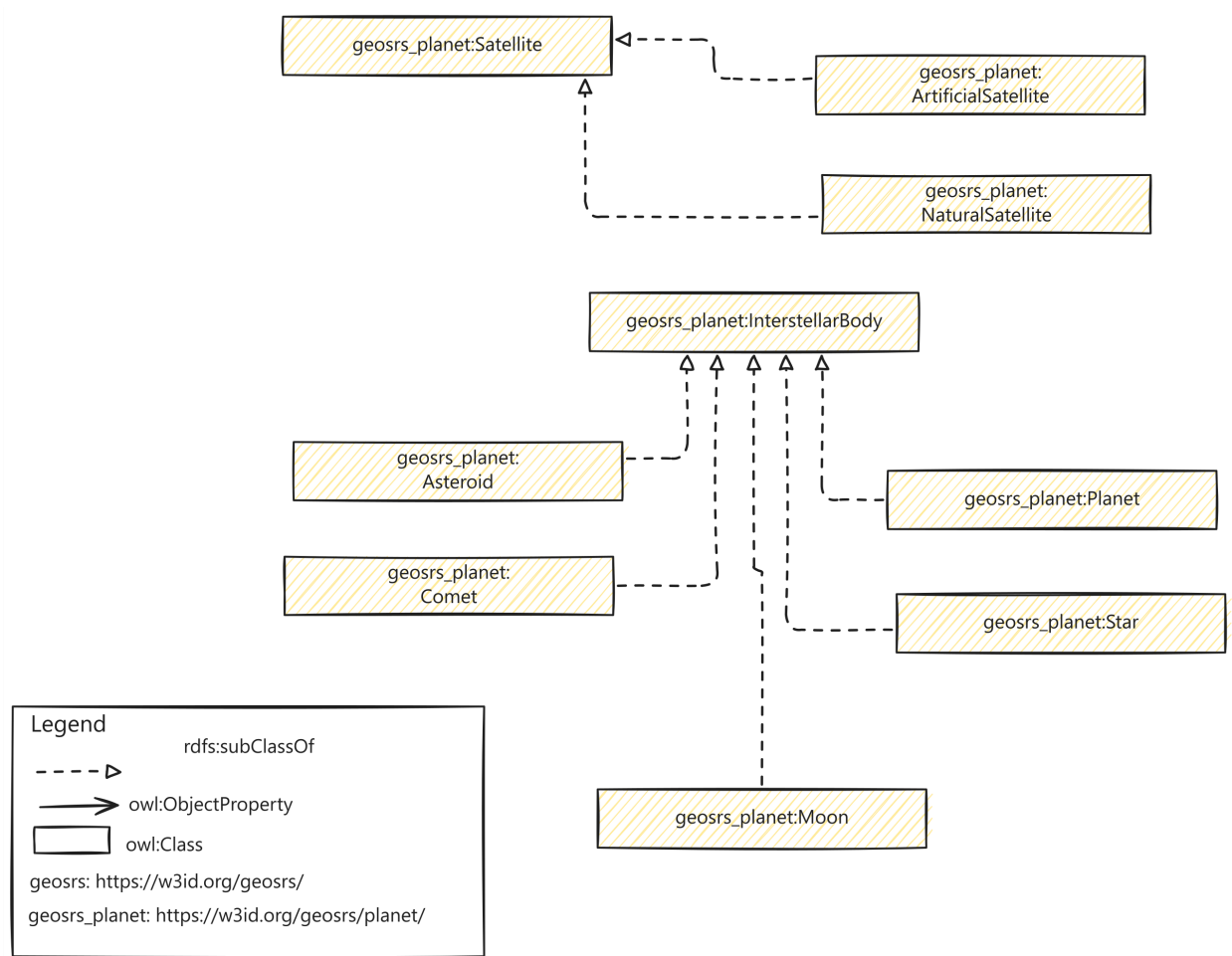


Figure 7

| REQUIREMENTS CLASS 7: 12-PLANET_MODULE.ADOC EXTENSION |                                     |
|-------------------------------------------------------|-------------------------------------|
| IDENTIFIER                                            | /req/planet                         |
| TARGET TYPE                                           | Implementation Specification        |
| CONFORMANCE CLASS                                     | Conformance class A.7: /conf/planet |
| REQUIREMENT                                           | /req/planet/Interstellar_Body       |

# 12.1. Interstellar Body

## REQUIREMENT 40: INTERSTELLAR BODY

|            |                                                                                                                                                                                                                                                                                                                   |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | /req/planet/Interstellar_Body                                                                                                                                                                                                                                                                                     |
| STATEMENT  | Implementations shall allow the RDFS classes geosrs:ArtificialSatellite, geosrs:Asteroid, geosrs:Comet, geosrs:DwarfPlanet, geosrs:InterstellarBody, geosrs:Moon, geosrs:NaturalSatellite, geosrs:Planet, geosrs:PlanetStatus, geosrs:Plutoid, geosrs:Star, geosrs:Satellite to be used in SPARQL graph patterns. |

### 12.1.1. Class: geosrs:ArtificialSatellite

Table 339 — geosrs:ArtificialSatellite

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/ArtificialSatellite">https://w3id.org/geosrs/planet/ArtificialSatellite</a> |
| Super-classes | <a href="#">Satellite</a>                                                                                           |

### 12.1.2. Class: geosrs:Asteroid

Table 340 — geosrs:Asteroid

|               |                                                                                                                                                                                                                                                                                                                             |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/Asteroid">https://w3id.org/geosrs/planet/Asteroid</a>                                                                                                                                                                                                                               |
| Definition    | Asteroid, any of a host of small bodies, about 1000 km (600 miles) or less in diameter, that orbit the Sun primarily between the orbits of Mars and Jupiter in a nearly flat ring called the asteroid belt (source: <a href="https://www.britannica.com/science/asteroid">https://www.britannica.com/science/asteroid</a> ) |
| Super-classes | <a href="#">InterstellarBody</a>                                                                                                                                                                                                                                                                                            |

### 12.1.3. Class: geosrs:Comet

**Table 341** — geosrs:Comet

|               |                                                                                         |
|---------------|-----------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/Comet">https://w3id.org/geosrs/planet/Comet</a> |
| Super-classes | <a href="#">InterstellarBody</a>                                                        |

#### 12.1.4. Class: geosrs:DwarfPlanet

**Table 342** — geosrs:DwarfPlanet

|     |                                                                                                     |
|-----|-----------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/planet/DwarfPlanet">https://w3id.org/geosrs/planet/DwarfPlanet</a> |
|-----|-----------------------------------------------------------------------------------------------------|

#### 12.1.5. Class: geosrs:InterstellarBody

**Table 343** — geosrs:InterstellarBody

|     |                                                                                                               |
|-----|---------------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/planet/InterstellarBody">https://w3id.org/geosrs/planet/InterstellarBody</a> |
|-----|---------------------------------------------------------------------------------------------------------------|

#### 12.1.6. Class: geosrs:Moon

**Table 344** — geosrs:Moon

|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/Moon">https://w3id.org/geosrs/planet/Moon</a> |
| Super-classes | <a href="#">InterstellarBody</a>                                                      |

#### 12.1.7. Class: geosrs:NaturalSatellite

**Table 345** — geosrs:NaturalSatellite

|               |                                                                                                               |
|---------------|---------------------------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/NaturalSatellite">https://w3id.org/geosrs/planet/NaturalSatellite</a> |
| Super-classes | <a href="#">Satellite</a>                                                                                     |

### 12.1.8. Class: geosrs:Planet

Table 346 — geosrs:Planet

|               |                                                                                           |
|---------------|-------------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/Planet">https://w3id.org/geosrs/planet/Planet</a> |
| Super-classes | <a href="#">InterstellarBody</a>                                                          |

### 12.1.9. Class: geosrs:PlanetStatus

Table 347 — geosrs:PlanetStatus

|     |                                                                                                       |
|-----|-------------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/planet/PlanetStatus">https://w3id.org/geosrs/planet/PlanetStatus</a> |
|-----|-------------------------------------------------------------------------------------------------------|

### 12.1.10. Class: geosrs:Plutoid

Table 348 — geosrs:Plutoid

|     |                                                                                             |
|-----|---------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/planet/Plutoid">https://w3id.org/geosrs/planet/Plutoid</a> |
|-----|---------------------------------------------------------------------------------------------|

### 12.1.11. Class: geosrs:Star

Table 349 — geosrs:Star

|               |                                                                                       |
|---------------|---------------------------------------------------------------------------------------|
| URI           | <a href="https://w3id.org/geosrs/planet/Star">https://w3id.org/geosrs/planet/Star</a> |
| Super-classes | <a href="#">InterstellarBody</a>                                                      |

### 12.1.12. Class: geosrs:Satellite

Table 350 — geosrs:Satellite

|     |                                                                                                 |
|-----|-------------------------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/planet/Satellite">https://w3id.org/geosrs/planet/Satellite</a> |
|-----|-------------------------------------------------------------------------------------------------|



13

# COMMON INSTANCES

---

This clause establishes common instances which are needed in CRS specifications as Requirement class **INSTANCES**, with IRI `/req/instances`, which has a corresponding Conformance Class, **INSTANCES**, with IRI `/conf/instances`.

#### REQUIREMENTS CLASS 8: 13-INSTANCES.ADOC EXTENSION

|                   |                                                     |
|-------------------|-----------------------------------------------------|
| IDENTIFIER        | <code>/req/instances</code>                         |
| TARGET TYPE       | Implementation Specification                        |
| CONFORMANCE CLASS | Conformance class A.8: <code>/conf/instances</code> |
| REQUIREMENT       | <code>/req/instances/Coordinate_System_Axis</code>  |
|                   | <code>/req/instances/Spheroids</code>               |
|                   | <code>/req/instances/SRS_Literal_Types</code>       |

### 13.1. Coordinate System Axis

#### REQUIREMENT 41: COORDINATE SYSTEM AXIS

|            |                                                                                                                                                                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER | <code>/req/instances/Coordinate_System_Axis</code>                                                                                                                                                                                                  |
| STATEMENT  | Implementations shall allow the RDFS instances <code>geosrs:Down</code> , <code>geosrs:East</code> , <code>geosrs:North</code> , <code>geosrs:South</code> , <code>geosrs:Up</code> , <code>geosrs:West</code> to be used in SPARQL graph patterns. |

#### 13.1.1. Instance: `geosrs:Down`

Table 351 — `geosrs:Down`

|      |                                                                         |
|------|-------------------------------------------------------------------------|
| URI  | <a href="https://w3id.org/geosrs/Down">https://w3id.org/geosrs/Down</a> |
| Type | <code>geosrs:AxisDirection</code>                                       |

|            |                          |
|------------|--------------------------|
| Definition | Downwards axis direction |
|------------|--------------------------|

### 13.1.2. Instance: geosrs:East

**Table 352** — geosrs:East

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/East">https://w3id.org/geosrs/East</a> |
| Type       | <a href="#">geosrs:AxisDirection</a>                                    |
| Definition | east axis direction                                                     |

### 13.1.3. Instance: geosrs:North

**Table 353** — geosrs:North

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/North">https://w3id.org/geosrs/North</a> |
| Type       | <a href="#">geosrs:AxisDirection</a>                                      |
| Definition | North axis direction                                                      |

### 13.1.4. Instance: geosrs:South

**Table 354** — geosrs:South

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/South">https://w3id.org/geosrs/South</a> |
| Type       | <a href="#">geosrs:AxisDirection</a>                                      |
| Definition | South axis direction                                                      |

### 13.1.5. Instance: geosrs:Up



**Table 355** — geosrs:Up

|            |                                                                     |
|------------|---------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Up">https://w3id.org/geosrs/Up</a> |
| Type       | <a href="#">geosrs:AxisDirection</a>                                |
| Definition | Up axis direction                                                   |

### 13.1.6. Instance: geosrs:West

**Table 356** — geosrs:West

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/West">https://w3id.org/geosrs/West</a> |
| Type       | <a href="#">geosrs:AxisDirection</a>                                    |
| Definition | West axis direction                                                     |

## 13.2. SRS Literal Types

### REQUIREMENT 42: SRS LITERAL TYPES

**IDENTIFIER**      /req/instances/SRS\_Literal\_Types

**STATEMENT**      Implementations shall allow the RDFS instances geosrs:proj4Literal, geosrs:projJSONLiteral, geosrs:wktLiteral to be used in SPARQL graph patterns.

### 13.2.1. Instance: geosrs:proj4Literal

**Table 357** — geosrs:proj4Literal

|            |                                                                                         |
|------------|-----------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/proj4Literal">https://w3id.org/geosrs/proj4Literal</a> |
| Type       | <a href="#">rdf:Datatype[rdf:Datatype]</a>                                              |
| Definition | A literal which stores a proj4 String                                                   |

|         |                                     |
|---------|-------------------------------------|
| Example | <a href="#">geosrs:proj4Literal</a> |
|---------|-------------------------------------|

### 13.2.2. Instance: geosrs:projJSONLiteral

**Table 358** — geosrs:projJSONLiteral

|            |                                                                                               |
|------------|-----------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/projJSONLiteral">https://w3id.org/geosrs/projJSONLiteral</a> |
| Type       | rdf:Datatype[rdf:Datatype]                                                                    |
| Definition | A literal which stores a projection JSON (ProjJSON) String                                    |
| Example    | <a href="#">geosrs:projJSONLiteral</a>                                                        |

### 13.2.3. Instance: geosrs:wktLiteral

**Table 359** — geosrs:wktLiteral

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/wktLiteral">https://w3id.org/geosrs/wktLiteral</a> |
| Type       | rdf:Datatype[rdf:Datatype]                                                          |
| Definition | A literal which stores a WKT for CRS String                                         |
| Example    | <a href="#">geosrs:wktLiteral</a>                                                   |

## 13.3. Spheroids

| REQUIREMENT 43: SPHEROIDS |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                | /req/instances/Spheroids                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| STATEMENT                 | Implementations shall allow the RDFS instances geosrs:GRS1980, geosrs:GRS67, geosrs:PZ90, geosrs:Airy1830, geosrs:AiryModified1849, geosrs:International1924, geosrs:AustralianNationalSpheroid, geosrs:Everest1930, geosrs:Clarke1866, geosrs:Plessis1817, geosrs:Danish1876, geosrs:Struve1860, geosrs:IAG1975, geosrs:Clarke1858, geosrs:Clarke1880, geosrs:Helmert1906, geosrs:CGCS2000, geosrs:GSK-2011, geosrs:Zach1812, geosrs:Clarke1880ARC, geosrs:Clarke1880IGN, |

## REQUIREMENT 43: SPHEROIDS

geosrs:WGS66, geosrs:WGS72, geosrs:WGS84, geosrs:Krassowsky1940 to be used in SPARQL graph patterns.

### 13.3.1. Instance: geosrs:GRS1980

Table 360 — geosrs:GRS1980

|            |                                                                               |
|------------|-------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/GRS1980">https://w3id.org/geosrs/GRS1980</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                              |
| Definition | GRS 1980 Ellipsoid                                                            |
| Example    | <a href="#">geosrs:GRS1980</a>                                                |

### 13.3.2. Instance: geosrs:GRS67

Table 361 — geosrs:GRS67

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/GRS67">https://w3id.org/geosrs/GRS67</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                          |
| Definition | GRS 67 Ellipsoid                                                          |
| Example    | <a href="#">geosrs:GRS67</a>                                              |

### 13.3.3. Instance: geosrs:PZ90

Table 362 — geosrs:PZ90

|            |                                                                         |
|------------|-------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/PZ90">https://w3id.org/geosrs/PZ90</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                        |
| Definition | PZ 90 Ellipsoid                                                         |

|         |                                                        |
|---------|--------------------------------------------------------|
| Example | <a href="https://w3id.org/geosrs/PZ90">geosrs:PZ90</a> |
|---------|--------------------------------------------------------|

### 13.3.4. Instance: geosrs:Airy1830

**Table 363** — geosrs:Airy1830

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Airy1830">https://w3id.org/geosrs/Airy1830</a> |
| Type       | <a href="https://w3id.org/geosrs/Ellipsoid">geosrs:Ellipsoid</a>                |
| Definition | Airy 1830 Ellipsoid                                                             |
| Example    | <a href="https://w3id.org/geosrs/Airy1830">geosrs:Airy1830</a>                  |

### 13.3.5. Instance: geosrs:AiryModified1849

**Table 364** — geosrs:AiryModified1849

|            |                                                                                                 |
|------------|-------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/AiryModified1849">https://w3id.org/geosrs/AiryModified1849</a> |
| Type       | <a href="https://w3id.org/geosrs/Ellipsoid">geosrs:Ellipsoid</a>                                |
| Definition | Airy 1849 Modified Ellipsoid                                                                    |
| Example    | <a href="https://w3id.org/geosrs/AiryModified1849">geosrs:AiryModified1849</a>                  |

### 13.3.6. Instance: geosrs:International1924

**Table 365** — geosrs:International1924

|            |                                                                                                   |
|------------|---------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/International1924">https://w3id.org/geosrs/International1924</a> |
| Type       | <a href="https://w3id.org/geosrs/Ellipsoid">geosrs:Ellipsoid</a>                                  |
| Definition | International 1924 Ellipsoid                                                                      |
| Example    | <a href="https://w3id.org/geosrs/International1924">geosrs:International1924</a>                  |

### 13.3.7. Instance: geosrs:AustralianNationalSpheroid

**Table 366** — geosrs:AustralianNationalSpheroid

|            |                                                                                                                     |
|------------|---------------------------------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/AustralianNationalSpheroid">https://w3id.org/geosrs/AustralianNationalSpheroid</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                                                    |
| Definition | Australian National Spheroid                                                                                        |
| Example    | <a href="#">geosrs:AustralianNationalSpheroid</a>                                                                   |

### 13.3.8. Instance: geosrs:Everest1930

**Table 367** — geosrs:Everest1930

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Everest1930">https://w3id.org/geosrs/Everest1930</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                      |
| Definition | Everest 1930 Spheroid                                                                 |

### 13.3.9. Instance: geosrs:Clarke1866

**Table 368** — geosrs:Clarke1866

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Clarke1866">https://w3id.org/geosrs/Clarke1866</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                    |
| Definition | Clarke 1866 Spheroid                                                                |
| Example    | <a href="#">geosrs:Clarke1866</a>                                                   |

### 13.3.10. Instance: geosrs:Plessis1817

**Table 369** — geosrs:Plessis1817

|            |                                                                                       |
|------------|---------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Plessis1817">https://w3id.org/geosrs/Plessis1817</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                      |
| Definition | Plessis 1817 Spheroid                                                                 |
| Example    | <a href="#">geosrs:Plessis1817</a>                                                    |

### 13.3.11. Instance: geosrs:Danish1876

**Table 370** — geosrs:Danish1876

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Danish1876">https://w3id.org/geosrs/Danish1876</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                    |
| Definition | Danish 1876 Spheroid                                                                |
| Example    | <a href="#">geosrs:Danish1876</a>                                                   |

### 13.3.12. Instance: geosrs:Struve1860

**Table 371** — geosrs:Struve1860

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Struve1860">https://w3id.org/geosrs/Struve1860</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                    |
| Definition | Struve 1860 Spheroid                                                                |
| Example    | <a href="#">geosrs:Struve1860</a>                                                   |

### 13.3.13. Instance: geosrs:IAG1975

**Table 372** — geosrs:IAG1975

|     |                                                                               |
|-----|-------------------------------------------------------------------------------|
| URI | <a href="https://w3id.org/geosrs/IAG1975">https://w3id.org/geosrs/IAG1975</a> |
|-----|-------------------------------------------------------------------------------|

|            |                                  |
|------------|----------------------------------|
| Type       | <a href="#">geosrs:Ellipsoid</a> |
| Definition | IAG 1975 Spheroid                |
| Example    | <a href="#">geosrs:IAG1975</a>   |

### 13.3.14. Instance: geosrs:Clarke1858

**Table 373** — geosrs:Clarke1858

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Clarke1858">https://w3id.org/geosrs/Clarke1858</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                    |
| Definition | Clarke 1858 Spheroid                                                                |
| Example    | <a href="#">geosrs:Clarke1858</a>                                                   |

### 13.3.15. Instance: geosrs:Clarke1880

**Table 374** — geosrs:Clarke1880

|            |                                                                                     |
|------------|-------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Clarke1880">https://w3id.org/geosrs/Clarke1880</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                    |
| Definition | Clarke 1880 Spheroid                                                                |
| Example    | <a href="#">geosrs:Clarke1880</a>                                                   |

### 13.3.16. Instance: geosrs:Helmert1906

**Table 375** — geosrs:Helmert1906

|      |                                                                                       |
|------|---------------------------------------------------------------------------------------|
| URI  | <a href="https://w3id.org/geosrs/Helmert1906">https://w3id.org/geosrs/Helmert1906</a> |
| Type | <a href="#">geosrs:Ellipsoid</a>                                                      |

|            |                                    |
|------------|------------------------------------|
| Definition | Helmert 1906 Spheroid              |
| Example    | <a href="#">geosrs:Helmert1906</a> |

### 13.3.17. Instance: geosrs:CGCS2000

**Table 376** — geosrs:CGCS2000

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/CGCS2000">https://w3id.org/geosrs/CGCS2000</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                |
| Definition | CGCS2000 Spheroid                                                               |
| Example    | <a href="#">geosrs:CGCS2000</a>                                                 |

### 13.3.18. Instance: geosrs:GSK-2011

**Table 377** — geosrs:GSK-2011

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/GSK-2011">https://w3id.org/geosrs/GSK-2011</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                |
| Definition | GSK-2011 Spheroid                                                               |

### 13.3.19. Instance: geosrs:Zach1812

**Table 378** — geosrs:Zach1812

|            |                                                                                 |
|------------|---------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Zach1812">https://w3id.org/geosrs/Zach1812</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                |
| Definition | Zach 1812 Spheroid                                                              |
| Example    | <a href="#">geosrs:Zach1812</a>                                                 |



### 13.3.20. Instance: geosrs:Clarke1880ARC

**Table 379** — geosrs:Clarke1880ARC

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Clarke1880ARC">https://w3id.org/geosrs/Clarke1880ARC</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                          |
| Definition | Clarke 1880 (Arc) Spheroid                                                                |
| Example    | <a href="#">geosrs:Clarke1880ARC</a>                                                      |

### 13.3.21. Instance: geosrs:Clarke1880IGN

**Table 380** — geosrs:Clarke1880IGN

|            |                                                                                           |
|------------|-------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Clarke1880IGN">https://w3id.org/geosrs/Clarke1880IGN</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                          |
| Definition | Clarke 1880 (Ing) Spheroid                                                                |
| Example    | <a href="#">geosrs:Clarke1880IGN</a>                                                      |

### 13.3.22. Instance: geosrs:WGS66

**Table 381** — geosrs:WGS66

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/WGS66">https://w3id.org/geosrs/WGS66</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                          |
| Definition | WGS 66 Spheroid                                                           |

### 13.3.23. Instance: geosrs:WGS72

**Table 382** — geosrs:WGS72

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/WGS72">https://w3id.org/geosrs/WGS72</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                          |
| Definition | WGS 72 Spheroid                                                           |
| Example    | <a href="#">geosrs:WGS72</a>                                              |

### 13.3.24. Instance: geosrs:WGS84

**Table 383** — geosrs:WGS84

|            |                                                                           |
|------------|---------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/WGS84">https://w3id.org/geosrs/WGS84</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                          |
| Definition | WGS 84 Spheroid                                                           |
| Example    | <a href="#">geosrs:WGS84</a>                                              |

### 13.3.25. Instance: geosrs:Krassowsky1940

**Table 384** — geosrs:Krassowsky1940

|            |                                                                                             |
|------------|---------------------------------------------------------------------------------------------|
| URI        | <a href="https://w3id.org/geosrs/Krassowsky1940">https://w3id.org/geosrs/Krassowsky1940</a> |
| Type       | <a href="#">geosrs:Ellipsoid</a>                                                            |
| Definition | Krassowsky 1940 Spheroid                                                                    |
| Example    | <a href="#">geosrs:Krassowsky1940</a>                                                       |

















# ANNEX A (NORMATIVE) ABSTRACT TEST SUITE

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# ANNEX A

## (NORMATIVE)

## ABSTRACT TEST SUITE

### A.0. Overview

This Annex lists tests for the Conformance Classes defined in the main body sections of this Specification with links to their Requirements and test purpose method and type. Conformance classes may be used to signify the compatibility of a given implementation to parts of the CRS Ontology standard. They may be stated as part of a SPARQL 1.1 Service Description [SPARQLSERVDESC] .

### A.1. Conformance Class: Core

| CONFORMANCE CLASS A.1: 06-CORE.ADOC |                                                                                                                                                                                                                 |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER                          | /conf/core                                                                                                                                                                                                      |
| REQUIREMENTS CLASS                  | Requirements class 1: /req/core                                                                                                                                                                                 |
| CONFORMANCE TESTS                   | Abstract test A.1: /conf/core/Coordinate_Reference_System_Parameters<br>Abstract test A.2: /conf/core/Coordinate_Reference_System_Types<br>Abstract test A.3: /conf/core/Coordinate_Reference_System_Properties |

#### A.1.1. Coordinate Reference System Parameters

| ABSTRACT TEST A.1 |                                                   |
|-------------------|---------------------------------------------------|
| IDENTIFIER        | /conf/core/Coordinate_Reference_System_Parameters |

## ABSTRACT TEST A.1

|                  |                                                                                                                                                                             |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| REQUIREMENT      | Requirement 1: /req/core/Coordinate_Reference_System_Parameters                                                                                                             |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                     |
| TEST METHOD      | Verify that queries involving geosrs:AreaOfUse geosrs:Extent geosrs:GeographicBoundingBox geosrs:AxesList geosrs:SingleCRSList return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                |
| REFERENCE        | geosrs:AreaOfUse geosrs:Extent geosrs:GeographicBoundingBox geosrs:AxesList geosrs:SingleCRSList                                                                            |

### A.1.2. Coordinate Reference System Types

## ABSTRACT TEST A.2

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/core/Coordinate_Reference_System_Types                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| REQUIREMENT      | Requirement 3: /req/core/Coordinate_Reference_System_Types                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| TEST METHOD      | Verify that queries involving geosrs:BoundCRS geosrs:CompoundCRS geosrs:CRS geosrs:EngineeringCRS geosrs:GeocentricCRS geosrs:GeodeticCRS geosrs:GeographicCRS geosrs:ParametricCRS geosrs:ProjectedCRS geosrs:SelenographicCRS geosrs:ReferenceSystem geosrs:SingleCRS geosrs:SpatialReferenceSystem geosrs:SpatioParametricCompoundCRS geosrs:SpatioParametricTemporalCompoundCRS geosrs:SpatioTemporalCompoundCRS geosrs:StaticCRS geosrs:TemporalCRS geosrs:VerticalCRS return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
| REFERENCE        | geosrs:BoundCRS geosrs:CompoundCRS geosrs:CRS geosrs:EngineeringCRS geosrs:GeocentricCRS geosrs:GeodeticCRS geosrs:GeographicCRS geosrs:ParametricCRS geosrs:ProjectedCRS geosrs:SelenographicCRS geosrs:ReferenceSystem geosrs:SingleCRS geosrs:SpatialReferenceSystem geosrs:SpatioParametricCompoundCRS geosrs:SpatioParametricTemporalCompoundCRS geosrs:SpatioTemporalCompoundCRS geosrs:StaticCRS geosrs:TemporalCRS geosrs:VerticalCRS                                                                            |

### A.1.3. Coordinate Reference System Properties

## ABSTRACT TEST A.3

|                  |                                                                                          |
|------------------|------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/core/Coordinate_Reference_System_Properties                                        |
| REQUIREMENT      | Requirement 2: /req/core/Coordinate_Reference_System_Properties                          |
| TEST PURPOSE     | Check conformance with this requirement                                                  |
| TEST METHOD      | Verify that queries involving geosrs:method return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                             |
| REFERENCE        | geosrs:method                                                                            |

## A.2. Conformance Class: Co

### CONFORMANCE CLASS A.2: 07-CO\_MODULE.ADOC

|                    |                                                                                                                                                                                                                                                       |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER         | /conf/co                                                                                                                                                                                                                                              |
| REQUIREMENTS CLASS | Requirements class 2: /req/co                                                                                                                                                                                                                         |
| CONFORMANCE TESTS  | Abstract test A.4: /conf/co/Coordinate_Operation_Methods<br>Abstract test A.5: /conf/co/Coordinate_Operation_Parameters<br>Abstract test A.6: /conf/co/Coordinate_Operation_Categories<br>Abstract test A.7: /conf/co/Coordinate_Operation_Properties |

### A.2.1. Coordinate Operation Methods

#### ABSTRACT TEST A.4

|              |                                                                                                                                                                                  |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER   | /conf/co/Coordinate_Operation_Methods                                                                                                                                            |
| REQUIREMENT  | Requirement 5: /req/co/Coordinate_Operation_Methods                                                                                                                              |
| TEST PURPOSE | Check conformance with this requirement                                                                                                                                          |
| TEST METHOD  | Verify that queries involving geosrs:CoordinateOperation geosrs:PassThroughOperation geosrs:ConcatenatedOperation geosrs:SingleOperation geosrs:Transformation geosrs:Conversion |

## ABSTRACT TEST A.4

|                  |                                                                                                                                                                                                       |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | geosrs:PointMotionOperation geosrs:OperationMethod return the correct result on a test dataset.                                                                                                       |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                          |
| REFERENCE        | geosrs:CoordinateOperation geosrs:PassThroughOperation geosrs:ConcatenatedOperation geosrs:SingleOperation geosrs:Transformation geosrs:Conversion geosrs:PointMotionOperation geosrs:OperationMethod |

## A.2.2. Coordinate Operation Parameters

### ABSTRACT TEST A.5

|                  |                                                                                                                                                                                                                                                              |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/co/Coordinate_Operation_Parameters                                                                                                                                                                                                                     |
| REQUIREMENT      | Requirement 6: /req/co/Coordinate_Operation_Parameters                                                                                                                                                                                                       |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                      |
| TEST METHOD      | Verify that queries involving geosrs:GeneralOperationParameter geosrs:OperationParameter Group geosrs:OperationParameter geosrs:GeneralParameterValue geosrs:ParameterValueGroup geosrs:OperationParameterValue return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                 |
| REFERENCE        | geosrs:GeneralOperationParameter geosrs:OperationParameterGroup geosrs:Operation Parameter geosrs:GeneralParameterValue geosrs:ParameterValueGroup geosrs:Operation ParameterValue                                                                           |

## A.2.3. Coordinate Operation Categories

### ABSTRACT TEST A.6

|              |                                                                                                                                                                                        |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER   | /conf/co/Coordinate_Operation_Categories                                                                                                                                               |
| REQUIREMENT  | Requirement 4: /req/co/Coordinate_Operation_Categories                                                                                                                                 |
| TEST PURPOSE | Check conformance with this requirement                                                                                                                                                |
| TEST METHOD  | Verify that queries involving geosrs:GeographicObject geosrs:RegisterOperations geosrs:Scale Operation geosrs:RotationOperation geosrs:IdentityOperation geosrs:ShearOperation geosrs: |

## ABSTRACT TEST A.6

|                  |                                                                                                                                                                                                                  |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | TranslationOperation geosrs:AffineTransformationOperation return the correct result on a test dataset.                                                                                                           |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                     |
| REFERENCE        | geosrs:GeographicObject geosrs:RegisterOperations geosrs:ScaleOperation geosrs:RotationOperation geosrs:IdentityOperation geosrs:ShearOperation geosrs:TranslationOperation geosrs:AffineTransformationOperation |

### A.2.4. Coordinate Operation Properties

## ABSTRACT TEST A.7

|                  |                                                                                                                                                         |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/co/Coordinate_Operation_Properties                                                                                                                |
| REQUIREMENT      | Requirement 7: /req/co/Coordinate_Operation_Properties                                                                                                  |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                 |
| TEST METHOD      | Verify that queries involving geosrs:derivingConversion geosrs:parameter geosrs:sourceCRS geosrs:targetCRS return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                            |
| REFERENCE        | geosrs:derivingConversion geosrs:parameter geosrs:sourceCRS geosrs:targetCRS                                                                            |

## A.3. Conformance Class: Cs

## CONFORMANCE CLASS A.3: 08-CS\_MODULE.ADOC

|                    |                                                                                                                                                                                                                                   |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER         | /conf/cs                                                                                                                                                                                                                          |
| REQUIREMENTS CLASS | Requirements class 3: /req/cs                                                                                                                                                                                                     |
| CONFORMANCE TESTS  | Abstract test A.8: /conf/cs/Temporal_Coordinate_Systems<br>Abstract test A.9: /conf/cs/3D_Coordinate_Systems<br>Abstract test A.10: /conf/cs/Coordinate_System_Types<br>Abstract test A.11: /conf/cs/Celestial_Coordinate_Systems |

## CONFORMANCE CLASS A.3: 08-CS\_MODULE.ADOC

Abstract test A.12: /conf/cs/Coordinate\_System\_Components

Abstract test A.13: /conf/cs/Coordinate\_System\_Properties

### A.3.1. Temporal Coordinate Systems

#### ABSTRACT TEST A.8

|                  |                                                                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/cs/Temporal_Coordinate_Systems                                                                                                                                                                                                                     |
| REQUIREMENT      | Requirement 13: /req/cs/Temporal_Coordinate_Systems                                                                                                                                                                                                      |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                  |
| TEST METHOD      | Verify that queries involving geosrs:1DCoordinateSystem geosrs:DateTimeTemporalCoordinateSystem geosrs:TemporalCountCoordinateSystem geosrs:TemporalCoordinateSystem geosrs:TemporalMeasureCoordinateSystem return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                             |
| REFERENCE        | geosrs:1DCoordinateSystem geosrs:DateTimeTemporalCoordinateSystem geosrs:TemporalCountCoordinateSystem geosrs:TemporalCoordinateSystem geosrs:TemporalMeasureCoordinateSystem                                                                            |

### A.3.2. 3D Coordinate Systems

#### ABSTRACT TEST A.9

|                  |                                                                                                                                                                                                                                            |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/cs/3D_Coordinate_Systems                                                                                                                                                                                                             |
| REQUIREMENT      | Requirement 8: /req/cs/3D_Coordinate_Systems                                                                                                                                                                                               |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                    |
| TEST METHOD      | Verify that queries involving geosrs:3DCoordinateSystem geosrs:ConicalCoordinateSystem geosrs:CylindricalCoordinateSystem geosrs:EllipsoidalCoordinateSystem geosrs:SphericalCoordinateSystem return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                               |
| REFERENCE        | geosrs:3DCoordinateSystem geosrs:ConicalCoordinateSystem geosrs:CylindricalCoordinateSystem geosrs:EllipsoidalCoordinateSystem geosrs:SphericalCoordinateSystem                                                                            |

### A.3.3. Coordinate System Types

#### ABSTRACT TEST A.10

**IDENTIFIER** /conf/cs/Coordinate\_System\_Types

**REQUIREMENT** Requirement 12: /req/cs/Coordinate\_System\_Types

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:CoordinateSystem geosrs:AffineCoordinateSystem geosrs:BarycentricCoordinateSystem geosrs:CartesianCoordinateSystem geosrs:CurvilinearCoordinateSystem geosrs:EngineeringCoordinateSystem geosrs:GeodeticCoordinateSystem geosrs:GeographicalCoordinateSystem geosrs:GridCoordinateSystem geosrs:HexagonalCoordinateSystem geosrs:LinearCoordinateSystem geosrs:LocalCoordinateSystem geosrs:ObliqueCoordinateSystem geosrs:OrdinalCoordinateSystem geosrs:OrthogonalCoordinateSystem geosrs:ParametricCoordinateSystem geosrs:PlanarCoordinateSystem geosrs:PolarCoordinateSystem geosrs:VerticalCoordinateSystem return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:CoordinateSystem geosrs:AffineCoordinateSystem geosrs:BarycentricCoordinateSystem geosrs:CartesianCoordinateSystem geosrs:CurvilinearCoordinateSystem geosrs:EngineeringCoordinateSystem geosrs:GeodeticCoordinateSystem geosrs:GeographicalCoordinateSystem geosrs:GridCoordinateSystem geosrs:HexagonalCoordinateSystem geosrs:LinearCoordinateSystem geosrs:LocalCoordinateSystem geosrs:ObliqueCoordinateSystem geosrs:OrdinalCoordinateSystem geosrs:OrthogonalCoordinateSystem geosrs:ParametricCoordinateSystem geosrs:PlanarCoordinateSystem geosrs:PolarCoordinateSystem geosrs:VerticalCoordinateSystem

### A.3.4. Celestial Coordinate Systems

#### ABSTRACT TEST A.11

**IDENTIFIER** /conf/cs/Celestial\_Coordinate\_Systems

**REQUIREMENT** Requirement 9: /req/cs/Celestial\_Coordinate\_Systems

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:CelestialCoordinateSystem geosrs:EclipticCoordinateSystem geosrs:EquatorialCoordinateSystem geosrs:GalacticCoordinateSystem geosrs:Horizontal



## ABSTRACT TEST A.11

|                  |                                                                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                  | CoordinateSystem geosrs:PerifocalCoordinateSystem geosrs:SuperGalacticCS return the correct result on a test dataset.                                                                                                        |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                 |
| REFERENCE        | geosrs:CelestialCoordinateSystem geosrs:EclipticCoordinateSystem geosrs:EquatorialCoordinateSystem geosrs:GalacticCoordinateSystem geosrs:HorizontalCoordinateSystem geosrs:PerifocalCoordinateSystem geosrs:SuperGalacticCS |

### A.3.5. Coordinate System Components

## ABSTRACT TEST A.12

|                  |                                                                                                        |
|------------------|--------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/cs/Coordinate_System_Components                                                                  |
| REQUIREMENT      | Requirement 10: /req/cs/Coordinate_System_Components                                                   |
| TEST PURPOSE     | Check conformance with this requirement                                                                |
| TEST METHOD      | Verify that queries involving geosrs:CoordinateSystemAxis return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                           |
| REFERENCE        | geosrs:CoordinateSystemAxis                                                                            |

### A.3.6. Coordinate System Properties

## ABSTRACT TEST A.13

|                  |                                                                                                             |
|------------------|-------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/cs/Coordinate_System_Properties                                                                       |
| REQUIREMENT      | Requirement 11: /req/cs/Coordinate_System_Properties                                                        |
| TEST PURPOSE     | Check conformance with this requirement                                                                     |
| TEST METHOD      | Verify that queries involving geosrs:axis geosrs:axisDirection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                |

## ABSTRACT TEST A.13

|           |                                  |
|-----------|----------------------------------|
| REFERENCE | geosrs:axis geosrs:axisDirection |
|-----------|----------------------------------|

## A.4. Conformance Class: Datum

### CONFORMANCE CLASS A.4: 09-DATUM\_MODULE.ADOC

|                    |                                                                                                                                                                                                                                                              |
|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER         | /conf/datum                                                                                                                                                                                                                                                  |
| REQUIREMENTS CLASS | Requirements class 4: /req/datum                                                                                                                                                                                                                             |
| CONFORMANCE TESTS  | Abstract test A.14: /conf/datum/Datum_Types<br>Abstract test A.15: /conf/datum/Datum_Parameters<br>Abstract test A.16: /conf/datum/Spheroid_Types<br>Abstract test A.17: /conf/datum/Spheroid_Properties<br>Abstract test A.18: /conf/datum/Datum_Properties |

### A.4.1. Datum Types

#### ABSTRACT TEST A.14

|                  |                                                                                                                                                                                                                                                                                             |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/datum/Datum_Types                                                                                                                                                                                                                                                                     |
| REQUIREMENT      | Requirement 16: /req/datum/Datum_Types                                                                                                                                                                                                                                                      |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                     |
| TEST METHOD      | Verify that queries involving geosrs:Datum geosrs:GeodeticDatum geosrs:DynamicGeodeticReferenceFrame geosrs:VerticalDatum geosrs:DynamicVerticalDatum geosrs:ParametricDatum geosrs:EngineeringDatum geosrs:TemporalDatum geosrs:DatumEnsemble return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                |
| REFERENCE        | geosrs:Datum geosrs:GeodeticDatum geosrs:DynamicGeodeticReferenceFrame geosrs:VerticalDatum geosrs:DynamicVerticalDatum geosrs:ParametricDatum geosrs:EngineeringDatum geosrs:TemporalDatum geosrs:DatumEnsemble                                                                            |

## A.4.2. Datum Parameters

### ABSTRACT TEST A.15

|                  |                                                                                                                          |
|------------------|--------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/datum/Datum_Parameters                                                                                             |
| REQUIREMENT      | Requirement 14: /req/datum/Datum_Parameters                                                                              |
| TEST PURPOSE     | Check conformance with this requirement                                                                                  |
| TEST METHOD      | Verify that queries involving geosrs:PrimeMeridian geosrs:DefiningParameter return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                             |
| REFERENCE        | geosrs:PrimeMeridian geosrs:DefiningParameter                                                                            |

## A.4.3. Spheroid Types

### ABSTRACT TEST A.16

|                  |                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/datum/Spheroid_Types                                                                                           |
| REQUIREMENT      | Requirement 18: /req/datum/Spheroid_Types                                                                            |
| TEST PURPOSE     | Check conformance with this requirement                                                                              |
| TEST METHOD      | Verify that queries involving geosrs:Ellipsoid geosrs:TriaxialEllipsoid return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                         |
| REFERENCE        | geosrs:Ellipsoid geosrs:TriaxialEllipsoid                                                                            |

## A.4.4. Spheroid Properties

## ABSTRACT TEST A.17

|                  |                                                                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/datum/Spheroid_Properties                                                                                                                                                   |
| REQUIREMENT      | Requirement 17: /req/datum/Spheroid_Properties                                                                                                                                    |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                           |
| TEST METHOD      | Verify that queries involving geosrs:eccentricity geosrs:inverseFlattening geosrs:isSphere geosrs:semiMajorAxis geosrs:semiMinorAxis return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                      |
| REFERENCE        | geosrs:eccentricity geosrs:inverseFlattening geosrs:isSphere geosrs:semiMajorAxis geosrs:semiMinorAxis                                                                            |

### A.4.5. Datum Properties

## ABSTRACT TEST A.18

|                  |                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/datum/Datum_Properties                                                                                                                   |
| REQUIREMENT      | Requirement 15: /req/datum/Datum_Properties                                                                                                    |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                        |
| TEST METHOD      | Verify that queries involving geosrs:datumDefiningParameter geosrs:ellipsoid geosrs:primeMeridian return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                   |
| REFERENCE        | geosrs:datumDefiningParameter geosrs:ellipsoid geosrs:primeMeridian                                                                            |

### A.5. Conformance Class: Srsapplication

## CONFORMANCE CLASS A.5: 10-SRSAPPLICATION\_MODULE.ADOC

|            |                      |
|------------|----------------------|
| IDENTIFIER | /conf/srsapplication |
|------------|----------------------|

## CONFORMANCE CLASS A.5: 10-SRSAPPLICATION\_MODULE.ADOC

|                    |                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------|
| REQUIREMENTS CLASS | Requirements class 5: /req/srsapplication                                                                            |
| CONFORMANCE TESTS  | Abstract test A.19: /conf/srsapplication/SRS_Application_Types<br>Abstract test A.20: /conf/srsapplication/Map_Types |

### A.5.1. SRS Application Types

#### ABSTRACT TEST A.19

|                  |                                                                                                                                                                                                                                                                                                                           |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/srsapplication/SRS_Application_Types                                                                                                                                                                                                                                                                                |
| REQUIREMENT      | Requirement 20: /req/srsapplication/SRS_Application_Types                                                                                                                                                                                                                                                                 |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                                                   |
| TEST METHOD      | Verify that queries involving geosrs:SRSApplication geosrs:SpatialReferencing geosrs:EngineeringSurvey geosrs:SatelliteSurvey geosrs:SatelliteNavigation geosrs:CoastalHydrography geosrs:OffshoreEngineering geosrs:Hydrography geosrs:Drilling geosrs:OilAndGasExploration return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                              |
| REFERENCE        | geosrs:SRSApplication geosrs:SpatialReferencing geosrs:EngineeringSurvey geosrs:SatelliteSurvey geosrs:SatelliteNavigation geosrs:CoastalHydrography geosrs:OffshoreEngineering geosrs:Hydrography geosrs:Drilling geosrs:OilAndGasExploration                                                                            |

### A.5.2. Map Types

#### ABSTRACT TEST A.20

|                  |                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/srsapplication/Map_Types                                                                                                                                                |
| REQUIREMENT      | Requirement 19: /req/srsapplication/Map_Types                                                                                                                                 |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                       |
| TEST METHOD      | Verify that queries involving geosrs:CadastreMap geosrs:NauticalChart geosrs:ThematicMap geosrs:TopographicMap geosrs:WeatherMap return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                  |

## ABSTRACT TEST A.20

|           |                                                                                                    |
|-----------|----------------------------------------------------------------------------------------------------|
| REFERENCE | geosrs:CadastreMap geosrs:NauticalChart geosrs:ThematicMap geosrs:TopographicMap geosrs:WeatherMap |
|-----------|----------------------------------------------------------------------------------------------------|

## A.6. Conformance Class: Projections

### CONFORMANCE CLASS A.6: 11-PROJECTIONS\_MODULE.ADOC

|                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER         | /conf/projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| REQUIREMENTS CLASS | Requirements class 6: /req/projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| CONFORMANCE TESTS  | <p>Abstract test A.21: /conf/projections/Lenticular_Projections<br/>Abstract test A.22: /conf/projections/Conformal_Projections<br/>Abstract test A.23: /conf/projections/Minimum_Error_Projections<br/>Abstract test A.24: /conf/projections/Pseudo_Azimuthal_Projections<br/>Abstract test A.25: /conf/projections/Equal_Area_Projections<br/>Abstract test A.26: /conf/projections/Pseudo_Conical_Projections<br/>Abstract test A.27: /conf/projections/Globular_Projections<br/>Abstract test A.28: /conf/projections/Pseudo_Cylindrical_Projections<br/>Abstract test A.29: /conf/projections/Archaic_Projections<br/>Abstract test A.30: /conf/projections/Cylindrical_Projections<br/>Abstract test A.31: /conf/projections/Compromise_Projections<br/>Abstract test A.32: /conf/projections/Polyhedral_Projections<br/>Abstract test A.33: /conf/projections/Equidistant_Projections<br/>Abstract test A.34: /conf/projections/Azimuthal_Projections<br/>Abstract test A.35: /conf/projections/Conical_Projections<br/>Abstract test A.36: /conf/projections/Perspective_Projections<br/>Abstract test A.37: /conf/projections/Stereographic_Projections<br/>Abstract test A.38: /conf/projections/Polyconic_Projections<br/>Abstract test A.39: /conf/projections/Projection</p> |

### A.6.1. Lenticular Projections

#### ABSTRACT TEST A.21

|             |                                                         |
|-------------|---------------------------------------------------------|
| IDENTIFIER  | /conf/projections/Lenticular_Projections                |
| REQUIREMENT | Requirement 30: /req/projections/Lenticular_Projections |

## ABSTRACT TEST A.21

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:A4Projection geosrs:BriesemeisterProjection geosrs:CiricIProjection geosrs:CupolaProjection geosrs:DedistortProjection geosrs:DietrichKitadaProjection geosrs:FranculaIIProjection geosrs:FranculaIVProjection geosrs:FranculaIXProjection geosrs:FranculaVIIIProjection geosrs:FranculaVProjection geosrs:FranculaXIIIProjection geosrs:FranculaXIIProjection geosrs:FranculaXIVProjection geosrs:HamusoidalProjection geosrs:KissProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:A4Projection geosrs:BriesemeisterProjection geosrs:CiricIProjection geosrs:CupolaProjection geosrs:DedistortProjection geosrs:DietrichKitadaProjection geosrs:FranculaIIIProjection geosrs:FranculaIVProjection geosrs:FranculaIXProjection geosrs:FranculaVIIIProjection geosrs:FranculaVProjection geosrs:FranculaXIIIProjection geosrs:FranculaXIIProjection geosrs:FranculaXIVProjection geosrs:HamusoidalProjection geosrs:KissProjection

## A.6.2. Conformal Projections

### ABSTRACT TEST A.22

**IDENTIFIER** /conf/projections/Conformal\_Projections

**REQUIREMENT** Requirement 24: /req/projections/Conformal\_Projections

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:AdamsProjection geosrs:AdamsWorldInASquareIIProjection geosrs:AdamsWorldInASquareIProjection geosrs:AugustEpicycloidalProjection geosrs:CoxConformalProjection geosrs:EisenlohrProjection geosrs:GS50Projection geosrs:PeirceQuincuncialProjection geosrs:StereographicProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:AdamsProjection geosrs:AdamsWorldInASquareIIProjection geosrs:AdamsWorldInASquareIProjection geosrs:AugustEpicycloidalProjection geosrs:CoxConformalProjection geosrs:EisenlohrProjection geosrs:GS50Projection geosrs:PeirceQuincuncialProjection geosrs:StereographicProjection

## A.6.3. Minimum Error Projections

## ABSTRACT TEST A.23

|                  |                                                                                                  |
|------------------|--------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Minimum_Error_Projections                                                      |
| REQUIREMENT      | Requirement 31: /req/projections/Minimum_Error_Projections                                       |
| TEST PURPOSE     | Check conformance with this requirement                                                          |
| TEST METHOD      | Verify that queries involving geosrs:AiryProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                     |
| REFERENCE        | geosrs:AiryProjection                                                                            |

## A.6.4. Pseudo Azimuthal Projections

### ABSTRACT TEST A.24

|                  |                                                                                                                                                                                                                                                                                   |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Pseudo_Azimuthal_Projections                                                                                                                                                                                                                                    |
| REQUIREMENT      | Requirement 36: /req/projections/Pseudo_Azimuthal_Projections                                                                                                                                                                                                                     |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                           |
| TEST METHOD      | Verify that queries involving geosrs:AitoffObliqueProjection geosrs:AitoffProjection geosrs:BartholomewProjection geosrs:HammerProjection geosrs:PseudoAzimuthalProjection geosrs:Strebe1995Projection geosrs:WinkelTripelProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                      |
| REFERENCE        | geosrs:AitoffObliqueProjection geosrs:AitoffProjection geosrs:BartholomewProjection geosrs:HammerProjection geosrs:PseudoAzimuthalProjection geosrs:Strebe1995Projection geosrs:WinkelTripelProjection                                                                            |

## A.6.5. Equal Area Projections

### ABSTRACT TEST A.25

|            |                                          |
|------------|------------------------------------------|
| IDENTIFIER | /conf/projections/Equal_Area_Projections |
|------------|------------------------------------------|



## ABSTRACT TEST A.25

**REQUIREMENT** Requirement 27: /req/projections/Equal\_Area\_Projections

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:AlbersEqualAreaProjection geosrs:AzimuthalEqualAreaProjection geosrs:CylindricalEqualArea geosrs:EqualAreaProjection geosrs:GallPetersProjection geosrs:HoboDyerProjection geosrs:LambertAzimuthalEqualArea geosrs:LambertCylindricalEqualAreaProjection geosrs:ObliqueCylindricalEqualAreaProjection geosrs:SlideAndDiceParallelSmallCircle geosrs:SliceAndDiceVertexGreatCircle geosrs:SmythEqualSurfaceProjection geosrs:SnyderEqualArea geosrs:ToblerWorldInASquareProjection geosrs:TransverseCylindricalEqualAreaProjection geosrs:TrystanEdwardsProjection geosrs:WiechelProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:AlbersEqualAreaProjection geosrs:AzimuthalEqualAreaProjection geosrs:CylindricalEqualArea geosrs:EqualAreaProjection geosrs:GallPetersProjection geosrs:HoboDyerProjection geosrs:LambertAzimuthalEqualArea geosrs:LambertCylindricalEqualAreaProjection geosrs:ObliqueCylindricalEqualAreaProjection geosrs:SlideAndDiceParallelSmallCircle geosrs:SliceAndDiceVertexGreatCircle geosrs:SmythEqualSurfaceProjection geosrs:SnyderEqualArea geosrs:ToblerWorldInASquareProjection geosrs:TransverseCylindricalEqualAreaProjection geosrs:TrystanEdwardsProjection geosrs:WiechelProjection

## A.6.6. Pseudo Conical Projections

### ABSTRACT TEST A.26

**IDENTIFIER** /conf/projections/Pseudo\_Conical\_Projections

**REQUIREMENT** Requirement 37: /req/projections/Pseudo\_Conical\_Projections

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:AmericanPolyconicProjection geosrs:BonneProjection geosrs:BottomleyProjection geosrs:NicolosiGlobularProjection geosrs:PseudoConicalProjection geosrs:PtolemyIIProjection geosrs:StabiusWernerIIProjection geosrs:WernerProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:AmericanPolyconicProjection geosrs:BonneProjection geosrs:BottomleyProjection geosrs:NicolosiGlobularProjection geosrs:PseudoConicalProjection geosrs:PtolemyIIProjection geosrs:StabiusWernerIIProjection geosrs:WernerProjection

## A.6.7. Globular Projections

### ABSTRACT TEST A.27

|                  |                                                                                                                                                                              |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Globular_Projections                                                                                                                                       |
| REQUIREMENT      | Requirement 29: /req/projections/Globular_Projections                                                                                                                        |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                      |
| TEST METHOD      | Verify that queries involving geosrs:ApianGlobularIProjection geosrs:BaconGlobularProjection geosrs:FournierGlobularIProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                 |
| REFERENCE        | geosrs:ApianGlobularIProjection geosrs:BaconGlobularProjection geosrs:FournierGlobularIProjection                                                                            |

## A.6.8. Pseudo Cylindrical Projections

### ABSTRACT TEST A.28

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER   | /conf/projections/Pseudo_Cylindrical_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| REQUIREMENT  | Requirement 38: /req/projections/Pseudo_Cylindrical_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| TEST PURPOSE | Check conformance with this requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| TEST METHOD  | Verify that queries involving geosrs:ApianIIProjection geosrs:AtlantisProjection geosrs:BaranyIIIProjection geosrs:BaranyIIProjection geosrs:BaranyIProjection geosrs:BaranyiIVProjection geosrs:BoggsEumorphicProjection geosrs:BromleyProjection geosrs:CabotProjection geosrs:CollignonProjection geosrs:CrasterParabolicProjection geosrs:DeakinMinimumErrorProjection geosrs:Eckert1Projection geosrs:Eckert2Projection geosrs:Eckert3Projection geosrs:Eckert4Projection geosrs:Eckert5Projection geosrs:Eckert6Projection geosrs:EqualEarthProjection geosrs:FaheyProjection geosrs:FoucautProjection geosrs:FoucautSinusoidalProjection geosrs:FournierIIProjection geosrs:GinzburgVIIIProjection geosrs:GoodeHomolosineProjection geosrs:HEALPixProjection geosrs:HatanoAsymmetricalEqualAreaProjection geosrs:HufnagelProjection geosrs:Kavrayskiy7Projection geosrs:LoximuthalProjection geosrs:MayrProjection geosrs:McBrydeThomasFlatPolarParabolicProjection geosrs:McBrydeThomasFlatPolarQuarticProjection geosrs:McBrydeThomasFlatPolarSinusoidalProjection geosrs:McBrydeThomasIIProjection geosrs:McBrydeThomasIProjection geosrs:NaturalEarth2Projection geosrs:NaturalEarthProjection geosrs:NellHammerProjection geosrs:NellProjection geosrs:OrteliusOvalProjection geosrs: |

## ABSTRACT TEST A.28

PseudoCylindricalProjection geosrs:PutninsP1Projection geosrs:PutninsP2Projection geosrs:PutninsP3Projection geosrs:PutninsP5Projection geosrs:PutninsP6Projection geosrs:QuarticAuthalicProjection geosrs:RobinsonProjection geosrs:SinusoidalProjection geosrs:TheTimesProjection geosrs:ToblerG1Projection geosrs:ToblerHyperellipticalProjection geosrs:WagnerIIIProjection geosrs:WagnerIIProjection geosrs:WagnerIProjection geosrs:WagnerIVProjection geosrs:WagnerVProjection geosrs:WagnerVProjection geosrs:WerenskioldIProjection geosrs:PutninsP3'Projection geosrs:PutninsP4'Projection geosrs:PutninsP5'Projection geosrs:PutninsP6'Projection return the correct result on a test dataset.

### TEST-METHOD-TYPE

Capabilities

### REFERENCE

geosrs:ApianIIProjection geosrs:AtlantisProjection geosrs:BaranyIIIProjection geosrs:BaranyiIIProjection geosrs:BaranyiIProjection geosrs:BaranyiIVProjection geosrs:BoggsEumorphicProjection geosrs:BromleyProjection geosrs:CabotProjection geosrs:CollignonProjection geosrs:CrasterParabolicProjection geosrs:DeakinMinimumErrorProjection geosrs:Eckert1Projection geosrs:Eckert2Projection geosrs:Eckert3Projection geosrs:Eckert4Projection geosrs:Eckert5Projection geosrs:Eckert6Projection geosrs:EqualEarthProjection geosrs:FaheyProjection geosrs:FoucautProjection geosrs:FoucautSinusoidalProjection geosrs:FournierIIProjection geosrs:GinzburgVIIIProjection geosrs:GoodeHomolosineProjection geosrs:HEALPixProjection geosrs:HatanoAsymmetricalEqualAreaProjection geosrs:HufnagelProjection geosrs:Kavrayskiy7Projection geosrs:LoximuthalProjection geosrs:MayrProjection geosrs:McBrydeThomasFlatPolarParabolicProjection geosrs:McBrydeThomasFlatPolarQuarticProjection geosrs:McBrydeThomasFlatPolarSinusoidalProjection geosrs:McBrydeThomasIIProjection geosrs:McBrydeThomasIProjection geosrs:NaturalEarth2Projection geosrs:NaturalEarthProjection geosrs:NellHammerProjection geosrs:NellProjection geosrs:OrteliusOvalProjection geosrs:PseudoCylindricalProjection geosrs:PutninsP1Projection geosrs:PutninsP2Projection geosrs:PutninsP3Projection geosrs:PutninsP5Projection geosrs:PutninsP6Projection geosrs:QuarticAuthalicProjection geosrs:RobinsonProjection geosrs:SinusoidalProjection geosrs:TheTimesProjection geosrs:ToblerG1Projection geosrs:ToblerHyperellipticalProjection geosrs:WagnerIIIProjection geosrs:WagnerIIProjection geosrs:WagnerIProjection geosrs:WagnerIVProjection geosrs:WagnerVProjection geosrs:WagnerVProjection geosrs:WerenskioldIProjection geosrs:PutninsP3'Projection geosrs:PutninsP4'Projection geosrs:PutninsP5'Projection geosrs:PutninsP6'Projection

## A.6.9. Archaic Projections

### ABSTRACT TEST A.29

IDENTIFIER /conf/projections/Archaic\_Projections

REQUIREMENT Requirement 21: /req/projections/Archaic\_Projections

TEST PURPOSE Check conformance with this requirement

## ABSTRACT TEST A.29

|                  |                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------|
| TEST METHOD      | Verify that queries involving geosrs:ArchaicProjection geosrs:PtolemyIProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                  |
| REFERENCE        | geosrs:ArchaicProjection geosrs:PtolemyIProjection                                                                            |

## A.6.10. Cylindrical Projections

### ABSTRACT TEST A.30

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Cylindrical_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| REQUIREMENT      | Requirement 26: /req/projections/Cylindrical_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TEST METHOD      | Verify that queries involving geosrs:ArdenCloseProjection geosrs:BSAMCylindricalProjection geosrs:BalthasartProjection geosrs:BehrmannProjection geosrs:BraunPerspectiveProjection geosrs:BraunStereographicProjection geosrs:CompactMillerProjection geosrs:CylindricalProjection geosrs:CylindricalStereographicProjection geosrs:KarchenkoShabanovaProjection geosrs:LabordeProjection geosrs:MercatorProjection geosrs:MillerProjection geosrs:PattersonCylindricalProjection geosrs:PavlovProjection geosrs:ToblerCylindricalIIIProjection geosrs:ToblerCylindricalIProjection geosrs:TransverseMercatorProjection geosrs:UrmayevIIIProjection geosrs:WebMercatorProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| REFERENCE        | geosrs:ArdenCloseProjection geosrs:BSAMCylindricalProjection geosrs:BalthasartProjection geosrs:BehrmannProjection geosrs:BraunPerspectiveProjection geosrs:BraunStereographicProjection geosrs:CompactMillerProjection geosrs:CylindricalProjection geosrs:CylindricalStereographicProjection geosrs:KarchenkoShabanovaProjection geosrs:LabordeProjection geosrs:MercatorProjection geosrs:MillerProjection geosrs:PattersonCylindricalProjection geosrs:PavlovProjection geosrs:ToblerCylindricalIIIProjection geosrs:ToblerCylindricalIProjection geosrs:TransverseMercatorProjection geosrs:UrmayevIIIProjection geosrs:WebMercatorProjection                                                                            |

## A.6.11. Compromise Projections

## ABSTRACT TEST A.31

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Compromise_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| REQUIREMENT      | Requirement 23: /req/projections/Compromise_Projections                                                                                                                                                                                                                                                                                                                                                                                                                          |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| TEST METHOD      | Verify that queries involving geosrs:ArmadilloProjection geosrs:BakerDinomicProjection geosrs:BertinProjection geosrs:ChamberlinTrimetricProjection geosrs:DenoyerSemiEllipticalProjection geosrs:FairgrieveProjection geosrs:LarriveeProjection geosrs:PetermannStarProjection geosrs:SpilhausOceanicProjection geosrs:VanDerGrintenIIIProjection geosrs:WinkelIIIProjection geosrs:WinkelProjection geosrs:WinkelSnyderProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| REFERENCE        | geosrs:ArmadilloProjection geosrs:BakerDinomicProjection geosrs:BertinProjection geosrs:ChamberlinTrimetricProjection geosrs:DenoyerSemiEllipticalProjection geosrs:FairgrieveProjection geosrs:LarriveeProjection geosrs:PetermannStarProjection geosrs:SpilhausOceanicProjection geosrs:VanDerGrintenIIIProjection geosrs:WinkelIIIProjection geosrs:WinkelProjection geosrs:WinkelSnyderProjection                                                                            |

## A.6.12. Polyhedral Projections

### ABSTRACT TEST A.32

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Polyhedral_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| REQUIREMENT      | Requirement 34: /req/projections/Polyhedral_Projections                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| TEST METHOD      | Verify that queries involving geosrs:AuthaGraphProjection geosrs:CahillKeyesProjection geosrs:CollignonButterflyProjection geosrs:DodecahedralProjection geosrs:DymaxionProjection geosrs:GnomonicButterflyProjection geosrs:GnomonicCubedSphereProjection geosrs:GnomonicIcosahedronProjection geosrs:GuyouProjection geosrs:IcosahedronProjection geosrs:LeeProjection geosrs:MyrahedalProjection geosrs:OctantProjection geosrs:PolyhedralProjection geosrs:QuadrilateralizedSphericalCubeProjection geosrs:WatermanButterflyProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| REFERENCE        | geosrs:AuthaGraphProjection geosrs:CahillKeyesProjection geosrs:CollignonButterflyProjection geosrs:DodecahedralProjection geosrs:DymaxionProjection geosrs:GnomonicButterflyProjection geosrs:GnomonicCubedSphereProjection geosrs:GnomonicIcosahedronProjection geosrs:Guyou                                                                                                                                                                                                                                                                                                          |

## ABSTRACT TEST A.32

Projection geosrs:IcosahedralProjection geosrs:LeeProjection geosrs:MyrahedalProjection geosrs:OctantProjection geosrs:PolyhedralProjection geosrs:QuadrilateralizedSphericalCubeProjection geosrs:WatermanButterflyProjection

## A.6.13. Equidistant Projections

### ABSTRACT TEST A.33

**IDENTIFIER** /conf/projections/Equidistant\_Projections

**REQUIREMENT** Requirement 28: /req/projections/Equidistant\_Projections

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:AzimuthalEquidistantProjection geosrs:BerghausStarProjection geosrs:CassiniProjection geosrs:EquidistantConicProjection geosrs:EquidistantCylindricalProjection geosrs:EquidistantProjection geosrs:EquirectangularProjection geosrs:ObliquePlateCarreeProjection geosrs:PlateCarreeProjection geosrs:TwoPointEquidistantProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:AzimuthalEquidistantProjection geosrs:BerghausStarProjection geosrs:CassiniProjection geosrs:EquidistantConicProjection geosrs:EquidistantCylindricalProjection geosrs:EquidistantProjection geosrs:EquirectangularProjection geosrs:ObliquePlateCarreeProjection geosrs:PlateCarreeProjection geosrs:TwoPointEquidistantProjection

## A.6.14. Azimuthal Projections

### ABSTRACT TEST A.34

**IDENTIFIER** /conf/projections/Azimuthal\_Projections

**REQUIREMENT** Requirement 22: /req/projections/Azimuthal\_Projections

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:AzimuthalProjection geosrs:BreusingGeometricProjection geosrs:BreusingHarmonicProjection geosrs:GinzburgIIProjection geosrs:GinzburgIProjection geosrs:GnomonicProjection geosrs:JamesAzimuthalProjection return the correct result on a test dataset.

## ABSTRACT TEST A.34

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:AzimuthalProjection geosrs:BreusingGeometricProjection geosrs:BreusingHarmonicProjection geosrs:GinzburgIIProjection geosrs:GinzburgIProjection geosrs:GnomonicProjection geosrs:JamesAzimuthalProjection

## A.6.15. Conical Projections

### ABSTRACT TEST A.35

**IDENTIFIER** /conf/projections/Conical\_Projections

**REQUIREMENT** Requirement 25: /req/projections/Conical\_Projections

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:BipolarObliqueConicConformalProjection geosrs:CentralConicProjection geosrs:HerschelConformalConicProjection geosrs:Krovak geosrs:LambertConformalConicProjection geosrs:MurdochIIIProjection geosrs:MurdochIIProjection geosrs:MurdochIProjection geosrs:SchjerningIProjection geosrs:VitkovskyIProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:BipolarObliqueConicConformalProjection geosrs:CentralConicProjection geosrs:HerschelConformalConicProjection geosrs:Krovak geosrs:LambertConformalConicProjection geosrs:MurdochIIIProjection geosrs:MurdochIIProjection geosrs:MurdochIProjection geosrs:SchjerningIProjection geosrs:VitkovskyIProjection

## A.6.16. Perspective Projections

### ABSTRACT TEST A.36

**IDENTIFIER** /conf/projections/Perspective\_Projections

**REQUIREMENT** Requirement 32: /req/projections/Perspective\_Projections

**TEST PURPOSE** Check conformance with this requirement

## ABSTRACT TEST A.36

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TEST METHOD      | Verify that queries involving geosrs:CentralCylindricalProjection geosrs:GeneralVerticalPerspectiveProjection geosrs:GilbertTwoWorldPerspectiveProjection geosrs:LaHireProjection geosrs:LorgnaProjection geosrs:LowryProjection geosrs:OrthographicProjection geosrs:PerspectiveConicProjection geosrs:PerspectiveProjection geosrs:TiltedPerspectiveProjection geosrs:VerticalPerspectiveProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| REFERENCE        | geosrs:CentralCylindricalProjection geosrs:GeneralVerticalPerspectiveProjection geosrs:GilbertTwoWorldPerspectiveProjection geosrs:LaHireProjection geosrs:LorgnaProjection geosrs:LowryProjection geosrs:OrthographicProjection geosrs:PerspectiveConicProjection geosrs:PerspectiveProjection geosrs:TiltedPerspectiveProjection geosrs:VerticalPerspectiveProjection                                                                            |

## A.6.17. Stereographic Projections

### ABSTRACT TEST A.37

|                  |                                                                                                                                                                                      |
|------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/projections/Stereographic_Projections                                                                                                                                          |
| REQUIREMENT      | Requirement 39: /req/projections/Stereographic_Projections                                                                                                                           |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                              |
| TEST METHOD      | Verify that queries involving geosrs:GallStereographicProjection geosrs:MillerOblatedStereographicProjection geosrs:RoussilheProjection return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                         |
| REFERENCE        | geosrs:GallStereographicProjection geosrs:MillerOblatedStereographicProjection geosrs:RoussilheProjection                                                                            |

## A.6.18. Polyconic Projections

### ABSTRACT TEST A.38

|              |                                                        |
|--------------|--------------------------------------------------------|
| IDENTIFIER   | /conf/projections/Polyconic_Projections                |
| REQUIREMENT  | Requirement 33: /req/projections/Polyconic_Projections |
| TEST PURPOSE | Check conformance with this requirement                |



## ABSTRACT TEST A.38

**TEST METHOD** Verify that queries involving geosrs:GinzburgIVProjection geosrs:GinzburgIXProjection geosrs:GinzburgVIPProjection geosrs:GinzburgVProjection geosrs:GottWagnerProjection geosrs:HillEucyclicProjection geosrs:LagrangeProjection geosrs:LaskowskiProjection geosrs:PolyconicProjection geosrs:RectangularPolyconicProjection geosrs:StabiusWernerIIIProjection geosrs:StabiusWernerIProjection geosrs:VanDerGrintenIIProjection geosrs:VanDerGrintenIProjection geosrs:VanDerGrintenIVProjection geosrs:WagnerIXProjection geosrs:WagnerVIIIProjection geosrs:WagnerVIIProjection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:GinzburgIVProjection geosrs:GinzburgIXProjection geosrs:GinzburgVIPProjection geosrs:GinzburgVProjection geosrs:GottWagnerProjection geosrs:HillEucyclicProjection geosrs:LagrangeProjection geosrs:LaskowskiProjection geosrs:PolyconicProjection geosrs:RectangularPolyconicProjection geosrs:StabiusWernerIIIProjection geosrs:StabiusWernerIProjection geosrs:VanDerGrintenIIProjection geosrs:VanDerGrintenIProjection geosrs:VanDerGrintenIVProjection geosrs:WagnerIXProjection geosrs:WagnerVIIIProjection geosrs:WagnerVIIProjection

## A.6.19. Projection

### ABSTRACT TEST A.39

**IDENTIFIER** /conf/projections/Projection

**REQUIREMENT** Requirement 35: /req/projections/Projection

**TEST PURPOSE** Check conformance with this requirement

**TEST METHOD** Verify that queries involving geosrs:Projection return the correct result on a test dataset.

**TEST-METHOD-TYPE** Capabilities

**REFERENCE** geosrs:Projection

## A.7. Conformance Class: Planet

## CONFORMANCE CLASS A.7: 12-PLANET\_MODULE.ADOC

|                    |                                                    |
|--------------------|----------------------------------------------------|
| IDENTIFIER         | /conf/planet                                       |
| REQUIREMENTS CLASS | Requirements class 7: /req/planet                  |
| CONFORMANCE TEST   | Abstract test A.40: /conf/planet/Interstellar_Body |

### A.7.1. Interstellar Body

#### ABSTRACT TEST A.40

|                  |                                                                                                                                                                                                                                                                                                   |
|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/planet/Interstellar_Body                                                                                                                                                                                                                                                                    |
| REQUIREMENT      | Requirement 40: /req/planet/Interstellar_Body                                                                                                                                                                                                                                                     |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                           |
| TEST METHOD      | Verify that queries involving geosrs:ArtificialSatellite geosrs:Asteroid geosrs:Comet geosrs:Dwarf Planet geosrs:InterstellarBody geosrs:Moon geosrs:NaturalSatellite geosrs:Planet geosrs:Planet Status geosrs:Plutoid geosrs:Star geosrs:Satellite return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                      |
| REFERENCE        | geosrs:ArtificialSatellite geosrs:Asteroid geosrs:Comet geosrs:DwarfPlanet geosrs:Interstellar Body geosrs:Moon geosrs:NaturalSatellite geosrs:Planet geosrs:PlanetStatus geosrs:Plutoid geosrs:Star geosrs:Satellite                                                                             |

## A.8. Conformance Class: Instances

#### CONFORMANCE CLASS A.8: 13-INSTANCES.ADOC

|                    |                                                                                                                                                                      |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER         | /conf/instances                                                                                                                                                      |
| REQUIREMENTS CLASS | Requirements class 8: /req/instances                                                                                                                                 |
| CONFORMANCE TESTS  | Abstract test A.41: /conf/instances/Coordinate_System_Axis<br>Abstract test A.42: /conf/instances/Spheroids<br>Abstract test A.43: /conf/instances/SRS_Literal_Types |

## A.8.1. Coordinate System Axis

### ABSTRACT TEST A.41

|                  |                                                                                                                                                    |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/instances/Coordinate_System_Axis                                                                                                             |
| REQUIREMENT      | Requirement 41: /req/instances/Coordinate_System_Axis                                                                                              |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                            |
| TEST METHOD      | Verify that queries involving geosrs:Down geosrs:East geosrs:North geosrs:South geosrs:Up geosrs:West return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                       |
| REFERENCE        | geosrs:Down geosrs:East geosrs:North geosrs:South geosrs:Up geosrs:West                                                                            |

## A.8.2. Spheroids

### ABSTRACT TEST A.42

|                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/instances/Spheroids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| REQUIREMENT      | Requirement 43: /req/instances/Spheroids                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| TEST METHOD      | Verify that queries involving geosrs:GRS1980 geosrs:GRS67 geosrs:PZ90 geosrs:Airy1830 geosrs:AiryModified1849 geosrs:International1924 geosrs:AustralianNationalSpheroid geosrs:Everest1930 geosrs:Clarke1866 geosrs:Plessis1817 geosrs:Danish1876 geosrs:Struve1860 geosrs:LAG1975 geosrs:Clarke1858 geosrs:Clarke1880 geosrs:Helmert1906 geosrs:CGCS2000 geosrs:GSK-2011 geosrs:Zach1812 geosrs:Clarke1880ARC geosrs:Clarke1880IGN geosrs:WGS66 geosrs:WGS72 geosrs:WGS84 geosrs:Krassowsky1940 return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| REFERENCE        | geosrs:GRS1980 geosrs:GRS67 geosrs:PZ90 geosrs:Airy1830 geosrs:AiryModified1849 geosrs:International1924 geosrs:AustralianNationalSpheroid geosrs:Everest1930 geosrs:Clarke1866 geosrs:Plessis1817 geosrs:Danish1876 geosrs:Struve1860 geosrs:LAG1975 geosrs:Clarke1858 geosrs:Clarke1880 geosrs:Helmert1906 geosrs:CGCS2000 geosrs:GSK-2011 geosrs:Zach1812                                                                                                                                                                                   |

## ABSTRACT TEST A.42

geosrs:Clarke1880ARC geosrs:Clarke1880IGN geosrs:WGS66 geosrs:WGS72 geosrs:WGS84  
geosrs:Krassowsky1940

### A.8.3. SRS Literal Types

## ABSTRACT TEST A.43

|                  |                                                                                                                                         |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| IDENTIFIER       | /conf/instances/SRS_Literal_Types                                                                                                       |
| REQUIREMENT      | Requirement 42: /req/instances/SRS_Literal_Types                                                                                        |
| TEST PURPOSE     | Check conformance with this requirement                                                                                                 |
| TEST METHOD      | Verify that queries involving geosrs:proj4Literal geosrs:projJSONLiteral geosrs:wktLiteral return the correct result on a test dataset. |
| TEST-METHOD-TYPE | Capabilities                                                                                                                            |
| REFERENCE        | geosrs:proj4Literal geosrs:projJSONLiteral geosrs:wktLiteral                                                                            |



# ANNEX B (INFORMATIVE) ALIGNMENTS

---

# B

## ANNEX B (INFORMATIVE) ALIGNMENTS

### Overview

The prefixes used for the ontologies mapped to in all following sections are given in the following table.

**Table B.1** — Alignment: Namespaces

|           |                                                                                                                                                                               |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ign:      | <a href="http://data.ign.fr/def/ignf#">http://data.ign.fr/def/ignf#</a>                                                                                                       |
| iso19111: | <a href="http://def.isotc211.org/iso19112/2019/SpatialReferencingByGeographicIdentifier#">http://def.isotc211.org/iso19112/2019/SpatialReferencingByGeographicIdentifier#</a> |
| geosrs:   | <a href="http://www.opengis.net/ont/geosparql#">http://www.opengis.net/ont/geosparql#</a>                                                                                     |
| ifc:      | <a href="https://standards.buildingsmart.org/IFC/DEV/IFC4/ADD2_TC1/OWL/">https://standards.buildingsmart.org/IFC/DEV/IFC4/ADD2_TC1/OWL/</a>                                   |
| owl:      | <a href="http://www.w3.org/2002/07/owl#">http://www.w3.org/2002/07/owl#</a>                                                                                                   |
| prov:     | <a href="http://www.w3.org/ns/prov#">http://www.w3.org/ns/prov#</a>                                                                                                           |
| rdf:      | <a href="http://www.w3.org/1999/02/22-rdf-syntax-ns#">http://www.w3.org/1999/02/22-rdf-syntax-ns#</a>                                                                         |
| rdfs:     | <a href="http://www.w3.org/2000/01/rdf-schema#">http://www.w3.org/2000/01/rdf-schema#</a>                                                                                     |

### B.1. IGN Ontology

**Table B.2 – Alignment: IGN Ontology**

| FROM ELEMENT                                       | MAPPING RELATION                    | TO ELEMENT                                  | NOTES |
|----------------------------------------------------|-------------------------------------|---------------------------------------------|-------|
| <a href="#">geosrs:CoordinateSystem</a>            | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CoordinateSystem</a>        | -     |
| <a href="#">geosrs:Datum</a>                       | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Datum</a>                   | -     |
| <a href="#">geosrs:Ellipsoid</a>                   | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Ellipsoid</a>               | -     |
| <a href="#">geosrs:Conversion</a>                  | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Conversion</a>              | -     |
| <a href="#">geosrs:CoordinateOperation</a>         | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CoordinateOperation</a>     | -     |
| <a href="#">geosrs:OperationMethod</a>             | <a href="#">owl:equivalentClass</a> | <a href="#">ign:OperationMethod</a>         | -     |
| <a href="#">geosrs:OperationParameter</a>          | <a href="#">owl:equivalentClass</a> | <a href="#">ign:OperationParameter</a>      | -     |
| <a href="#">geosrs:OperationParameterValue</a>     | <a href="#">owl:equivalentClass</a> | <a href="#">ign:OperationParameterValue</a> | -     |
| <a href="#">geosrs:SingleOperation</a>             | <a href="#">owl:equivalentClass</a> | <a href="#">ign:SingleOperation</a>         | -     |
| <a href="#">geosrs:Transformation</a>              | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Transformation</a>          | -     |
| <a href="#">geosrs:CartesianCoordinateSystem</a>   | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CartesianCS</a>             | -     |
| <a href="#">geosrs:CoordinateSystem</a>            | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CoordinateSystem</a>        | -     |
| <a href="#">geosrs:CoordinateSystemAxis</a>        | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CoordinateSystemAxis</a>    | -     |
| <a href="#">geosrs:EllipsoidalCoordinateSystem</a> | <a href="#">owl:equivalentClass</a> | <a href="#">ign:EllipsoidalICS</a>          | -     |
| <a href="#">geosrs:VerticalCoordinateSystem</a>    | <a href="#">owl:equivalentClass</a> | <a href="#">ign:VerticalCS</a>              | -     |
| <a href="#">geosrs:Datum</a>                       | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Datum</a>                   | -     |
| <a href="#">geosrs:Ellipsoid</a>                   | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Ellipsoid</a>               | -     |
| <a href="#">geosrs:GeodeticDatum</a>               | <a href="#">owl:equivalentClass</a> | <a href="#">ign:GeodeticDatum</a>           | -     |
| <a href="#">geosrs:PrimeMeridian</a>               | <a href="#">owl:equivalentClass</a> | <a href="#">ign:PrimeMeridian</a>           | -     |
| <a href="#">geosrs:VerticalDatum</a>               | <a href="#">owl:equivalentClass</a> | <a href="#">ign:VerticalDatum</a>           | -     |
| <a href="#">geosrs:AxesList</a>                    | <a href="#">owl:equivalentClass</a> | <a href="#">ign:AxesList</a>                | -     |

| FROM ELEMENT                                 | MAPPING RELATION                    | TO ELEMENT                                | NOTES |
|----------------------------------------------|-------------------------------------|-------------------------------------------|-------|
| <a href="#">geosrs:CRS</a>                   | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CRS</a>                   | -     |
| <a href="#">geosrs:CompoundCRS</a>           | <a href="#">owl:equivalentClass</a> | <a href="#">ign:CompoundCRS</a>           | -     |
| <a href="#">geosrs:Extent</a>                | <a href="#">owl:equivalentClass</a> | <a href="#">ign:Extent</a>                | -     |
| <a href="#">geosrs:GeodeticCRS</a>           | <a href="#">owl:equivalentClass</a> | <a href="#">ign:GeodeticCRS</a>           | -     |
| <a href="#">geosrs:GeographicBoundingBox</a> | <a href="#">owl:equivalentClass</a> | <a href="#">ign:GeographicBoundingBox</a> | -     |
| <a href="#">geosrs:ProjectedCRS</a>          | <a href="#">owl:equivalentClass</a> | <a href="#">ign:ProjectedCRS</a>          | -     |
| <a href="#">geosrs:SingleCRS</a>             | <a href="#">owl:equivalentClass</a> | <a href="#">ign:SingleCRS</a>             | -     |
| <a href="#">geosrs:SingleCRSList</a>         | <a href="#">owl:equivalentClass</a> | <a href="#">ign:SingleCRSList</a>         | -     |
| <a href="#">geosrs:VerticalCRS</a>           | <a href="#">owl:equivalentClass</a> | <a href="#">ign:VerticalCRS</a>           | -     |

## B.2. ISO19111 Ontology

**Table B.3** – Alignment: ISO19111 Ontology

| FROM ELEMENT                            | MAPPING RELATION                    | TO ELEMENT                                | NOTES |
|-----------------------------------------|-------------------------------------|-------------------------------------------|-------|
| <a href="#">geosrs:CoordinateSystem</a> | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:CoordinateSystem</a> | -     |
| <a href="#">geosrs:Datum</a>            | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:Datum</a>            | -     |
| <a href="#">geosrs:Ellipsoid</a>        | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:Ellipsoid</a>        | -     |
| <a href="#">geosrs:CRS</a>              | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:CRS</a>              | -     |
| <a href="#">geosrs:CompoundCRS</a>      | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:CompoundCRS</a>      | -     |
| <a href="#">geosrs:EngineeringCRS</a>   | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:EngineeringCRS</a>   | -     |
| <a href="#">geosrs:GeodeticCRS</a>      | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:GeodeticCRS</a>      | -     |



| FROM ELEMENT                         | MAPPING RELATION                    | TO ELEMENT                             | NOTES |
|--------------------------------------|-------------------------------------|----------------------------------------|-------|
| <a href="#">geosrs:GeographicCRS</a> | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:GeographicCRS</a> | -     |
| <a href="#">geosrs:ParametricCRS</a> | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:ParametricCRS</a> | -     |
| <a href="#">geosrs:ProjectedCRS</a>  | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:ProjectedCRS</a>  | -     |
| <a href="#">geosrs:SingleCRS</a>     | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:SingleCRS</a>     | -     |
| <a href="#">geosrs:TemporalCRS</a>   | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:TemporalCRS</a>   | -     |
| <a href="#">geosrs:VerticalCRS</a>   | <a href="#">owl:equivalentClass</a> | <a href="#">iso19111:VerticalCRS</a>   | -     |

## B.3. IFC Ontology

**Table B.4** — Alignment: IFC Ontology

| FROM ELEMENT                               | MAPPING RELATION                       | TO ELEMENT                                       | NOTES |
|--------------------------------------------|----------------------------------------|--------------------------------------------------|-------|
| <a href="#">geosrs:AxisDirection</a>       | <a href="#">owl:equivalentClass</a>    | <a href="#">ifc:IfcDirection</a>                 | -     |
| <a href="#">geosrs:CRS</a>                 | <a href="#">owl:equivalentClass</a>    | <a href="#">ifc:IfcCoordinateReferenceSystem</a> | -     |
| <a href="#">geosrs:CoordinateOperation</a> | <a href="#">owl:equivalentClass</a>    | <a href="#">ifc:IfcCoordinateOperation</a>       | -     |
| <a href="#">geosrs:ProjectedCRS</a>        | <a href="#">owl:equivalentClass</a>    | <a href="#">ifc:IfcProjectedCRS</a>              | -     |
| <a href="#">geosrs:axis</a>                | <a href="#">owl:equivalentProperty</a> | <a href="#">ifc:axis_IfcAxis1Placement</a>       | -     |
| <a href="#">geosrs:sourceCRS</a>           | <a href="#">owl:equivalentProperty</a> | <a href="#">ifc:sourceCRS</a>                    | -     |
| <a href="#">geosrs:targetCRS</a>           | <a href="#">owl:equivalentProperty</a> | <a href="#">ifc:targetCRS</a>                    | -     |



# ANNEX C (INFORMATIVE) SHACL SHAPES

---



## ANNEX C (INFORMATIVE) SHACL SHAPES

This section introduces SHACL shapes which can be used to verify graphs encoded using the vocabulary defined in this specification.

### Overview

SHACL shapes in this specification are subdivided by the same module designations as used previously. In order to verify a graph a single validation file of SHACL shapes is provided alongside this specification.

### C.1. SHACL Shapes: Core

Table C.1 — Core

| LABEL    | TARGETNODE | PROPERTY                | CLASS                   | MINCOUNT | MAXCOUNT | COMMENT                                                                 |
|----------|------------|-------------------------|-------------------------|----------|----------|-------------------------------------------------------------------------|
| Shape S1 | geosrs:CRS | geosrs:coordinateSystem | geosrs:CoordinateSystem | 1        | 1        | A coordinate reference system should have exactly one coordinate system |
| Shape S2 | geosrs:CRS | geosrs:domainOfValidity | geosrs:AreaOfUse        | 1        | -        | A coordinate reference system should have at least one area of use      |
| Shape S3 | geosrs:CRS | geosrs:datum            | geosrs:Datum            | -        | 1        | A coordinate reference                                                  |

| LABEL    | TARGETNODE           | PROPERTY                 | CLASS                               | MINCOUNT | MAXCOUNT | COMMENT                                                                                                  |
|----------|----------------------|--------------------------|-------------------------------------|----------|----------|----------------------------------------------------------------------------------------------------------|
|          |                      |                          |                                     |          |          | system should have exactly one datum                                                                     |
| Shape S4 | geosrs:CRS           | geosrs:datum Ensemble    | geosrs:DatumEnsemble                | -        | 1        | A coordinate reference system may have exactly one datum ensemble                                        |
| Shape S5 | geosrs:CompoundCRS   | geosrs:includes SRS      | geosrs:SingleCRS                    | 1        | -        | A compound coordinate reference system should consist of at least one single coordinate reference system |
| Shape S6 | geosrs:GeodeticCRS   | geosrs:coordinate System | geosrs:GeodeticCoordinate System    | 1        | 1        | A geodetic coordinate reference system should have exactly one geodetic coordinate system                |
| Shape S7 | geosrs:GeographicCRS | geosrs:datum             | geosrs:GeodeticDatum                | 1        | 1        | A geographic coordinate reference system should have exactly one geodetic datum                          |
| Shape S8 | geosrs:GeographicCRS | geosrs:coordinate System | geosrs:EllipsoidalCoordinate System | 1        | 1        | A geographic coordinate reference system should have                                                     |

| LABEL     | TARGETNODE           | PROPERTY                 | CLASS                   | MINCOUNT | MAXCOUNT | COMMENT                                                                           |
|-----------|----------------------|--------------------------|-------------------------|----------|----------|-----------------------------------------------------------------------------------|
|           |                      |                          |                         |          |          | exactly one ellipsoidal coordinate system                                         |
| Shape S9  | geosrs:ParametricCRS | geosrs:datum             | geosrs:ParametricDatum  | 1        | 1        | A parametric coordinate reference system should have exactly one parametric datum |
| Shape S10 | geosrs:ProjectedCRS  | geosrs:conversion        | geosrs:Conversion       | 1        | -        | A projected coordinate reference system should have at least one conversion       |
| Shape S11 | geosrs:Single CRS    | geosrs:coordinate System | geosrs:CoordinateSystem | 1        | 1        | A single coordinate reference system should have exactly one coordinate system    |
| Shape S12 | geosrs:Single CRS    | geosrs:datum             | geosrs:Datum            | 1        | 1        | A single coordinate reference system should have exactly one datum                |
| Shape S13 | geosrs:TemporalCRS   | geosrs:datum             | geosrs:TemporalDatum    | 1        | 1        | A projected coordinate reference system should have exactly one temporal datum    |

## C.2. SHACL Shapes: Datum

Table C.2 – Datum

| LABEL    | TARGETNODE             | PROPERTY                 | CLASS                    | MINCOUNT | MAXCOUNT | COMMENT                                                        |
|----------|------------------------|--------------------------|--------------------------|----------|----------|----------------------------------------------------------------|
| Shape S1 | geosrs:ParametricDatum | geosrs:definingParameter | geosrs:DefiningParameter | 1        | -        | A parametric datum should have at least one defining parameter |

## C.3. SHACL Shapes: Cs

Table C.3 – Cs

| LABEL    | TARGETNODE                     | PROPERTY    | CLASS                       | MINCOUNT | MAXCOUNT | COM                                                    |
|----------|--------------------------------|-------------|-----------------------------|----------|----------|--------------------------------------------------------|
| Shape S1 | geosrs:1DCoordinateSystem      | geosrs:axis | geosrs:CoordinateSystemAxis | 1        | 1        | A 1D coordinate system should have exactly one axis    |
| Shape S2 | geosrs:3DCoordinateSystem      | geosrs:axis | geosrs:CoordinateSystemAxis | 3        | -        | A 3D coordinate system should have at least three axes |
| Shape S3 | geosrs:ConicalCoordinateSystem | geosrs:axis | geosrs:CoordinateSystemAxis | 3        | -        | A conical coordinate system                            |

| LABEL    | TARGETNODE                         | PROPERTY             | CLASS                       | MINCOUNT | MAXCOUNT | COM                                                                                                         |
|----------|------------------------------------|----------------------|-----------------------------|----------|----------|-------------------------------------------------------------------------------------------------------------|
|          |                                    |                      |                             |          |          | should have at least three axes                                                                             |
| Shape S4 | geosrs:CoordinateSystem            | geosrs:axis          | geosrs:CoordinateSystemAxis | 1        | -        | A coordinate system should have at least one axis                                                           |
| Shape S5 | geosrs:CoordinateSystemAxis        | geosrs:axisDirection | geosrs:AxisDirection        | 1        | 1        | A coordinate system axis should have exactly one axis direction                                             |
| Shape S6 | geosrs:CurvilinearCoordinateSystem | geosrs:axis          | geosrs:CoordinateSystemAxis | 3        | -        | A curvilinear coordinate system is defined in Euclidean space and should therefore have at least three axes |
| Shape S7 | geosrs:CylindricalCoordinateSystem | geosrs:axis          | geosrs:CoordinateSystemAxis | 3        | -        | A cylindrical coordinate                                                                                    |

| LABEL     | TARGETNODE                              | PROPERTY    | CLASS                       | MINCOUNT | MAXCOUNT | COM                                                                 |
|-----------|-----------------------------------------|-------------|-----------------------------|----------|----------|---------------------------------------------------------------------|
|           |                                         |             |                             |          |          | system should have at least three axes                              |
| Shape S8  | geosrs:DateTimeTemporalCoordinateSystem | geosrs:axis | geosrs:CoordinateSystemAxis | 1        | 1        | A date time temporal coordinate system should have exactly one axis |
| Shape S9  | geosrs:PlanarCoordinateSystem           | geosrs:axis | geosrs:CoordinateSystemAxis | 2        | -        | A planar coordinate system should have at least two axes            |
| Shape S10 | geosrs:TemporalCoordinateSystem         | geosrs:axis | geosrs:CoordinateSystemAxis | 1        | 1        | A temporal coordinate system should have exactly one axis           |
| Shape S11 | geosrs:TemporalCountCoordinateSystem    | geosrs:axis | geosrs:CoordinateSystemAxis | 1        | 1        | A temporal count coordinate system should have exactly              |



| LABEL     | TARGETNODE                              | PROPERTY    | CLASS                        | MINCOUNT | MAXCOUNT | COM                                                               |
|-----------|-----------------------------------------|-------------|------------------------------|----------|----------|-------------------------------------------------------------------|
|           |                                         |             |                              |          |          | one axis                                                          |
| Shape S12 | geosrs:TemporalMeasureCoordinate System | geosrs:axis | geosrs:Coordinate SystemAxis | 1        | 1        | A temporal measure coordinate system should have exactly one axis |



# ANNEX D (INFORMATIVE) APPLICATION EXAMPLES

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## ANNEX D (INFORMATIVE) APPLICATION EXAMPLES

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### Overview

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#### D.1. Minimum Example

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#### D.2. Elaborate Example

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# ANNEX E (INFORMATIVE) JSON-LD CONTEXT

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## ANNEX E (INFORMATIVE) JSON-LD CONTEXT

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We provide JSON-LD contexts to be compatible with other JSON-based formats which provide coordinate reference system data.

### Overview

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#### E.1. Compatibility to PROJJSON

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PROJJSON is an established format to share geospatial data which has emerge from the PROJ library and encodes the WKT encoding of coordiante references systems. By adding a JSON-LD context to the PROJJSON standard we achieve an immediate compatibility with an established standard simply by extending it by one simple statement.

```
{
  "@context": "https://opengeospatial.github.io/ontology-crs/context/geosrs-
context.json",
  "$schema": "https://proj.org/schemas/v0.7/projjson.schema.json",
  ...
}
```

Listing E.1

We provide examples of application of this JSON-LD context with the distribution of this standard.

#### E.2. Compatibility to OGCJSON

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The OGC CRS working group is aiming towards the creation of their own JSON format for CRS. The JSON-LD context we provide aims to be compatible with both PROJJSON and OGCJSON.



# ANNEX F (INFORMATIVE) REVISION HISTORY

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# ANNEX F

## (INFORMATIVE)

### REVISION HISTORY

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| DATE       | RELEASE | AUTHOR    | PRIMARY CLAUSES MODIFIED | DESCRIPTION     |
|------------|---------|-----------|--------------------------|-----------------|
| 2016-04-28 | 0.1     | G. Editor | all                      | initial version |



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